

Supporting Information

Halogenated Metal-Organic Framework Glasses and Liquids

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Experimental

Synthesis

ZIF-UC-2 (6-Cl-5-FbIm): Zinc nitrate hexahydrate (220 mg, 0.74 mmol) and imidazole (1188 mg, 16.42 mmol) were dissolved in 10 mL *N,N*-dimethylformamide (DMF) within a screw top vial. After being stirred for 1 hour, 6-chloro-5-fluorobenzimidazole (297.4 mg, 1.74 mmol) was added to the mixture. The solution was heated at 403 K for 48 hours and then cooled at 5 K hour⁻¹.

ZIF-UC-3 (5-Cl-2-mbIm): Imidazole (60 mg, 0.89 mmol), 5-chloro-2-methylbenzimidazole (293 mg, 1.76 mmol) and zinc nitrate hexahydrate (262 mg, 0.88 mmol) were fully dissolved in a solution of 3 mL DMF and 4 mL *N,N*-diethylformamide (DEF), to which 0.25 mL of 2.5 mol dm⁻³ NaOH was added. The mixture was stirred for 1 hour and subsequently heated to 383 K for 5 days. The crystalline product was collected by filtration with dichloromethane.

ZIF-UC-4 (5-FbIm) and ZIF-UC-5 (5-ClbIm): Imidazole (120 mg, 1.77 mmol) and 5-fluorobenzimidazole (112 mg, 0.82 mmol) or 5-chlorobenzimidazole (130 mg, 0.85 mmol) were fully dissolved in a solution of 8.3 mL DMF and 5.7 mL DEF. Then zinc nitrate hexahydrate (120 mg, 0.40 mmol) was added to the solution and stirred for 1 hour. Subsequently, NaOH (0.52 mL, 2.5 mol dm⁻³) (10 wt%) was added to the mixture, and the turbid solution was then heated to 363 K for 5 days. To obtain a pure phase of product, the cooling rate was set at 5 K hour⁻¹.

Prior to subsequent experiments, all ZIF crystals were centrifuged and washed with dichloromethane for three times, and then placed in a vacuum oven at 450 K overnight for activation.

Single crystal X-ray measurement

Single crystal X-ray diffraction data were collected at 298 K on a Gemini E Ultra diffractometer from Oxford Diffraction with an area detector Atlas CCD. Data collection was performed using Cu K α (λ = 1.540598 Å) radiation. The structure was solved and refined using the program WinGX version 2018.2.¹ More details on data collection and structural refinement are presented in the results section.

Powder X-ray diffraction

Powder X-ray diffraction was carried out under ambient conditions with a Bruker-AXS D8 diffractometer and a Cu K α (λ = 1.540598 Å) radiation source. The detector was a LynxEye position-sensitive detector in Bragg-Brentano parafocusing geometry. The 2 θ range was 5° to 40°, with a step size of 0.02° and a 10s step rate. Pawley refinement was performed with the program jEdit 4.3 and TOPAS academic V5.² The unit cell parameters obtained by single crystal X-ray diffraction were used in the Pawley refinement.³

Liquid-state nuclear magnetic resonance spectroscopy

Liquid-state ¹H and ¹⁹F nuclear magnetic resonance (NMR) measurements were conducted with ca. 6 mg of sample, digested in a mixture of DCl/D₂O (35%; 0.1 mL) and DMSO-d₆ (0.5 mL). The ¹H NMR measurement was performed with a Bruker Avance III 400 MHz spectrometer at ambient conditions. Chemical shifts were referenced to the residual protio-solvent signals of DMSO-d₆. The ¹⁹F spectra were recorded using Varian Inova 300 MHz spectrometer and referenced relative to external CFC1₃.

Thermogravimetric and calorimetric analysis

A TA Instruments SDT Q600 was used for thermogravimetric (TG) analysis. Samples (*ca.* 10 mg) were held in an alumina pan under a continuous flow of argon gas at a rate of 100 ml min⁻¹. For the melting experiments, the heating rate was controlled at 20 K min⁻¹. The melting and glass transition temperatures of the sample were measured with a Netzsch 214 Polyma instrument. The sample (*ca.* 4 mg) was sealed in an aluminium crucible, with an *ca.* 0.5 mm hole on the lid. The temperature ramp was set at 20 K min⁻¹ for the first heating (melting) cycle and then 10 K min⁻¹ for the downscan and second heating cycle. To determine the glass transition temperature, the samples were heated to 693 K for ZIF-UC-2 (6-Cl-5-FbIm), 683 K for ZIF-UC-3 (5-Cl-2-mbIm), 708 K for ZIF-UC-4 (5-FbIm) and 723 K for ZIF-UC-5 (5-ClbIm) in the first cycle, before being cooled back to 310 K for the second heating cycle. An empty pan was run with identical settings and this was subtracted from the original trace, in order to baseline correct all scans.

Scanning electron microscopy

The surface morphologies of the crystal and glass samples were examined with a high-resolution scanning electron microscope (FEI Nova Nano SEM 450, accelerating voltage 10 kV). After being vacuum dried overnight, all samples were coated with a layer of gold prior to imaging.

Synchrotron X-ray total scattering and pair distribution function analysis

XRD data were collected on the I15-1 beamline at the Diamond Light Source, UK using an X-ray wavelength $\lambda = 0.161669 \text{ \AA}$ (76.7 keV). Crystalline and glass samples were loaded into borosilicate glass capillaries of 1.17 mm (inner) diameter. Data from the samples, empty instrument and empty capillary were collected in the region of $\sim 0.4 < Q < \sim 26 \text{ \AA}^{-1}$. Corrections for background, multiple scattering, container scattering, Compton scattering and absorption were performed using the GudrunX program.⁴ The refinement was carried out with PDFGUI software.⁵

Solid-state nuclear magnetic resonance (NMR) spectroscopy

Solid-state NMR measurements were performed on a 600 MHz Varian NMR system equipped with a 1.6 mm HXY magic angle spinning (MAS) probe. For all the experiments, the sample spinning frequency was 40 kHz. ¹⁹F MAS NMR spectra were recorded using Hahn-echo sequence with inter-pulse delays of 25 μ s (one rotation period). The 90° pulse length was 1.7 μ s, 180° pulse length was 3.4 μ s, and 160 scans were accumulated with the repetition delay of 5 s. The ¹H-¹³C cross-polarization (CP) MAS experiments consisted of excitation of protons with a 90° pulse of 1.5 μ s, followed by the CP block of 5 ms, and signal acquisition with high-power XiX proton decoupling. A total of 40,000 scans were accumulated with the repetition delay of 1 s. For the ¹H-¹³C Lee-Goldburg cross-polarization (LG-CP) experiment, the LG-CP scheme with duration of 50 μ s was used. The ¹⁹F-¹³C CPMAS experiment consisted of ¹⁹F excitation pulse of 1.7 μ s and the CP block of 1 ms. The number of scans was 10,000 and recycling delay was 1 s. For the selective ¹⁹F-¹³C CPMAS experiment the ¹⁹F transmitting frequency was set to the selected peak position and a low-power ¹⁹F excitation pulse with duration of 10 μ s was applied prior to the CP block. The contact time was 5 ms, the repetition delay 2.5 s and 3300 scans were accumulated. The two-dimensional ¹⁹F spin-diffusion measurement on ZIF-UC-4 (5-FbIm) consisted of 50 increments along the indirect dimension with 32 scans per increment and repetition delay of 1 s; spectral width in the indirect dimension was 40 kHz. The spin-diffusion mixing time was 100 ms. The wide-range ¹⁹F spin-diffusion experiment on *a*_gZIF-UC-4 (5-FbIm) was performed with 80 increments and 64 scans for each increment; spectral width in the indirect dimension was 160 kHz. Spin-diffusion mixing times

varied between 0 and 100 ms. ^{19}F spin-lattice (T_1) and spin-spin (T_2) relaxation-time measurements were carried out under MAS using the inversion-recovery and spin-echo pulse sequences, respectively. Larmor frequencies for ^{19}F and ^{13}C nuclei were 564.03 MHz and 150.73 MHz, respectively. ^{19}F chemical shifts were referenced to the signal of CFCl_3 and ^{13}C chemical shifts to the signal of TMS.

Gas sorption and gas pycnometry (density) tests

Nitrogen and carbon dioxide gas sorption isotherm tests were carried out with a Micromeritics ASAP 2020 instrument. For each measurement, at least 100 mg sample was used. All samples were degassed overnight at 450 K under vacuum. For measurement, the equilibration interval was 45 s and the minimum equilibration delay at $p/p^* \geq 0.995$ was 600s. The BET surface area was calculated using pyGAPS v2.0.2 (Python General Adsorption Processing Suite) framework on Python3.5.⁶ A Micromeritics Accupyc 1340 helium pycnometer was used for the density measurement. For each test, a minimum of 100 mg of sample was applied and a cycle of 10 measurements was conducted. The volumetric gas uptake (cm^3 (STP) / cm^3 adsorbent) was calculated with the measured sample density.

Nanoindentation measurement

The mechanical properties of the glass samples were measured with an MTS Nanoindenter XP system. All glass samples were mounted within epoxy resin followed by polishing with fine diamond suspension liquids. Nanoindentation experiments were carried out at ambient conditions under the dynamic displacement controlled mode, with a constant strain rate at 0.05 s^{-1} . The testing penetration depth of 1000 nm using a three-sided pyramidal (Berkovich) diamond indenter tip. A Poisson's ratio of 0.4 was applied. Each sample was tested for at least 20 different locations. The load-displacement data were analyzed using the Oliver and Pharr method, with the gradient of unloading curves measured at 90 % of the maximum load.⁷

Computational methodology for bonding strength and charge density

DFT calculations were performed using projected augmented wave (PAW)^{8,9} potentials as implemented in VASP code^{10,11} with following electrons treated explicitly: Zn $3d^{10}4s^2$, O $2s^22p^4$, C $2s^22p^2$ N $2s^22p^3$ and H $1s^1$. Van der Waals functional (optB86b-vdW)¹² was used to treat the exchange-correlation part. A 500 eV kinetic energy cut-off and a gamma point only k-point mesh were used for all calculations. Calculations were performed on the clusters of $\text{Zn}(\text{ImH})_3(5\text{-ClImH})$ and $\text{Zn}(\text{ImH})_3(5\text{-FbImH})$ where all the bridging nitrogens within Im and bIm are connected to a H atom. Clusters were put at the center of a $30 \times 30 \times 30 \text{ \AA}^3$ box and relaxed until interatomic forces were smaller than $0.01 \text{ eV/\AA}$.

Pelletized bulk glass

200 mg ZIF-UC-3 (5-Cl-2-mbIm) was vitrified at 683 K in a tube furnace at a ramp rate of 20 K min^{-1} . Then the sample was held at 683 K for 14 min, before being cooled to room temperature. PXRD was performed to ensure the sample was fully amorphized. The quenched sample was ball milled at 25 Hz for 5 minutes with two 7 mm stainless steel balls to produce a fine powder. Then the powder was pressed into a 13 mm diameter pellet with the application of 10 tons for 1 min. The pellet was then heated to 653 K (i.e., above T_g but below the decomposition temperature of the crystal) and held for 5 hours. The resultant sample was referred to as $\text{ag-ZIF-UC-3 (5-Cl-2-mbIm)}$, where 'gr' refers to remelted glass.

Synchrotron TeraHz Far-Infrared (THz/Far-IR) absorption spectroscopy

THz/Far-IR absorption spectra of all samples were collected at the THz/Far-IR beamline at the Australian Synchrotron. The THz spectroscopy was performed using a Bruker IFS 125/HR

Fourier Transform (FT) spectrometer. The measurements were performed with a 6 μm thick Multilayer Mylar beamsplitter. In order to obtain a high signal-to-noise ratio, a liquid helium-cooled bolometer was deployed.

The Attenuated Total Reflection (ATR) method was applied for sample analysis: powdered samples were mounted on top of the diamond ATR crystal, then held in position by applying pressure with the lever arm. For each test, around 20 mg sample was used to ensure the sample thickness was higher than the penetration depth of the beam, which can be of the order of 20-200 μm . Prior to each sample measurement, the diamond crystal was cleaned with isopropanol, and background spectra were collected. For *in situ* heating/cooling measurements, the background spectra were recorded at each tested temperature prior to recording the corresponding sample spectra in order to obtain accurate data. The ATR heating stage and sample were enclosed in an environment of flowing argon gas. The collected spectral data were then corrected via Extended ATR correction algorithms in the OPUS 8.0 software. The peak positions in the spectra during the *in situ* heating test were assigned using NumPy module v1.15 together with Python v3.5.¹³

Computational details for THz/Far-IR spectra

Standard computational chemistry calculations were carried out using Gaussian 16 for the calculation of THz/Far-IR spectra. The detailed procedure can be found in the literature.¹⁴ Briefly, we used a series of molecular species to model the key component of interest of the materials system. They were complexes of Zn with three Im and one bIm derivatives, with an overall charge of minus two; they essentially corresponded to Zn(II) complexes. Geometries of these molecular models were optimized with the B3-LYP method¹⁵ in conjunction with the 6-31+G(d,p) basis set.¹⁶ Harmonic vibrational frequencies were obtained at the same level of theory and they were inspected to ensure that the optimized structures correspond to minima on their potential energy surface. Fundamental frequencies reported in the text correspond to scaled B3-LYP/6-31+G(d,p) harmonic frequencies using the literature scale factor of 0.9648.¹⁷ Simulated infrared spectra were obtained by assuming a Gaussian band for each peak with a full width at half maximum (FWHM) of 10 cm^{-1} .

Results

ZIF-UC-2 (6-Cl-5-FbIm) (**cag** — Orthorhombic)

Experimental and Refinement Details for **ZIF-UC-2 (6-Cl-5-FbIm)**

(CCDC Deposition Number 1951694)

A colourless polyhedral crystal ($0.10 \times 0.07 \times 0.05 \text{ mm}^3$) of ZIF-UC-2 (6-Cl-5-FbIm) was placed in a 0.2 mm diameter borosilicate loop and mounted on a Gemini E Ultra diffractometer from Oxford Diffraction, equipped with an Atlas CCD area detector and operated at 35 kV and 35 mA, to generate Cu K α radiation ($\lambda = 1.54184 \text{ \AA}$) at 293(2) K. A total of 10790 reflections were collected, of which 3641 were unique and 3021 of these were greater than $2\sigma(I)$. The range of θ was from 2.44 to 63.99°. Analysis of the data showed negligible decay during collection. The structure was solved in the orthorhombic *Pbca* space group with $Z = 8$ using the SUPERFLIP method.¹⁸ Modeling of electron density led to the identification of an asymmetric unit with two independent Zn atoms and four organic linkers. From those four linkers, three are unsubstituted Imidazoles with no disorder, and the fourth is disordered between 6-Cl-5-FbIm (with occupancy 0.260(7)) and Imidazole (with occupancy 0.740(7)). Note that in this fourth position the 6-Cl-5-FbIm linker is shifted away from the complementary Im position by 54.80°, which results in partial occupancy of sites normally occupied by solvent molecules. This shift avoids a steric clash with neighboring Im linkers in the fully occupied sites. This feature, in addition to a previous desolvation of the crystal caused the electron density within the void of the frameworks to be too low to model the solvent. With the exception of disordered Cl, F and four partial occupancy benzene carbon atoms, all non-hydrogen atoms have been refined anisotropically. All hydrogen atoms have been placed in geometrically located positions and their displacement parameters are tied to those of the attached carbon atoms. The unit cell of ZIF-UC-2 (6-Cl-5-FbIm) contains 1.87 Im and 0.13 6-Cl-5-FbIm per Zn. Final full matrix least squares refinement on F^2 converged to $R_1 = 0.0459$ ($F > 2\sigma F$) and $wR_2 = 0.1511$ (all data), with GooF = 1.074.

The structure shares the same *cag* network topology as ZIF-4 and ZIF-62.¹⁹

Table S1. Crystal data and structure refinement details for ZIF-UC-2 (6-Cl-5-FbIm).

Chemical formula	C _{13.04} H ₁₂ N ₈ Cl _{0.26} F _{0.26} Zn ₂
Formula weight	425.71 g/mol
Temperature	293(2)
Wavelength	1.54184 Å
Crystal system	Orthorhombic
Space group	<i>Pbca</i>
Unit cell dimensions	$a = 15.5560 (5) \text{ Å}$ $\alpha = 90^\circ$ $b = 15.6538 (5) \text{ Å}$ $\beta = 90^\circ$ $c = 18.0875 (6) \text{ Å}$ $\gamma = 90^\circ$
Volume	4404.5(2) Å ³
Density (calculated)	1.283 g/cm ³
Absorption coefficient	3.066
F(000)	1892.48
Crystal size	0.10 × 0.07 × 0.05 mm ³
Theta range for data collection	2.4428 - 63.9874°
Index ranges	-16 ≤ h ≤ 18, -17 ≤ k ≤ 18, -20 ≤ l ≤ 21
Reflections collected	10790
Independent reflections	3641
Completeness to theta = 66.97°	92.81 %
Absorption correction	multi-scan
Refinement method	Full-matrix least-squares on F^2
Data / restraints / parameters	3641 / 10 / 251
Goodness-of-fit on F^2	1.074
Final R indices [$I > 2\sigma(I)$]	$R_1 = 0.0459$, $wR_2 = 0.1408$
R indices (all data)	$R_1 = 0.0535$, $wR_2 = 0.1511$
Largest diff. peak and hole	0.757 and -0.482 e. Å ⁻³

Table S2. Additional refinement details for ZIF-UC-2 (6-Cl-5-FbIm): atomic positions (fractions), isotropic atomic displacement parameter (U iso or equivalent), type of atomic displacement parameters used for the site (Adp type) and corresponding site occupancy factors (Occup).

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occup
		x	y	z			
Zn1	Zn	0.69842(3)	0.14509(3)	0.09462(3)	0.0684(2)	Uani	1
Zn2	Zn	1.00115(3)	0.34160(3)	0.18728(3)	0.0673(2)	Uani	1
N4	N	0.8216(2)	0.1686(2)	0.11123(19)	0.0756(8)	Uani	1
N1	N	0.5567(2)	0.3167(2)	0.21662(17)	0.0746(8)	Uani	1
N5	N	0.6631(2)	0.0450(2)	0.15522(19)	0.0753(8)	Uani	1
N2	N	0.6318(2)	0.2442(2)	0.13360(17)	0.0732(8)	Uani	1
N3	N	0.9366(2)	0.2417(2)	0.14796(19)	0.0750(8)	Uani	1
N6	N	0.5823(2)	-0.0629(2)	0.19484(17)	0.0720(8)	Uani	1
C7	C	0.5990(3)	-0.0081(2)	0.1416(2)	0.0737(10)	Uani	1
H7	H	0.567937	-0.006823	0.097704	0.088	Uiso	1
C4	C	0.8522(2)	0.2408(3)	0.1385(2)	0.0748(10)	Uani	1
H4	H	0.817364	0.287222	0.15002	0.09	Uiso	1
C1	C	0.5915(3)	0.2437(3)	0.1981(2)	0.0778(10)	Uani	1
H1	H	0.588236	0.195412	0.227884	0.093	Uiso	1
C6	C	0.9620(3)	0.1650(3)	0.1222(4)	0.116(2)	Uani	1
H6	H	1.018639	0.146142	0.119591	0.139	Uiso	1
C8	C	0.6903(4)	0.0211(4)	0.2228(3)	0.116(2)	Uani	1
H8	H	0.73612	0.045209	0.248268	0.139	Uiso	1
C3	C	0.5758(5)	0.3677(3)	0.1597(3)	0.124(2)	Uani	1
H3	H	0.560992	0.425129	0.155855	0.149	Uiso	1
C2	C	0.6193(5)	0.3233(4)	0.1093(3)	0.126(2)	Uani	1
H2	H	0.638076	0.344364	0.06401	0.152	Uiso	1
C9	C	0.6405(4)	-0.0436(3)	0.2477(3)	0.1093(17)	Uani	1
H9	H	0.645537	-0.070232	0.293491	0.131	Uiso	1
C5	C	0.8926(4)	0.1200(4)	0.1009(4)	0.118(2)	Uani	1
H5	H	0.893289	0.064694	0.082134	0.142	Uiso	1
C10A	C	0.5949(9)	0.1453(8)	-0.0429(8)	0.064(2)	Uani	0.740(4)
H10A	H	0.550036	0.172077	-0.018037	0.077	Uiso	0.740(4)
N7A	N	0.6668(2)	0.1229(2)	-0.00937(18)	0.0801(9)	Uani	0.740(4)
N8A	N	0.5909(2)	0.1271(2)	-0.11374(19)	0.0816(9)	Uani	0.740(4)
C12A	C	0.6721(6)	0.1020(9)	-0.1298(4)	0.178(6)	Uani	0.740(4)
H12A	H	0.69301	0.089661	-0.176788	0.214	Uiso	0.740(4)
C11A	C	0.7176(6)	0.0979(10)	-0.0663(5)	0.187(7)	Uani	0.740(4)
H11A	H	0.774594	0.080655	-0.062311	0.225	Uiso	0.740(4)
C10B	C	0.6127(19)	0.161(2)	-0.053(2)	0.047(5)	Uani	0.260(4)
H10B	H	0.589977	0.214166	-0.039684	0.057	Uiso	0.260(4)
N7B	N	0.6668(2)	0.1229(2)	-0.00937(18)	0.0801(9)	Uani	0.260(4)

Continued

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occ
x	y	z					
N8B	N	0.5909(2)	0.1271(2)	-0.11374(19)	0.0816(9)	Uani	0.260(4)
C12B	C	0.6233(7)	0.0406(8)	-0.1068(6)	0.058(3)	Uani	0.260(4)
C11B	C	0.6723(7)	0.0385(8)	-0.0423(7)	0.061(3)	Uani	0.260(4)
C13	C	0.6170(11)	-0.0317(11)	-0.1518(9)	0.084(4)	Uiso	0.260(4)
H13	H	0.584561	-0.030954	-0.194931	0.101	Uiso	0.260(4)
C15	C	0.7152(13)	-0.1151(14)	-0.0696(12)	0.101(6)	Uiso	0.260(4)
C16	C	0.7175(12)	-0.0354(12)	-0.0219(11)	0.094(5)	Uiso	0.260(4)
H16	H	0.749531	-0.035492	0.021508	0.113	Uiso	0.260(4)
C14	C	0.6655(16)	-0.1127(18)	-0.1276(14)	0.123(7)	Uiso	0.260(4)
Cl	Cl	0.6661(6)	-0.1930(6)	-0.1833(4)	0.155(3)	Uiso	0.260(4)
F	F	0.7584(10)	-0.1829(10)	-0.0580(7)	0.135(5)	Uiso	0.260(4)

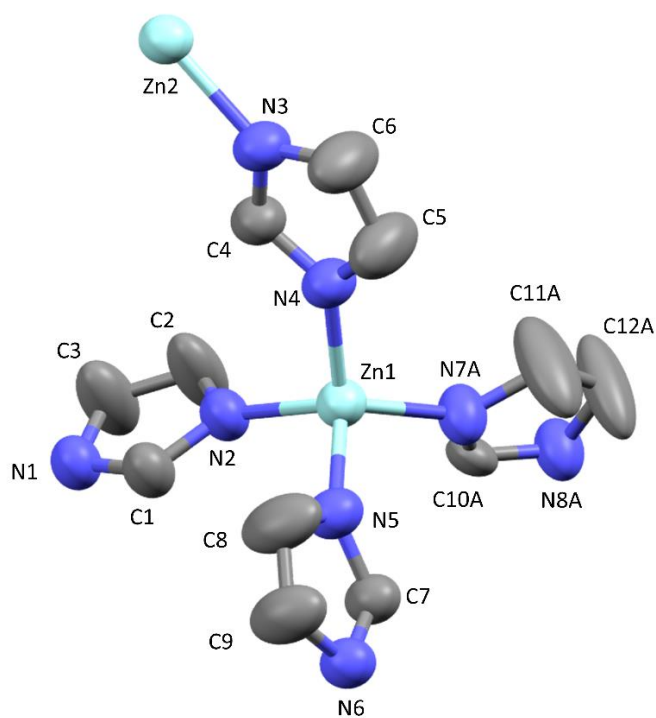


Figure S1. Graphical representation (thermal ellipsoids) of the first of two possible configurations for the asymmetric unit of ZIF-UC-2 (6-Cl-5-FbIm): two crystallographically independent Zn atoms, one of them surrounded by four independent Im linkers. This configuration represents 74% of the coordination environments for Zn atoms. H atoms are excluded for clarity.

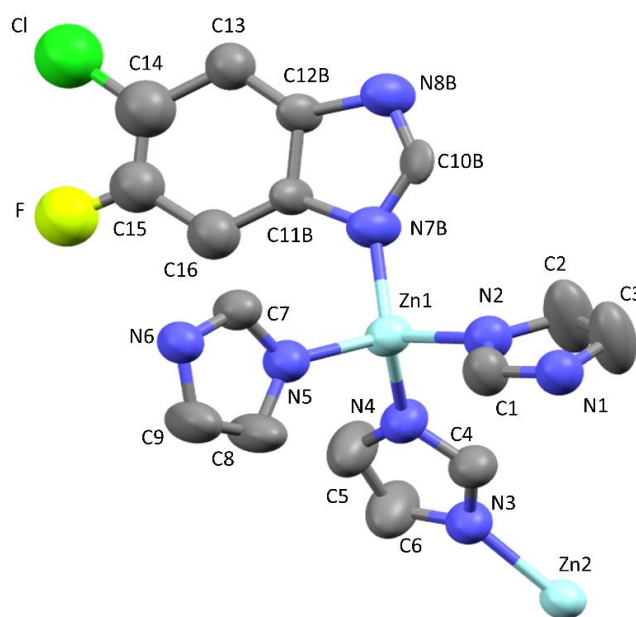


Figure S2. Graphical representation (thermal ellipsoids) of the second of two possible configurations for the asymmetric unit of ZIF-UC-2 (6-Cl-5-FbIm): two crystallographically independent Zn atoms, one of them surrounded by four independent linkers: three Im and one 6-Cl-5-FbIm. The 6-Cl-5-FbIm linker is shifted away from the complementary Im position by 54.80° to avoid clashing with neighboring Im linkers in totally occupied sites. This configuration represents 26% of the coordination environments for Zn atoms. H atoms are excluded for clarity.

ZIF-UC-3 (5-Cl-2-mbIm) (**cag** — Orthorhombic)
Experimental and Refinement Details for **ZIF-UC-3 (5-Cl-2-mbIm)**
(CCDC Deposition Number 1951695)

A colourless polyhedral crystal ($0.05 \times 0.04 \times 0.04$ mm³) of ZIF-UC-3 (5-Cl-2-mbIm) was placed in a 0.1 mm diameter borosilicate loop and mounted on a Gemini E Ultra diffractometer from Oxford Diffraction, equipped with an Atlas CCD area detector and operated at 35 kV and 35 mA, to generate Cu K α radiation ($\lambda = 1.54184$ Å) at 293(2) K. A total of 11002 reflections were collected, of which 3879 were unique and 2794 of these were greater than $2\sigma(I)$. The range of θ was from 2.45 to 64.19°. Analysis of the data showed negligible decay during collection. The structure was solved in the orthorhombic *Pbca* space group with $Z = 8$ using the SUPERFLIP method.¹⁸ Modeling of electron density led to identification of 0.78 DMF molecules in the asymmetric unit. The asymmetric unit is also integrated by two Zn atoms and four linkers. From those four linkers, three are Im in positions with no disorder and the fourth position is disordered between 5-Cl-2-mbIm (occupancy 0.375(10)) and Im (occupancy 0.625(12)). This fourth position can contain 5-Cl-2-mbIm with a maximum occupancy of a maximum of 50%, as the center of symmetry in this space group would bring the benzene rings related by this center into contact with each other. All non-H atoms, except the ones from the solvent, have been refined anisotropically. All H atoms have been placed in geometrically located positions and their displacement parameters are tied to those of the attached C atoms. The crystal was synthesized using a combination of DMF, DEF and NaOH solvents. Nevertheless the structure was refined just considering DEF because the other two were used in smaller quantities. This accounts for the residual density reported and resulting decrease of refinement stability. The unit cell of ZIF-UC-3 (5-Cl-2-mbIm) contains 1.81 Im and 0.19 5-Cl-2-mbIm per Zn. Final full matrix least -50 squares refinement on F^2 converged to $R_1 = 0.0968$ ($F > 2\sigma(F)$) and $wR_2 = 0.3257$ (all data) with GooF = 1.205.

The structure shares the same *cag* network topology as ZIF-4 and ZIF-62.¹⁹

Table S3. Crystal data and structure refinement for ZIF-UC-3 (5-Cl-2-mbIm).

Chemical formula	C _{16.22} H _{18.58} Cl _{0.38} N _{8.78} O _{0.78} Zn ₂
Formula weight	493.27 g/mol
Temperature	293(2)
Wavelength	1.54184 Å
Crystal system	Orthorhombic
Space group	<i>Pbca</i>
Unit cell dimensions	$a = 15.8749(6) \text{ Å}$ $\alpha = 90^\circ$ $b = 16.2805(6) \text{ Å}$ $\beta = 90^\circ$ $c = 18.0681(6) \text{ Å}$ $\gamma = 90^\circ$
Volume	4669.7(3) Å ³
Density (calculated)	1.401 g/cm ³
Absorption coefficient	3.110
F(000)	2225.76
Crystal size	0.05 × 0.04 × 0.04 mm ³
Theta range for data collection	2.445 - 64.188 °
Index ranges	-18 ≤ h ≤ 18, -18 ≤ k ≤ 18, -21 ≤ l ≤ 21
Reflections collected	11002
Independent reflections	3879
Completeness to theta = 66.97°	93.23 %
Absorption correction	multi-scan
Refinement method	Full-matrix least-squares on F^2
Data / restraints / parameters	3879 / 19 / 232
Goodness-of-fit on F^2	1.205
Final R indices [$I > 2\sigma(I)$]	$R_1 = 0.0968$, $wR_2 = 0.2950$
R indices (all data)	$R_1 = 0.1257$, $wR_2 = 0.3257$
Largest diff. peak and hole	0.932 and -1.143 e. Å ⁻³

Table S4. Additional refinement details for ZIF-UC-3 (5-Cl-2-mbIm): atomic positions (fractions), isotropic atomic displacement parameter (U iso or equivalent), type of atomic displacement parameters used for the site (Adp type) and corresponding site occupancy factors (Occup).

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occup
		x	y	z			
Zn1	Zn	0.68367(7)	0.35207(7)	0.09704(6)	0.0742(5)	Uani	1
Zn2	Zn	0.48797(7)	0.16158(7)	0.31624(6)	0.0708(5)	Uani	1
N3	N	0.6130(5)	0.2616(5)	0.1377(4)	0.089(2)	Uani	1
N4	N	0.5409(5)	0.1903(5)	0.2197(4)	0.0815(18)	Uani	1
N5	N	0.6568(5)	0.3653(5)	-0.0090(4)	0.0805(19)	Uani	1
N8	N	0.9190(4)	0.2535(4)	0.1454(4)	0.0768(17)	Uani	1
N7	N	0.8033(5)	0.3251(5)	0.1151(4)	0.0824(19)	Uani	1
N1	N	0.5852(6)	0.5649(6)	0.1893(5)	0.1117(16)	Uani	1
N6	N	0.5807(4)	0.3604(5)	-0.1141(4)	0.0797(19)	Uani	1
N2	N	0.6569(6)	0.4540(6)	0.1534(5)	0.1117(16)	Uani	1
C6	C	0.5765(6)	0.2604(5)	0.2026(5)	0.079(2)	Uani	1
H6	H	0.575918	0.305631	0.234025	0.095	Uiso	1
C10	C	0.8351(5)	0.2552(5)	0.1401(5)	0.076(2)	Uani	1
C5	C	0.5539(11)	0.1450(7)	0.1580(7)	0.137(6)	Uani	1
H5	H	0.534894	0.091399	0.151461	0.165	Uiso	1
C3	C	0.5969(7)	0.5096(7)	0.1395(6)	0.1117(16)	Uani	1
H3	H	0.565418	0.508349	0.096163	0.134	Uiso	1
C9	C	0.5849(5)	0.3448(5)	-0.0414(4)	0.077(2)	Uani	1
H9	H	0.540437	0.321202	-0.015522	0.092	Uiso	1
C7	C	0.6577(9)	0.3912(14)	-0.1292(8)	0.176(6)	Uani	1
H7	H	0.676296	0.408164	-0.175522	0.212	Uiso	1
C8	C	0.7041(9)	0.3936(13)	-0.0655(7)	0.176(6)	Uani	1
H8	H	0.759417	0.41181	-0.061743	0.212	Uiso	1
C4	C	0.5965(10)	0.1861(7)	0.1089(7)	0.139(6)	Uani	1
H4	H	0.612701	0.167339	0.062418	0.167	Uiso	1
C2	C	0.6846(7)	0.4789(7)	0.2193(6)	0.1117(16)	Uani	1
H2	H	0.730044	0.455487	0.243979	0.134	Uiso	1
C1	C	0.6391(7)	0.5415(7)	0.2456(6)	0.1117(16)	Uani	1
H1	H	0.642987	0.564739	0.292455	0.134	Uiso	1
C11B	C	0.9437(5)	0.3268(6)	0.1193(6)	0.111(4)	Uani	0.375(11)
H11B	H	0.998823	0.345858	0.116808	0.134	Uiso	0.375(11)
C12B	C	0.8721(5)	0.3692(6)	0.0968(5)	0.116(4)	Uani	0.375(11)
H12B	H	0.871751	0.419796	0.072963	0.139	Uiso	0.375(11)
C11A	C	0.9437(5)	0.3268(6)	0.1193(6)	0.111(4)	Uani	0.625(11)
C12A	C	0.8721(5)	0.3692(6)	0.0968(5)	0.116(4)	Uani	0.625(11)
C13	C	0.8800(10)	0.4423(7)	0.0578(9)	0.124(11)	Uani	0.375(11)
H13	H	0.832164	0.470681	0.042682	0.149	Uiso	0.375(11)

Continued

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occ
x	y	z					
C14	C	0.9595(12)	0.4731(9)	0.0413(12)	0.17(2)	Uani	0.375(11)
H14	H	0.964819	0.52207	0.01514	0.204	Uiso	0.375(11)
C15	C	1.0311(9)	0.4307(12)	0.0638(12)	0.152(17)	Uani	0.375(11)
Cl1	Cl	1.1210(8)	0.4631(6)	0.0534(5)	0.154(6)	Uani	0.375(11)
C17	C	0.7862(19)	0.1944(17)	0.165(2)	0.159(17)	Uani	0.375(11)
H17A	H	0.820866	0.149316	0.180858	0.238	Uiso	0.375(11)
H17B	H	0.753604	0.213835	0.206467	0.238	Uiso	0.375(11)
H17C	H	0.749085	0.176563	0.126564	0.238	Uiso	0.375(11)
C16	C	1.0232(5)	0.3576(11)	0.1028(10)	0.126(13)	Uani	0.375(11)
H16	H	1.071029	0.329237	0.117916	0.151	Uiso	0.375(11)
C1S	C	0.752(3)	0.413(2)	0.378(3)	0.4	Uiso	0.78(3)
H1S	H	0.751429	0.45919	0.347881	0.48	Uiso	0.78(3)
N1S	N	0.8262(17)	0.3328(18)	0.4005(15)	0.224(14)	Uiso	0.78(3)
C2S	C	0.863(4)	0.398(4)	0.452(3)	0.52(5)	Uiso	0.78(3)
H2S1	H	0.868188	0.375444	0.500818	0.78	Uiso	0.78(3)
H2S2	H	0.918198	0.413126	0.43415	0.78	Uiso	0.78(3)
H2S3	H	0.827332	0.44483	0.452541	0.78	Uiso	0.78(3)
O1S	O	0.6875(19)	0.369(3)	0.432(2)	0.37(2)	Uiso	0.78(3)
C3S	C	0.793(4)	0.289(4)	0.336(3)	0.49(4)	Uiso	0.78(3)
H3S1	H	0.777133	0.234875	0.350273	0.738	Uiso	0.78(3)
H3S2	H	0.745394	0.318302	0.317116	0.738	Uiso	0.78(3)
H3S3	H	0.836259	0.286598	0.298724	0.738	Uiso	0.78(3)

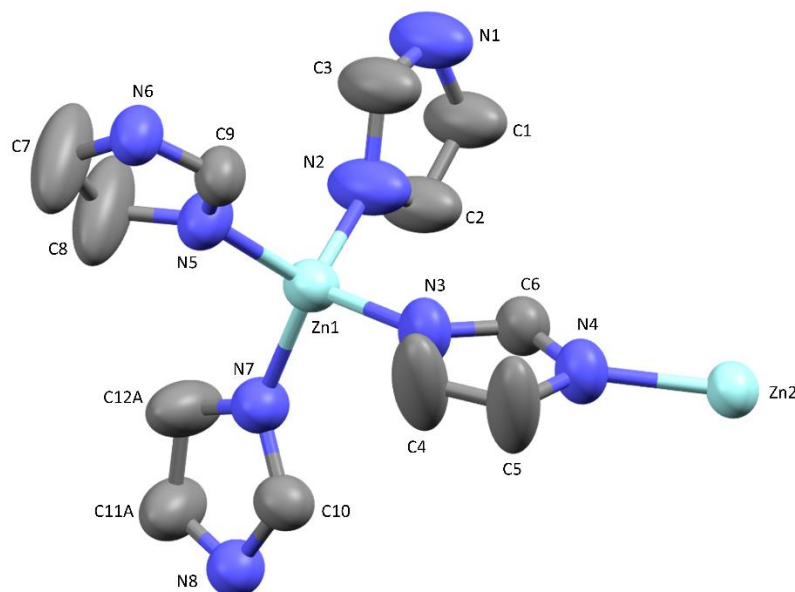


Figure S3. Graphical representation (thermal ellipsoids) of the first of two possible configurations for the asymmetric unit of ZIF-UC-3 (5-Cl-2-mbIm): two crystallographically independent Zn atoms, one of them surrounded by four independent Im linkers. This configuration represents 62% of the coordination environments for Zn atoms. H atoms are excluded for clarity.

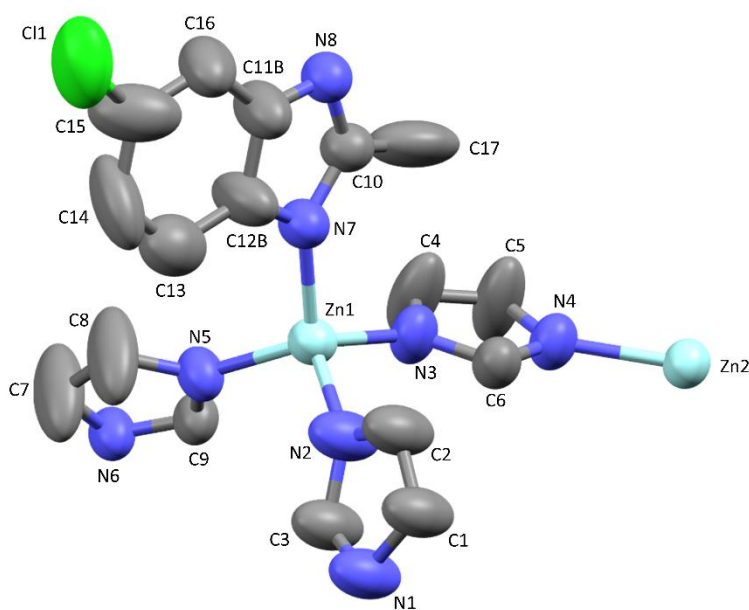


Figure S4. Graphical representation (thermal ellipsoids) of the second of two possible configurations for the asymmetric unit of ZIF-UC-3 (5-Cl-2-mbIm): two crystallographically independent Zn atoms, one of them surrounded by four independent linkers: three Im and one 5-Cl-2-mbIm. This configuration represents 38% of the coordination environments for Zn atoms. H atoms are excluded for clarity.

ZIF-UC-4 (5-FbIm) (**cag** — Orthorhombic)
Experimental and Refinement Details for **ZIF-UC-4 (5-FbIm)**
(CCDC Deposition Number 1951697)

A rose coloured polyhedral crystal ($0.05 \times 0.04 \times 0.03 \text{ mm}^3$) of ZIF-UC-4 (5-FbIm) was placed in a 0.1 mm diameter borosilicate loop and mounted on a Gemini E Ultra diffractometer from Oxford Diffraction, equipped with an Atlas CCD area detector and operated at 35 kV and 35 mA, to generate Cu K α radiation ($\lambda = 1.54184 \text{ \AA}$) at 293(2) K. A total of 11053 reflections were collected, of which 3717 were unique and 2405 of these were greater than $2\sigma(I)$. The range of θ was from 2.82 to 64.17°. Analysis of the data showed negligible decay during collection. The structure was solved in the orthorhombic *Pbca* space group with $Z = 8$ using the SUPERFLIP method.¹⁸ Modeling of electron density led to the identification of an asymmetric unit with two independent Zn and four independent linkers. From those four linkers, two are unsubstituted Im with no disorder, a third linker is 5-FbIm (occupancy 0.425(9)) and Im (occupancy 0.575(9)), and the fourth position is 5-FbIm (occupancy 0.257(11)) and Im (occupancy 0.743(11)). The third position can contain a maximum of 50% 5-FbIm because the center of symmetry in this space group would bring the benzene rings related by this center into contact with each other. The 5-FbIm linker at the fourth position is shifted away from the complementary Im position by 58.27°, which results in partial occupancy of sites normally occupied by solvent molecules. The shift happens to avoid clashing with one of the two neighboring Im linkers in totally occupied sites. This feature, in addition to a previous desolvation of the crystals, caused the electron density within the void of the frameworks to be too low to model the solvent. All non-H atoms have been refined anisotropically (isotropic refinements led to very high thermal ellipsoids). All H atoms, except H18 have been placed in geometrically located positions and their displacement parameters are tied to those of the attached carbon atoms. H18 was fixed in the geometrically equivalent position. There was a residual electronic density in the vicinity which indicates that possibly this site is partially occupied by another F atom with an occupancy too low to significantly change the final chemical composition of the material (occupancy less than 0.06 considering NMR results). Attempts to partition the electronic density to refine this extra configuration resulted in either an unstable refinement, or a non-physical structural solution. The unit cell of ZIF-UC-4 (5-FbIm) contains 1.66 Im and 0.34 5-FbIm per Zn. Final full matrix least -50 squares refinement on F^2 converged to $R_1 = 0.0663$ ($F > 2\sigma(F)$) and $wR_2 = 0.2174$ (all data) with $\text{GooF} = 1.065$.

The structure shares the same *cag* network topology as ZIF-4 and ZIF-62.¹⁹

Table S5. Crystal data and structure refinement for ZIF-UC-4 (5-FbIm).

Chemical formula	C _{14.73} H ₁₂ N ₈ F _{0.68} Zn ₂
Formula weight	444.77 g/mol
Temperature	293(2)
Wavelength	1.54184 Å
Crystal system	Orthorhombic
Space group	<i>Pbca</i>
Unit cell dimensions	$a = 15.6970(8) \text{ Å}$ $\alpha = 90^\circ$ $b = 15.9694(10) \text{ Å}$ $\beta = 90^\circ$ $c = 17.9282(16) \text{ Å}$ $\gamma = 90^\circ$
Volume	4494.1(5) Å ³
Density (calculated)	1.314 g/cm ³
Absorption coefficient	2.782
F(000)	1983.28
Crystal size	0.05 × 0.04 × 0.03 mm ³
Theta range for data collection	2.815 - 64.173°
Index ranges	-18 ≤ h ≤ 18, -13 ≤ k ≤ 18, -20 ≤ l ≤ 20
Reflections collected	11053
Independent reflections	3717
Completeness to theta = 66.97°	92.61 %
Absorption correction	multi-scan
Refinement method	Full-matrix least-squares on F^2
Data / restraints / parameters	3717 / 11 / 232
Goodness-of-fit on F^2	1.065
Final R indices [$I > 2\sigma(I)$]	$R_1 = 0.0663$, $wR_2 = 0.1901$
R indices (all data)	$R_1 = 0.1017$, $wR_2 = 0.2174$
Largest diff. peak and hole	0.606 and -0.575 e. Å ⁻³

Table S6. Additional refinement details for ZIF-UC-4 (5-FbIm): atomic positions (fractions), isotropic atomic displacement parameter (U iso or equivalent), type of atomic displacement parameters used for the site (Adp type) and corresponding site occupancy factors (Occup).

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occup
		x	y	z			
Zn1	Zn	0.30725(5)	0.35588(6)	0.09601(5)	0.0821(4)	Uani	1
Zn2	Zn	0.50575(5)	0.16039(6)	0.31335(5)	0.0796(3)	Uani	1
N2	N	0.4500(4)	0.1877(4)	0.2176(3)	0.0881(16)	Uani	1
N8	N	0.0706(3)	0.2575(3)	0.1466(3)	0.0853(16)	Uani	1
N7	N	0.1853(3)	0.3296(4)	0.1107(4)	0.0873(17)	Uani	1
N1	N	0.3739(4)	0.2598(4)	0.1357(3)	0.0844(15)	Uani	1
N4	N	0.4134(4)	0.5653(3)	0.1950(3)	0.0838(15)	Uani	1
N3	N	0.3389(4)	0.4546(4)	0.1561(4)	0.0875(16)	Uani	1
C6	C	0.3994(4)	0.5100(5)	0.1417(4)	0.088(2)	Uani	1
H6	H	0.429964	0.509921	0.097236	0.105	Uiso	1
C1	C	0.4171(4)	0.2589(5)	0.1990(4)	0.088(2)	Uani	1
H1	H	0.423486	0.306577	0.228367	0.105	Uiso	1
C5	C	0.3124(6)	0.4775(6)	0.2247(6)	0.135(4)	Uani	1
H5	H	0.26895	0.451606	0.251405	0.162	Uiso	1
C2	C	0.4247(8)	0.1365(6)	0.1625(6)	0.136(4)	Uani	1
H2	H	0.436074	0.079428	0.159996	0.164	Uiso	1
C14	C	0.1550(4)	0.2601(4)	0.1399(4)	0.0828(18)	Uani	1
H14	H	0.18989	0.216082	0.15475	0.099	Uiso	1
C4	C	0.3586(7)	0.5443(6)	0.2489(6)	0.130(3)	Uani	1
H4	H	0.353096	0.57071	0.294935	0.157	Uiso	1
C3	C	0.3799(7)	0.1808(6)	0.1107(5)	0.129(4)	Uani	1
H3	H	0.357672	0.16011	0.066316	0.155	Uiso	1
C15A	C	0.0450(5)	0.3316(5)	0.1161(5)	0.1082(17)	Uani	0.575(8)
H15A	H	-0.011224	0.349015	0.110718	0.13	Uiso	0.575(8)
C16A	C	0.1140(5)	0.3751(5)	0.0952(5)	0.1082(17)	Uani	0.575(8)
H16A	H	0.113385	0.427958	0.073515	0.13	Uiso	0.575(8)
N6A	N	0.4155(4)	0.3693(4)	-0.1136(4)	0.0989(19)	Uani	0.743(10)
N5A	N	0.3389(4)	0.3749(4)	-0.0092(4)	0.103(2)	Uani	0.743(10)
C7A	C	0.4103(7)	0.3539(8)	-0.0421(7)	0.079(3)	Uani	0.743(10)
H7A	H	0.455028	0.32942	-0.016003	0.094	Uiso	0.743(10)
C9A	C	0.2899(9)	0.3919(14)	-0.0705(11)	0.216(11)	Uani	0.743(10)
H9A	H	0.231868	0.403607	-0.068665	0.259	Uiso	0.743(10)
C8A	C	0.3350(8)	0.3896(13)	-0.1323(8)	0.204(10)	Uani	0.743(10)
H8A	H	0.314946	0.399995	-0.180167	0.245	Uiso	0.743(10)
C15B	C	0.0450(5)	0.3316(5)	0.1161(5)	0.1082(17)	Uani	0.425(8)
C16B	C	0.1140(5)	0.3751(5)	0.0952(5)	0.1082(17)	Uani	0.425(8)
C17	C	0.1071(11)	0.4510(12)	0.0499(12)	0.1082(17)	Uani	0.425(8)

Continued

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occ
x	y	z					
H17	H	0.154798	0.479804	0.033057	0.13	Uiso	0.425(8)
C19	C	-0.0416(12)	0.4377(12)	0.0589(13)	0.1082(17)	Uani	0.425(8)
C18	C	0.0199(11)	0.4791(12)	0.0327(13)	0.1082(17)	Uani	0.425(8)
C20	C	-0.0409(12)	0.3587(11)	0.0989(12)	0.1082(17)	Uani	0.425(8)
H20	H	-0.089636	0.329037	0.11194	0.13	Uiso	0.425(8)
F2	F	-0.1228(8)	0.4657(10)	0.0502(11)	0.188(7)	Uani	0.425(8)
N5B	N	0.3389(4)	0.3749(4)	-0.0092(4)	0.103(2)	Uani	0.257(10)
N6B	N	0.4155(4)	0.3693(4)	-0.1136(4)	0.0989(19)	Uani	0.257(10)
C7B	C	0.390(2)	0.341(3)	-0.055(2)	0.079(14)	Uiso	0.257(10)
H7B	H	0.411684	0.289262	-0.041175	0.094	Uiso	0.257(10)
C8B	C	0.3813(13)	0.4635(17)	-0.1082(17)	0.165(9)	Uani	0.257(10)
C9B	C	0.3335(16)	0.4669(18)	-0.0430(16)	0.165(9)	Uani	0.257(10)
C10	C	0.2949(15)	0.541(2)	-0.0216(13)	0.165(9)	Uani	0.257(10)
H10	H	0.262929	0.543694	0.021983	0.198	Uiso	0.257(10)
C11	C	0.3040(16)	0.6126(18)	-0.0655(17)	0.165(9)	Uani	0.257(10)
H11	H	0.278199	0.662536	-0.051237	0.198	Uiso	0.257(10)
C12	C	0.3518(18)	0.6093(18)	-0.1307(16)	0.165(9)	Uani	0.257(10)
C13	C	0.3905(14)	0.535(2)	-0.1521(13)	0.165(9)	Uani	0.257(10)
H13	H	0.42242	0.532458	-0.195692	0.198	Uiso	0.257(10)
F1	F	0.3588(16)	0.6685(15)	-0.1755(14)	0.165(9)	Uani	0.257(10)
H18	H	0.021255	0.547853	0.006863	0.166	Uiso	0.425(8)

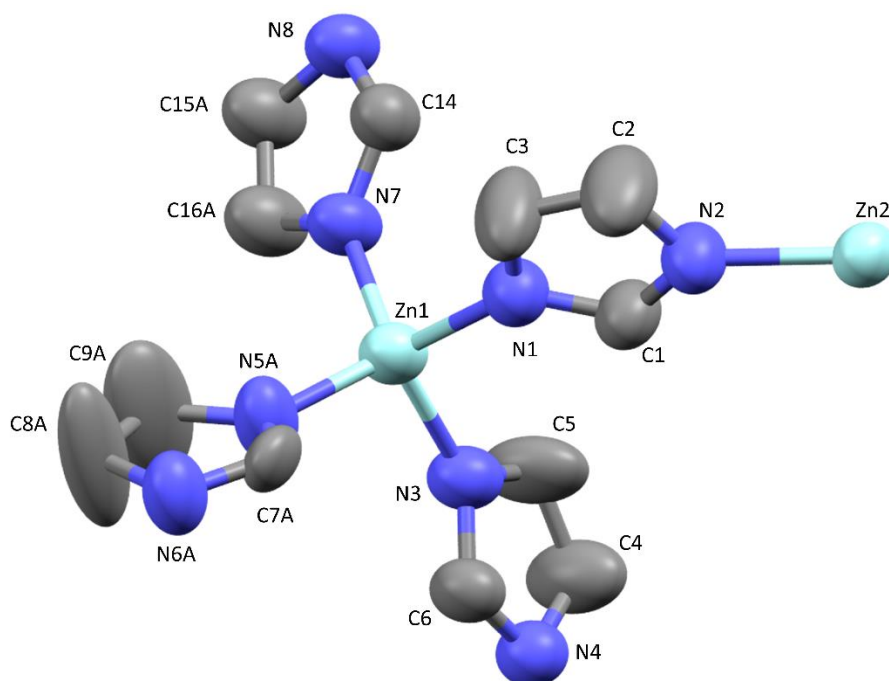


Figure S5. Graphical representation (thermal ellipsoids) of the first of four possible configurations for the asymmetric unit of ZIF-UC-4 (5-FbIm): two crystallographically independent Zn atoms, one of them surrounded by four independent Im linkers. This configuration represents 32% of the coordination environments for Zn atoms. H atoms are excluded for clarity.

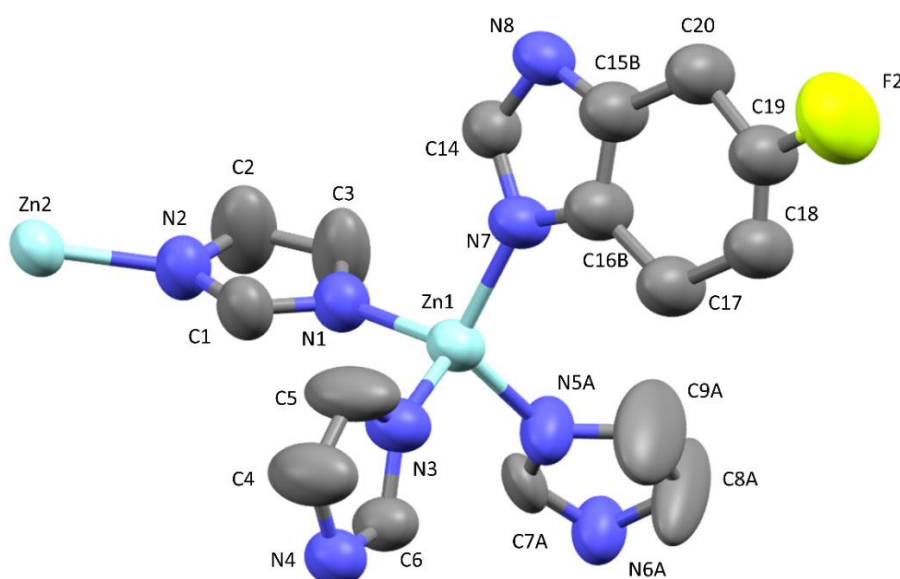


Figure S6. Graphical representation (thermal ellipsoids) of the second of four possible configurations for the asymmetric unit of ZIF-UC-4 (5-FbIm): two independent Zn atoms, one of them surrounded by four independent linkers: three Im and one 5-FbIm. The crystallographic position for 5-FbIm has a maximum occupancy of 50% because the center of symmetry in this space group would bring the 5-FbIm linkers related by this center into contact with each other. This configuration is one of three possible coordination environments for Zn atoms, that in total represents 68% of the tetrahedral units in the structure. H atoms are excluded for clarity.

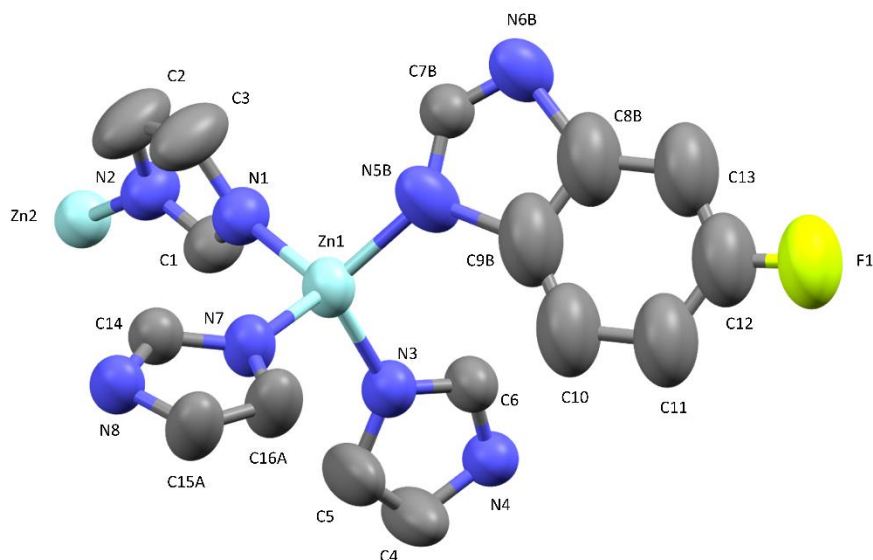


Figure S7. Graphical representation (thermal ellipsoids) of the third of four possible configurations for the asymmetric unit of ZIF-UC-4 (5-FbIm): two independent Zn atoms, one of them surrounded by four independent linkers: three Im and one 5-FbIm. The 5-FbIm linker is shifted away from the complementary Im position by 58.27° to avoid clashing with one of the two neighbors Im linkers in totally occupied sites. This configuration is one of three possible coordination environments for Zn atoms, that in total represents 68% of the tetrahedral units in the structure. H atoms are excluded for clarity.

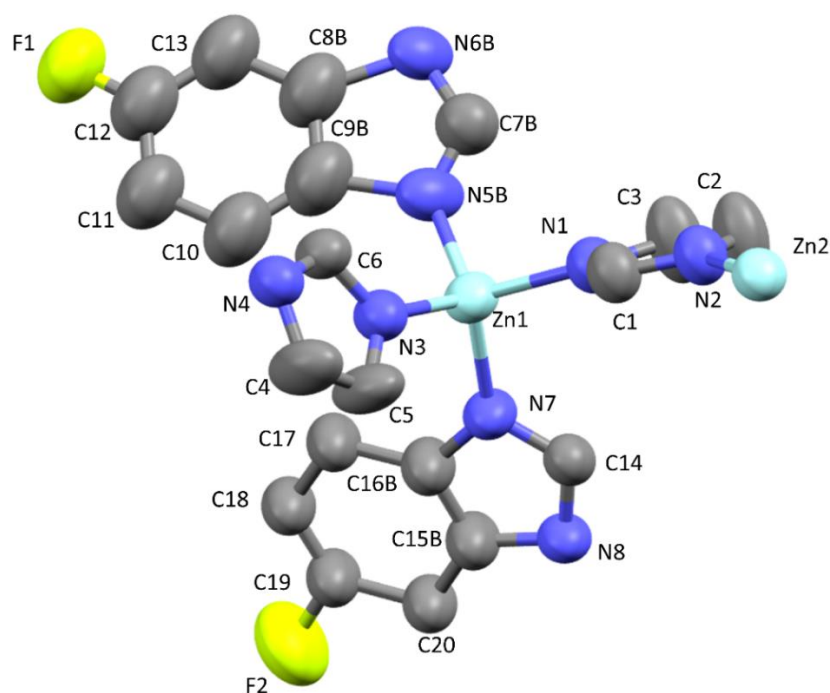


Figure S8. Graphical representation (thermal ellipsoids) of the fourth of four possible configurations for the asymmetric unit of ZIF-UC-4 (5-FbIm): two independent Zn atoms, one of them surrounded by four independent linkers: two Im and two 5-FbIm. Crystallographic details for both 5-FbIm linkers were described in Figures S6 and S7. This configuration is one of three possible coordination environments for Zn atoms, that in total represents 68% of the tetrahedral units in the structure. H atoms are excluded for clarity.

ZIF-UC-5 (5-ClbIm) (*cag* — Orthorhombic)

Experimental and Refinement Details for **ZIF-UC-5 (5-ClbIm)**

(Deposition Number 1951696)

A colourless tetrahedral crystal ($0.05 \times 0.04 \times 0.03 \text{ mm}^3$) of ZIF-UC-5 (5-ClbIm) was placed in a 0.1 mm diameter borosilicate loop and mounted on a Gemini E Ultra diffractometer from Oxford Diffraction equipped with an Atlas CCD area detector and operated at 35 kV and 35 mA, to generate Cu K α radiation ($\lambda = 1.54184 \text{ \AA}$) at 293(2) K. A total of 11094 reflections were collected, of which 3786 were unique and 2684 of these were greater than $2\sigma(I)$. The range of θ was from 2.43 to 64.01°. Analysis of the data showed negligible decay during collection. The structure was solved in the orthorhombic *Pbca* space group with $Z = 8$ using the SUPERFLIP method.¹⁸ Modeling of electron density led to the identification of 0.39 DMF molecules in the asymmetric unit (atoms with no crystallographic symmetry). The asymmetric unit is also integrated by two Zn atoms and four linkers. From those four linkers, two are unsubstituted Im with no disorder, a third linker in a position disordered between 5-ClbIm (0.397(10) occupancy) and Im (0.603(10) occupancy); and the fourth linker is in a position disordered between 5-ClbIm (0.171(11) occupancy) and Im (0.829(11) occupancy). The third position can contain a maximum of 50% 5-ClbIm because the center of symmetry in this space group would bring the 5-ClbIm linkers related by this center into contact with each other. The 5-ClbIm linker at the fourth position is shifted away from the complementary Im position by 57.47°, which results in partial occupancy of sites normally occupied by solvent molecules. The shift happens to avoid clashing with one of the two neighboring Im linker in totally occupied sites. This feature, in addition to a previous desolvation of the crystal is responsible for the low occupancy for solvent molecules. All non-H atoms, except the solvent molecules, the four C atoms from benzene rings in both 5-ClbIm linkers and corresponding Cl atoms, have been refined anisotropically. All H atoms, have been placed in geometrically located positions and their displacement parameters are tied to those of the attached C atoms. The positions for both Cl atoms and solvent atoms were fixed to be able to separate their contributions to the residual electronic density. The unit cell of ZIF-UC-5 (5-ClbIm) contains 1.72 Im and 0.28 ClbIm per Zn. Final full matrix least - squares refinement on F^2 converged to $R_1 = 0.0646$ ($F > 2\sigma F$) and $wR_2 = 0.2107$ (all data) with GooF = 1.103.

The structure shares the same *cag* network topology as ZIF-4 and ZIF-62.¹⁹

Table S7. Crystal data and structure refinement for ZIF-UC-5 (5-ClbIm).

Chemical formula	C _{15.44} H _{14.73} Cl _{0.57} N _{8.39} O _{0.39} Zn ₂
Formula weight	475.04 g/mol
Temperature	293(2)
Wavelength	1.54184 Å
Crystal system	Orthorhombic
Space group	<i>Pbca</i>
Unit cell dimensions	$a = 15.7260(8) \text{ Å}$ $\alpha = 90^\circ$ $b = 16.0184(11) \text{ Å}$ $\beta = 90^\circ$ $c = 18.1617(10) \text{ Å}$ $\gamma = 90^\circ$
Volume	4575.0(5) Å ³
Density (calculated)	1.375 g/cm ³
Absorption coefficient	3.331
F(000)	2125.60
Crystal size	0.05 × 0.04 × 0.03 mm ³
Theta range for data collection	2.433 - 64.013°
Index ranges	-17 ≤ h ≤ 18, -17 ≤ k ≤ 18, -8 ≤ l ≤ 21
Reflections collected	11094
Independent reflections	3786
Completeness to theta = 66.97°	92.90 %
Absorption correction	multi-scan
Refinement method	Full-matrix least-squares on F^2
Data / restraints / parameters	3786 / 8 / 244
Goodness-of-fit on F^2	1.103
Final R indices [$I > 2\sigma(I)$]	$R_1 = 0.0646$, $wR_2 = 0.1888$
R indices (all data)	$R_1 = 0.0872$, $wR_2 = 0.2107$
Largest diff. peak and hole	0.844 and -0.614 e. Å ⁻³

Table S8. Additional refinement details for ZIF-UC-5 (5-ClbIm): atomic positions (fractions), isotropic atomic displacement parameter (U iso or equivalent), type of atomic displacement parameters used for the site (Adp type) and corresponding site occupancy factors (Occup).

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occup
x	y	z					
Zn1	Zn	0.30911(5)	0.35723(5)	0.09849(5)	0.0769(3)	Uani	1
Zn2	Zn	0.49584(5)	0.66387(5)	0.18221(5)	0.0749(3)	Uani	1
N2	N	0.4487(3)	0.1914(3)	0.2224(3)	0.0830(14)	Uani	1
N6	N	0.0707(3)	0.2596(3)	0.1447(3)	0.0773(13)	Uani	1
N1	N	0.3746(3)	0.2613(3)	0.1402(3)	0.0789(13)	Uani	1
N5	N	0.1864(3)	0.3324(3)	0.1136(3)	0.0799(14)	Uani	1
N3	N	0.3412(4)	0.4582(4)	0.1553(3)	0.0870(15)	Uani	1
C4	C	0.4006(4)	0.5125(4)	0.1403(4)	0.0819(17)	Uani	1
H4	H	0.43019	0.511534	0.095992	0.098	Uiso	1
N4	N	0.4163(3)	0.5682(3)	0.1909(3)	0.0810(14)	Uani	1
C1	C	0.4151(4)	0.2624(4)	0.2035(4)	0.0829(17)	Uani	1
H1	H	0.41946	0.310156	0.232351	0.1	Uiso	1
C7	C	0.1558(4)	0.2619(4)	0.1398(3)	0.0746(15)	Uani	1
H7	H	0.190294	0.217371	0.153626	0.09	Uiso	1
C3	C	0.4267(7)	0.1401(5)	0.1667(5)	0.125(3)	Uani	1
H3	H	0.439419	0.083447	0.16382	0.15	Uiso	1
C6	C	0.3182(7)	0.4807(6)	0.2247(5)	0.143(4)	Uani	1
H6	H	0.277279	0.453693	0.252939	0.171	Uiso	1
C2	C	0.3837(6)	0.1836(5)	0.1163(5)	0.120(3)	Uani	1
H2	H	0.363467	0.162862	0.07182	0.144	Uiso	1
C5	C	0.3627(7)	0.5467(6)	0.2464(5)	0.139(4)	Uani	1
H5	H	0.357995	0.573492	0.291618	0.167	Uiso	1
N7A	N	0.3394(4)	0.3732(4)	-0.0060(3)	0.0928(16)	Uani	0.829(8)
N8A	N	0.4138(4)	0.3640(4)	-0.1107(3)	0.0919(16)	Uani	0.829(8)
C14A	C	0.4090(4)	0.3499(5)	-0.0389(4)	0.0717(19)	Uani	0.829(8)
H14A	H	0.453232	0.324454	-0.013322	0.086	Uiso	0.829(8)
C15A	C	0.3327(9)	0.3879(13)	-0.1288(7)	0.176(7)	Uani	0.829(8)
H15A	H	0.311643	0.397761	-0.175855	0.211	Uiso	0.829(8)
C16A	C	0.2901(9)	0.3942(15)	-0.0658(9)	0.220(11)	Uani	0.829(8)
H16A	H	0.23359	0.410937	-0.062602	0.264	Uiso	0.829(8)
C8A	C	0.1161(5)	0.3791(5)	0.0979(5)	0.107(3)	Uani	0.603(8)
H8A	H	0.11655	0.432373	0.077472	0.128	Uiso	0.603(8)
C9A	C	0.0462(5)	0.3359(5)	0.1168(5)	0.099(2)	Uani	0.603(8)
H9A	H	-0.009531	0.354408	0.111799	0.119	Uiso	0.603(8)
C8B	C	0.1161(5)	0.3791(5)	0.0979(5)	0.107(3)	Uani	0.397(8)
C9B	C	0.0462(5)	0.3359(5)	0.1168(5)	0.099(2)	Uani	0.397(8)
C10	C	-0.0373(11)	0.3620(10)	0.0974(8)	0.075(4)	Uiso	0.397(8)

Continued

Atom label	Atom sym	Atom site fraction coordinates			U iso or equivalent	Adp type	Occ
x	y	z					
H10	H	-0.086152	0.331599	0.108209	0.09	Uiso	0.397(8)
C13	C	0.1124(11)	0.4566(10)	0.0526(8)	0.080(4)	Uiso	0.397(8)
H13	H	0.160799	0.483255	0.034956	0.096	Uiso	0.397(8)
C11	C	-0.0378(12)	0.4426(11)	0.0581(9)	0.088(5)	Uiso	0.397(8)
Cl1	Cl	-0.1186	0.4693	0.0443	0.318(9)	Uiso	0.397(8)
C12	C	0.0288(12)	0.4863(12)	0.0386(9)	0.092(5)	Uiso	0.397(8)
H12	H	0.02136	0.537493	0.015232	0.11	Uiso	0.397(8)
N7B	N	0.3394(4)	0.3732(4)	-0.0060(3)	0.0928(16)	Uani	0.171(8)
N8B	N	0.4138(4)	0.3640(4)	-0.1107(3)	0.0919(16)	Uani	0.171(8)
C14B	C	0.3849(8)	0.3309(17)	-0.0505(9)	0.0717(19)	Uani	0.171(8)
H14B	H	0.397587	0.275496	-0.039748	0.086	Uiso	0.171(8)
C19	C	0.340(2)	0.5998(16)	-0.1255(19)	0.136	Uiso	0.171(8)
C20	C	0.3831(19)	0.528(2)	-0.1469(14)	0.129(18)	Uiso	0.171(8)
H20	H	0.413124	0.527458	-0.191051	0.154	Uiso	0.171(8)
C15B	C	0.3815(15)	0.4576(16)	-0.1025(14)	0.071(10)	Uani	0.171(8)
C16B	C	0.3366(16)	0.4586(16)	-0.0365(13)	0.080(11)	Uani	0.171(8)
C17	C	0.2933(18)	0.530(2)	-0.0151(15)	0.127(18)	Uiso	0.171(8)
H17	H	0.263314	0.530972	0.029028	0.152	Uiso	0.171(8)
C18	C	0.295(2)	0.6009(16)	-0.060(2)	0.14(2)	Uiso	0.171(8)
H18	H	0.265994	0.648805	-0.045212	0.166	Uiso	0.171(8)
Cl2	Cl	0.3427	0.668015	-0.1816	0.34(2)	Uiso	0.171(8)
C2S	C	0.385571	0.623075	-0.097095	0.231	Uiso	0.389(9)
H2S1	H	0.37928	0.581093	-0.059769	0.347	Uiso	0.389(9)
H2S2	H	0.398631	0.596913	-0.143287	0.347	Uiso	0.389(9)
H2S3	H	0.430846	0.660316	-0.083743	0.347	Uiso	0.389(9)
C1S	C	0.302164	0.713021	-0.137766	0.231	Uiso	0.389(9)
H1S	H	0.26359	0.727006	-0.174568	0.278	Uiso	0.389(9)
O1S	O	0.376005	0.750863	-0.086349	0.231	Uiso	0.389(9)
N1S	N	0.325212	0.658916	-0.1023	0.231	Uiso	0.389(9)
C3S	C	0.2658	0.6766	-0.042	0.231	Uiso	0.389(9)
H3S1	H	0.218689	0.708448	-0.060463	0.347	Uiso	0.389(9)
H3S2	H	0.245383	0.625017	-0.021711	0.347	Uiso	0.389(9)
H2S3	H	0.430846	0.660316	-0.083743	0.347	Uiso	0.389(9)

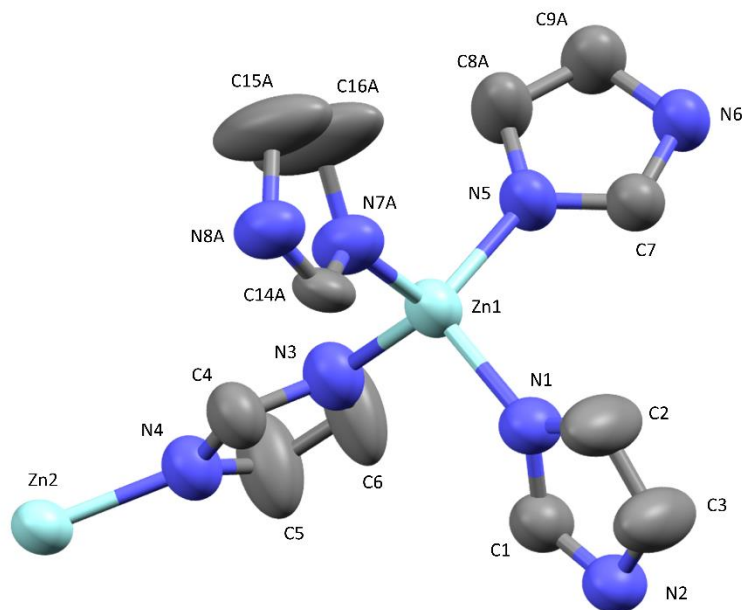


Figure S9. Graphical representation (thermal ellipsoids) of the first of four possible configurations for the asymmetric unit of ZIF-UC-5 (5-ClbIm): two independent Zn atoms, one of them is surrounded by four independent Im linkers. This configuration represents 43% of the coordination environments for Zn atoms. H atoms are excluded for clarity.

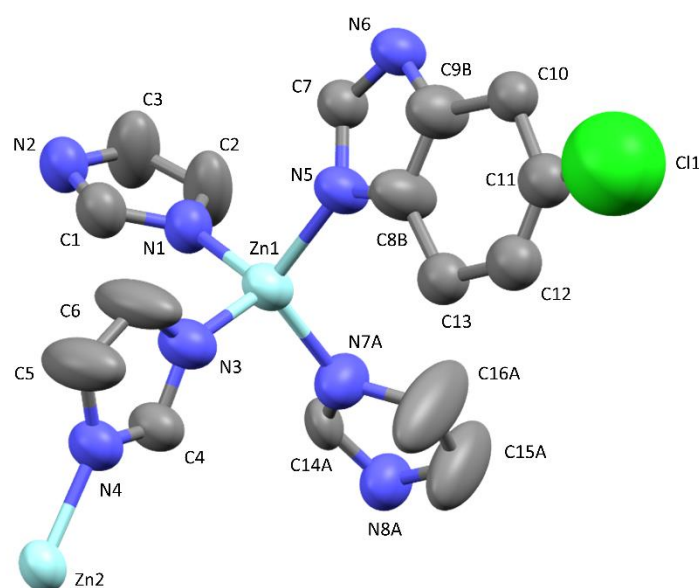


Figure S10. Graphical representation (thermal ellipsoids) of the second of four possible configurations for the asymmetric unit of ZIF-UC-5 (5-ClbIm): two independent Zn atoms, one of them surrounded by four independent linkers: three Im and one 5-ClbIm. The crystallographic position for 5-ClbIm has a maximum occupancy of 50% because the center of symmetry in this space group would bring the 5-ClbIm linkers related by this center into contact with each other. This configuration is one of three possible coordination environments for Zn atoms that in total represents 57% of the tetrahedral units in the structure. H atoms are excluded for clarity.

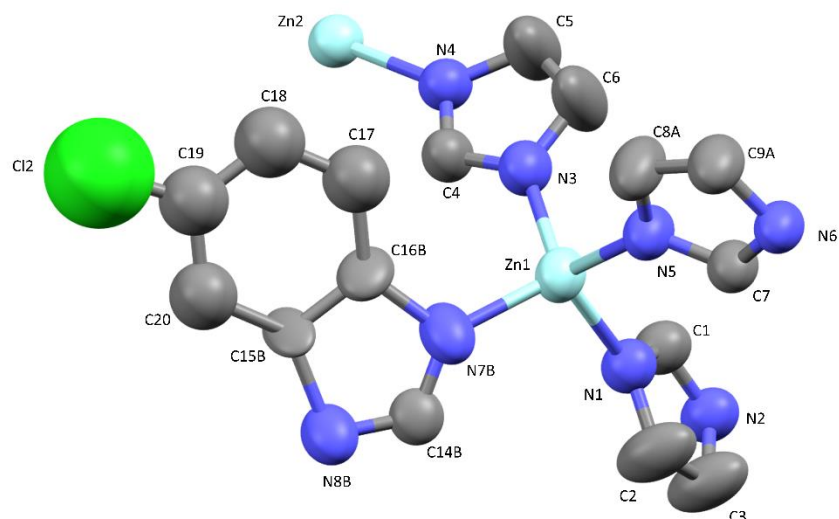


Figure S11. Graphical representation (thermal ellipsoids) of the third of four possible configurations for the asymmetric unit of ZIF-UC-5 (5-ClbIm): two independent Zn atoms, one of them surrounded by four independent linkers: three Im and one 5-ClbIm. The 5-ClbIm is shifted away from the complementary Im position by 57.47° to avoid clashing with one of the two neighboring Im linker in totally occupied sites. This configuration is one of three possible coordination environments for Zn atoms that in total represents 57% of the tetrahedral units in the structure. H atoms are excluded for clarity.

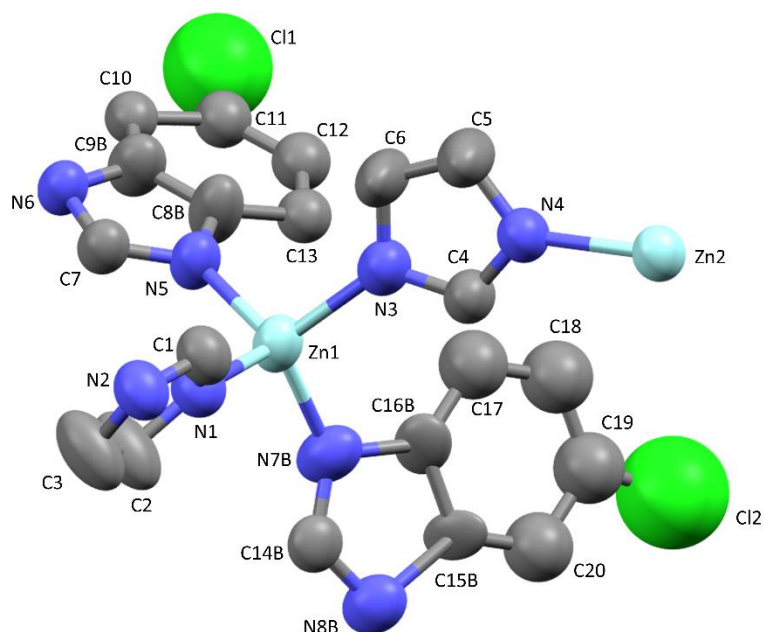


Figure S12. Graphical representation (thermal ellipsoids) of the fourth of four possible configurations for the asymmetric unit of ZIF-UC-5 (5-ClbIm): two crystallographically independent Zn atoms, one of them surrounded by four independent linkers: two Im and two 5-ClbIm. Crystallographic details for both 5-ClbIm linkers were described in Figures S10 and S11. This configuration is one of three possible coordination environments for Zn atoms that in total represents 57% of the tetrahedral units in the structure. H atoms are excluded for clarity.

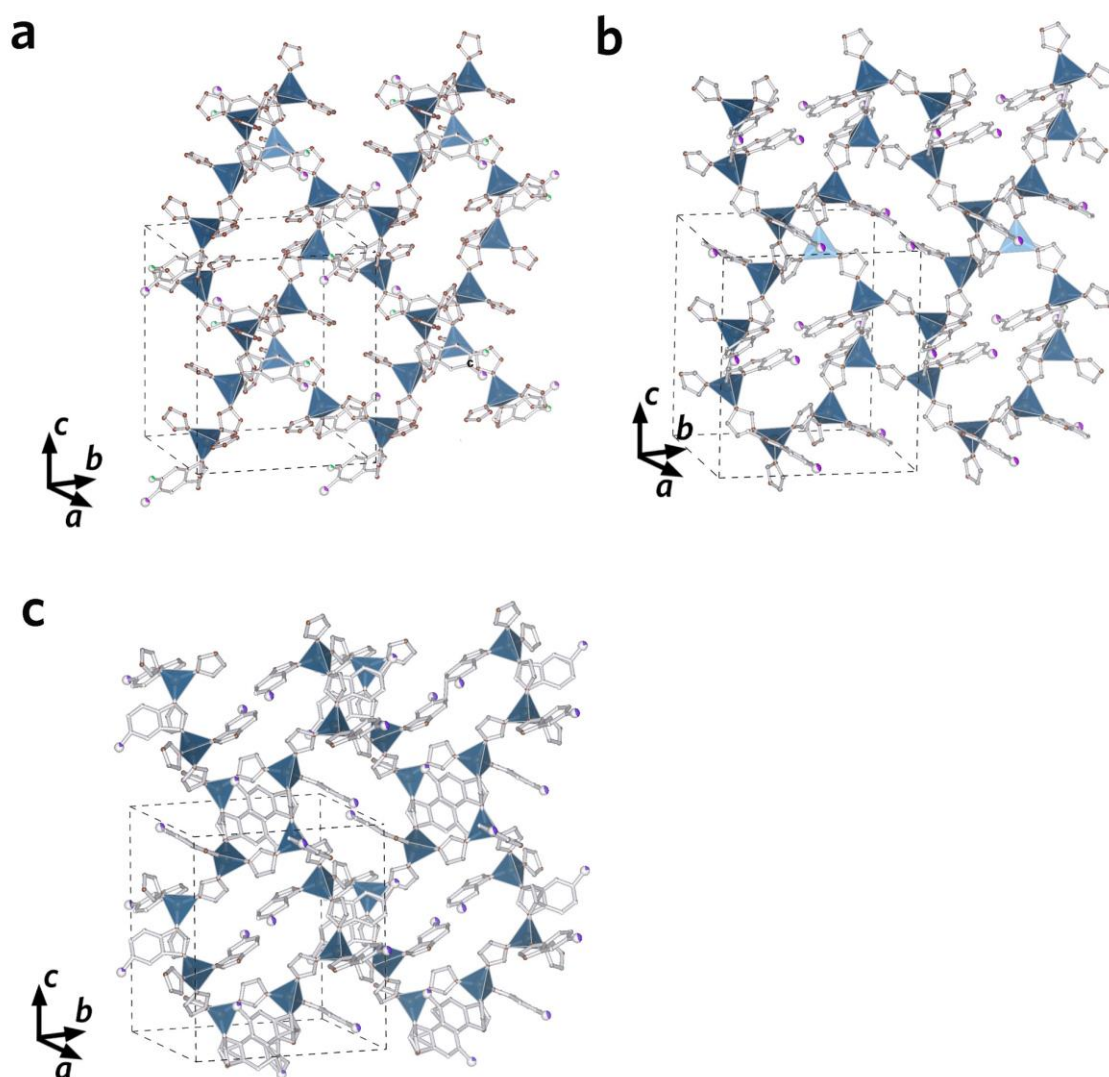


Figure S13. Schematic representative slice of (a) ZIF-UC-2 (6-Cl-5-FbIm), (b) ZIF-UC-3 (5-Cl-2-mbIm), and (c) ZIF-UC-5 (5-ClbIm) framework structures (N, orange; C, white; ZnN₄ tetrahedra, blue; Cl, purple; F, green), showing the six- and four-membered rings. For clarity, crystal structure within a range of fractional coordinates (0.3 to 0.8) along [100] direction are shown. H and solvent atoms are omitted.

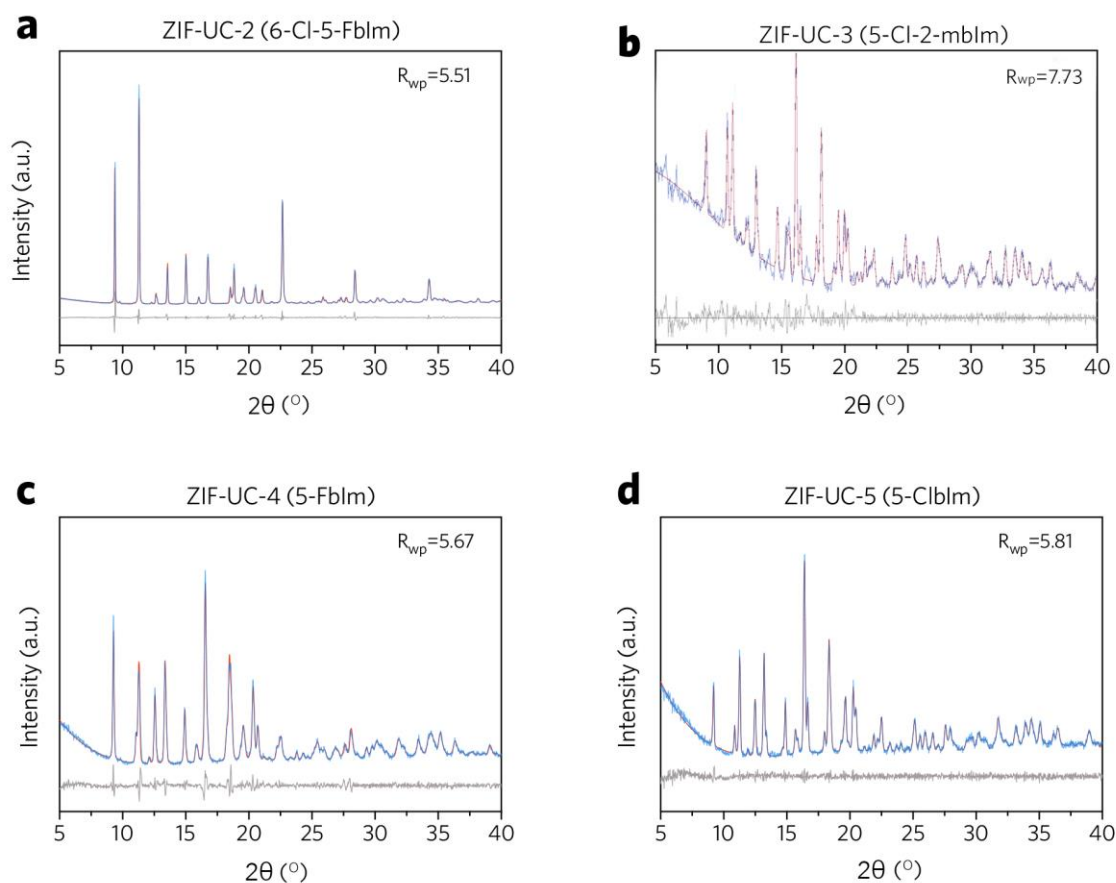


Figure S14. Pawley refinement results of the halogenated ZIFs synthesized in this study. The figures show the experimental powder X-ray diffraction (blue), fitted (red) and difference (grey) patterns.

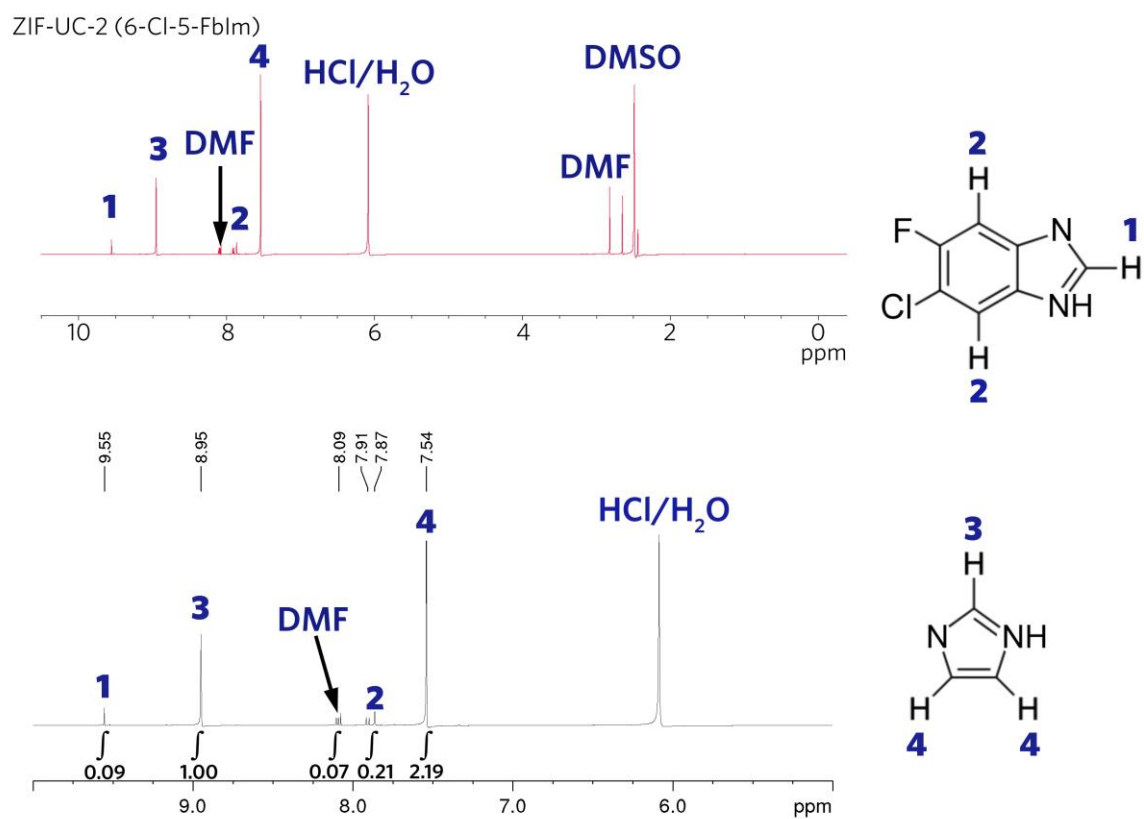


Figure S15. ^1H NMR spectrum of as-synthesized ZIF-UC-2 (6-Cl-5-FbIm) digested in $\text{DCl}/\text{DMSO}-d_6$. Top figure is the full spectrum and the bottom is a partially enlarged region.

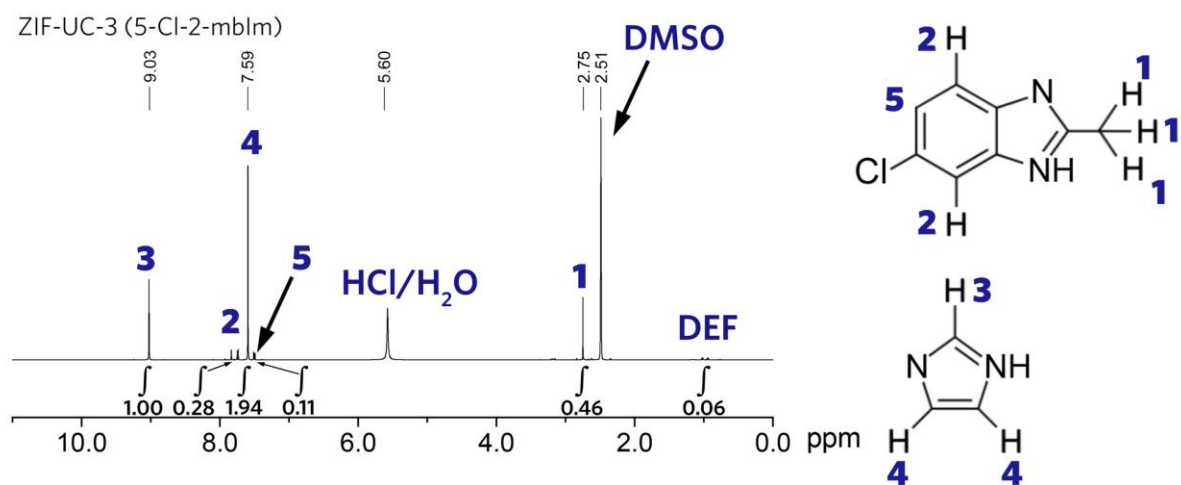


Figure S16. ¹H NMR spectrum of as-synthesized ZIF-UC-3 (5-Cl-2-mblm) digested in DCl/DMSO-*d*₆.

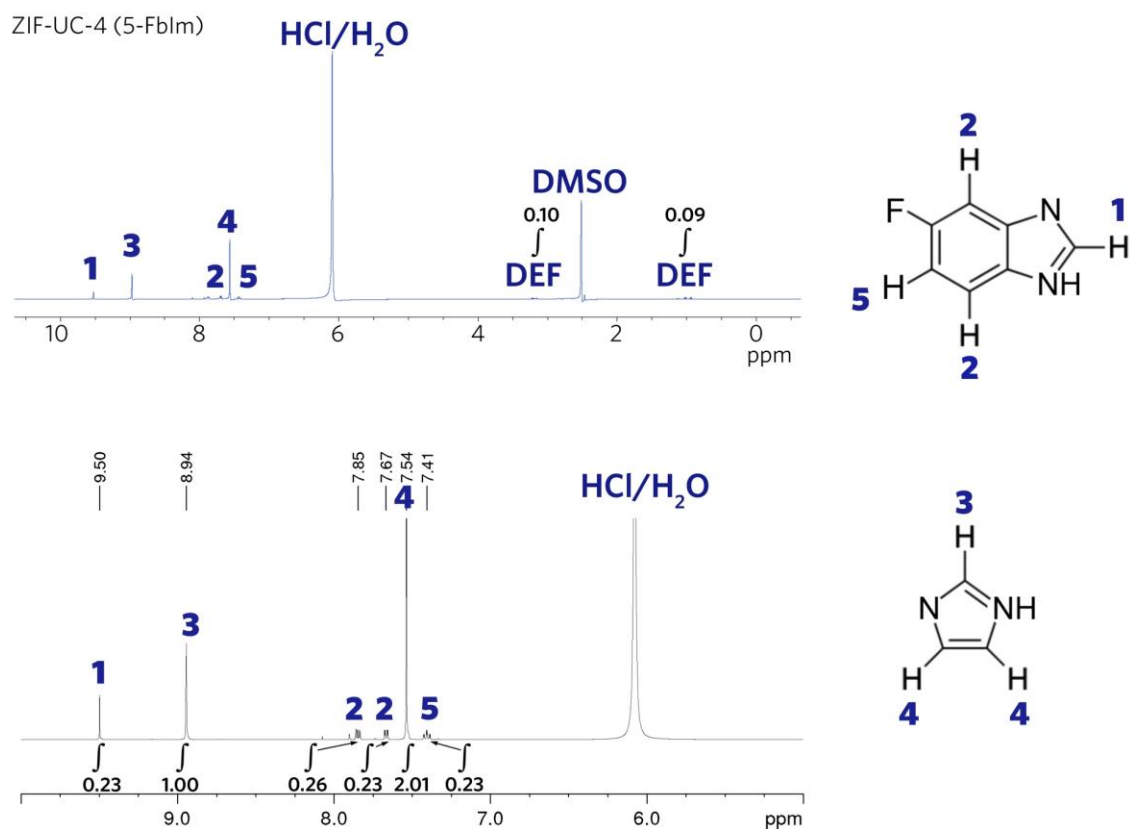


Figure S17. ^1H NMR spectrum of as-synthesized ZIF-UC-4 (5-FbIm) digested in DCl/DMSO- d_6 . Top figure is the full spectrum and the bottom is a partially enlarged region.

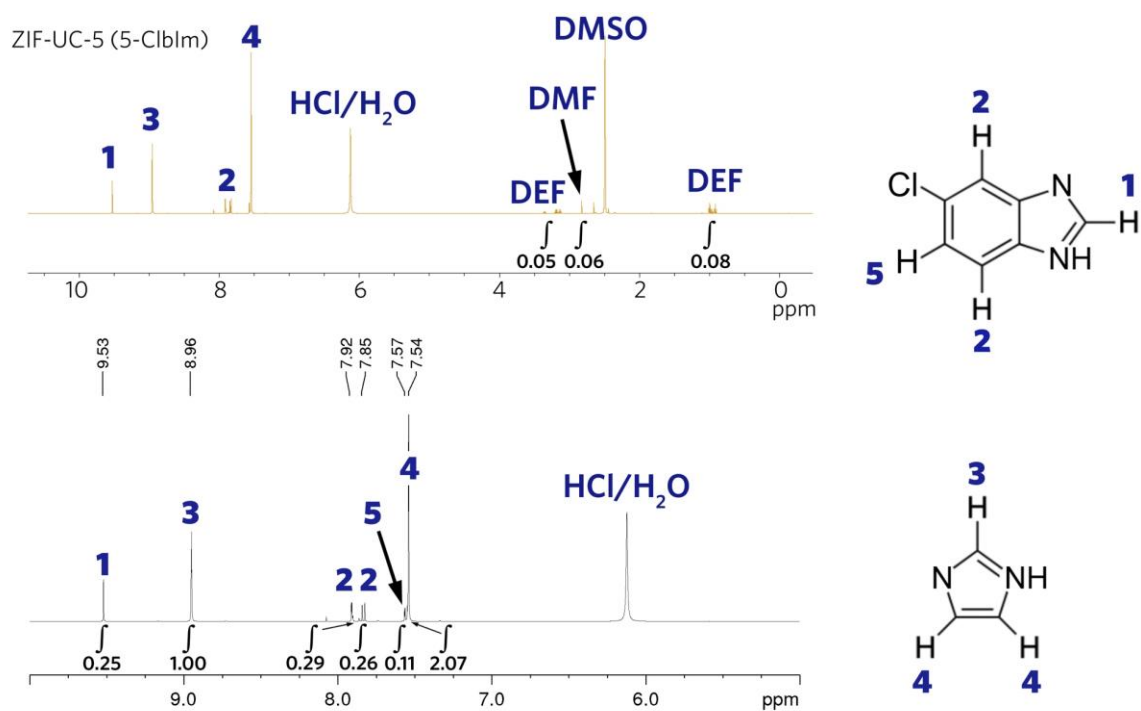


Figure S18. ^1H NMR spectrum of as-synthesized ZIF-UC-5 (5-ClbIm) digested in $\text{DCl}/\text{DMSO}-d_6$. Top figure is the full spectrum and the bottom is a partially enlarged region.

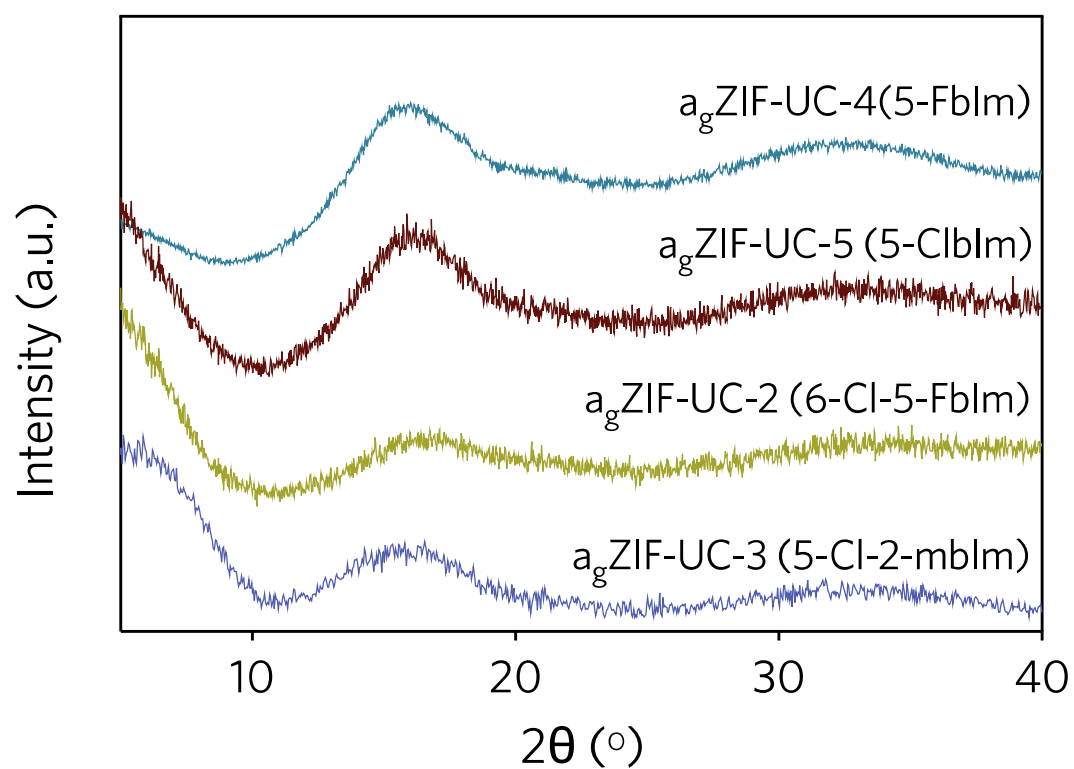


Figure S19. Powder X-ray diffraction patterns of the four halogenated glass samples.

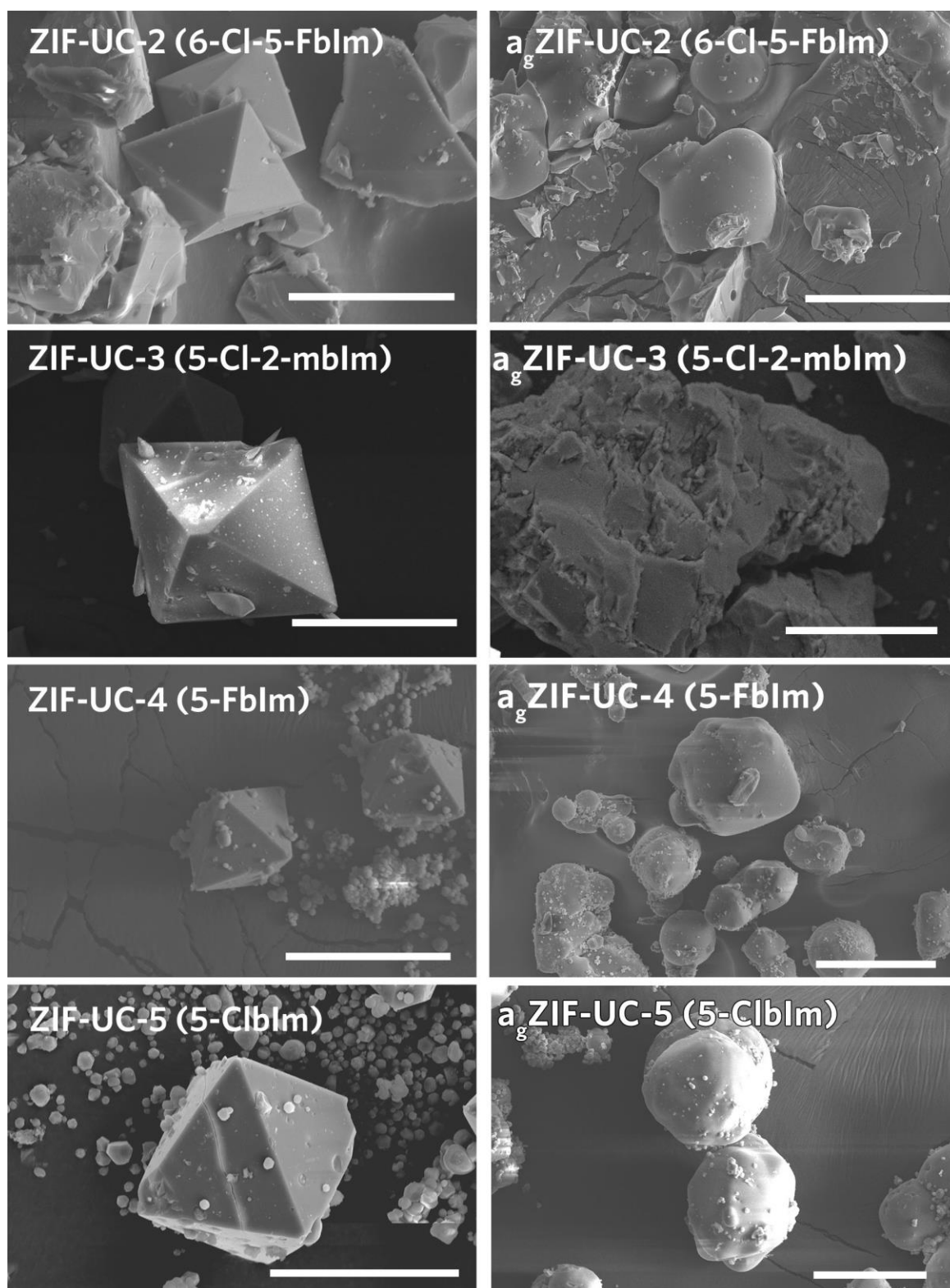


Figure S20. SEM images of the halogenated ZIF crystal precursors and the melt-quenched glasses. Scale bars are 50 μm for crystal samples and 100 μm for glass samples.

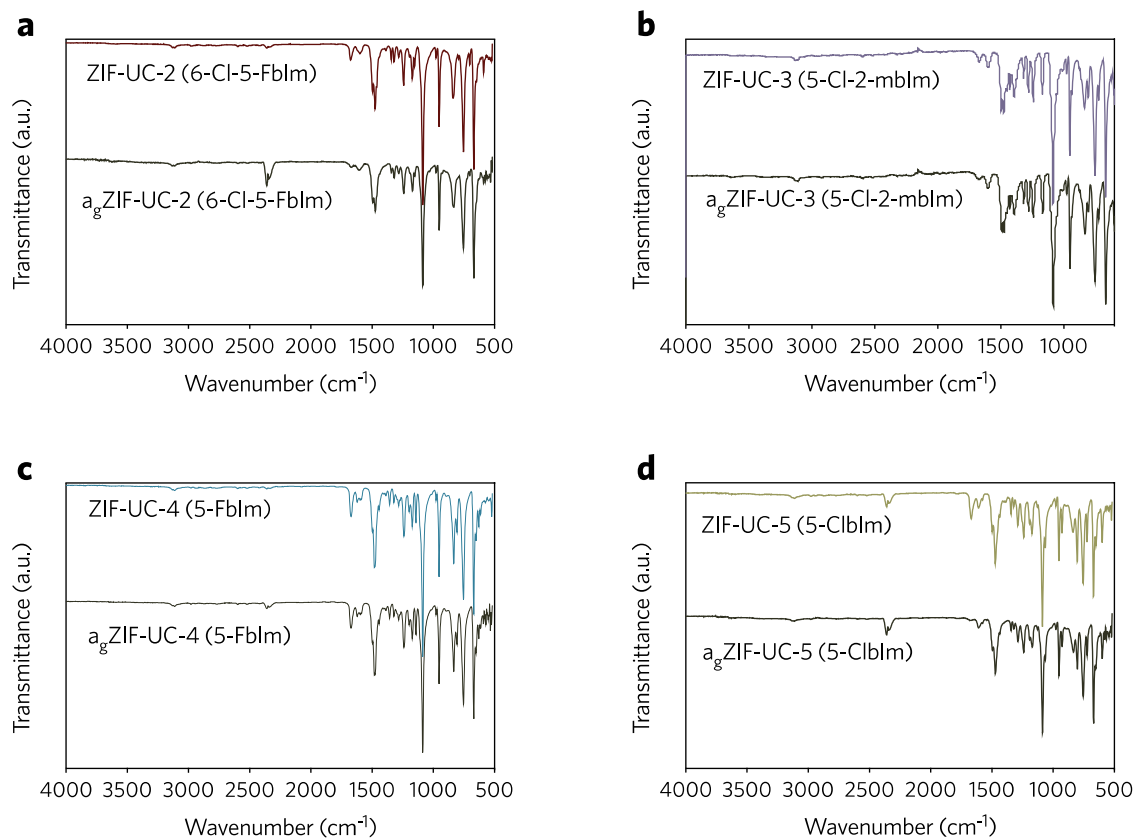


Figure S21. FTIR spectra of the crystal and glass samples.

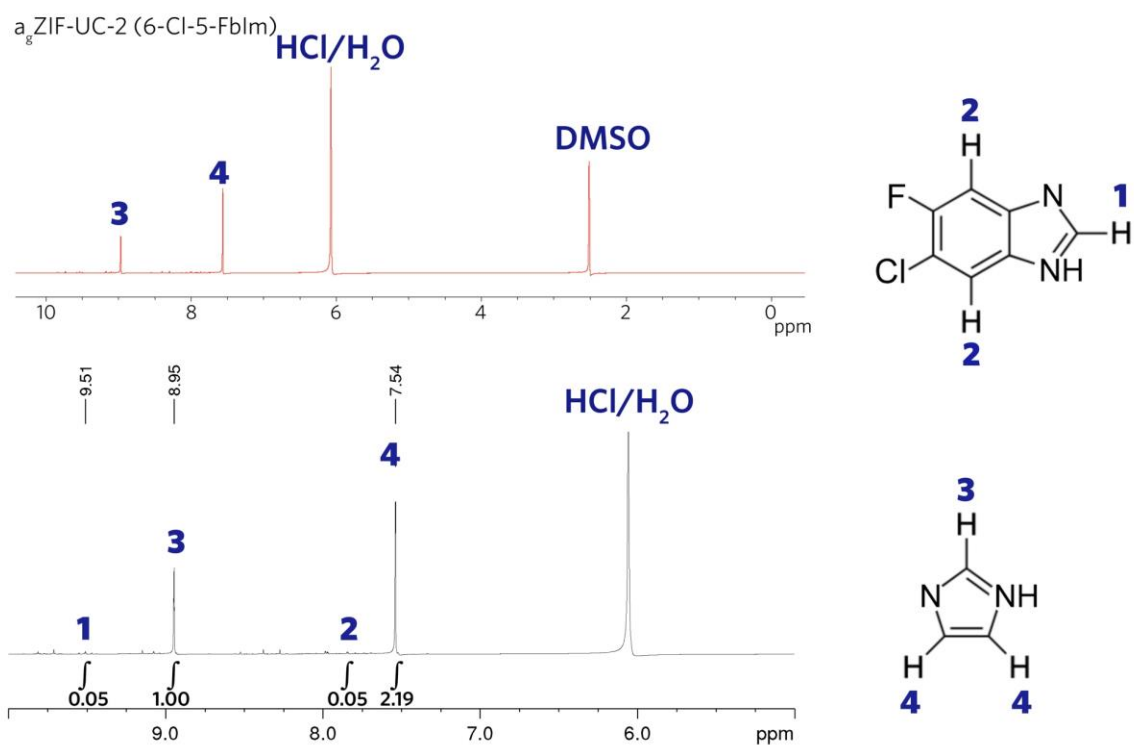


Figure S22. ^1H NMR spectrum of a_g ZIF-UC-2 (6-Cl-5-FbIm) digested in $\text{DCl}/\text{DMSO}-d_6$. Top figure is the full spectrum and the bottom is a partially enlarged region.

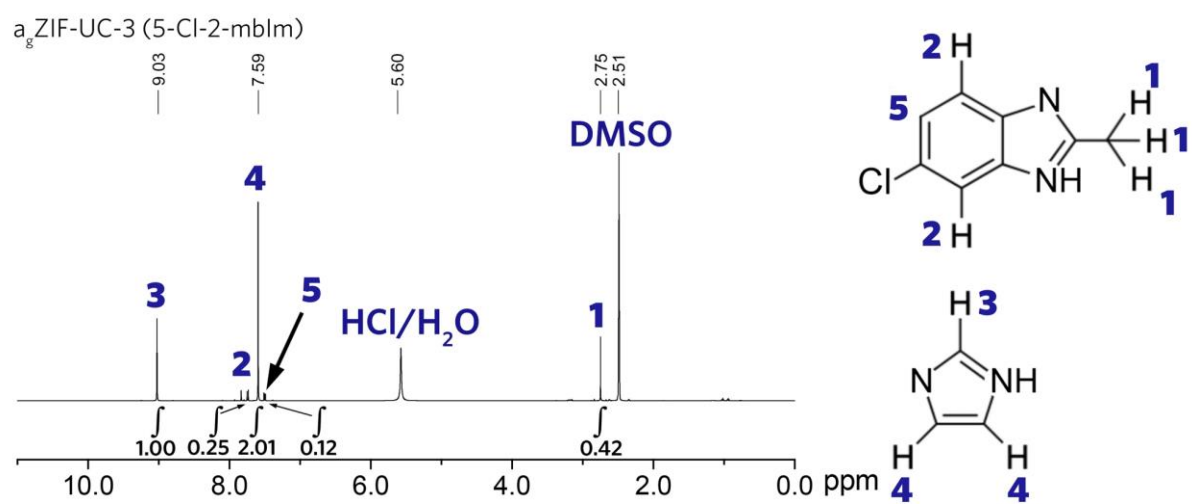


Figure S23. ^1H NMR spectrum of a_g ZIF-UC-3 (5-Cl-2-mbIm) digested in DCl/DMSO- d_6 .

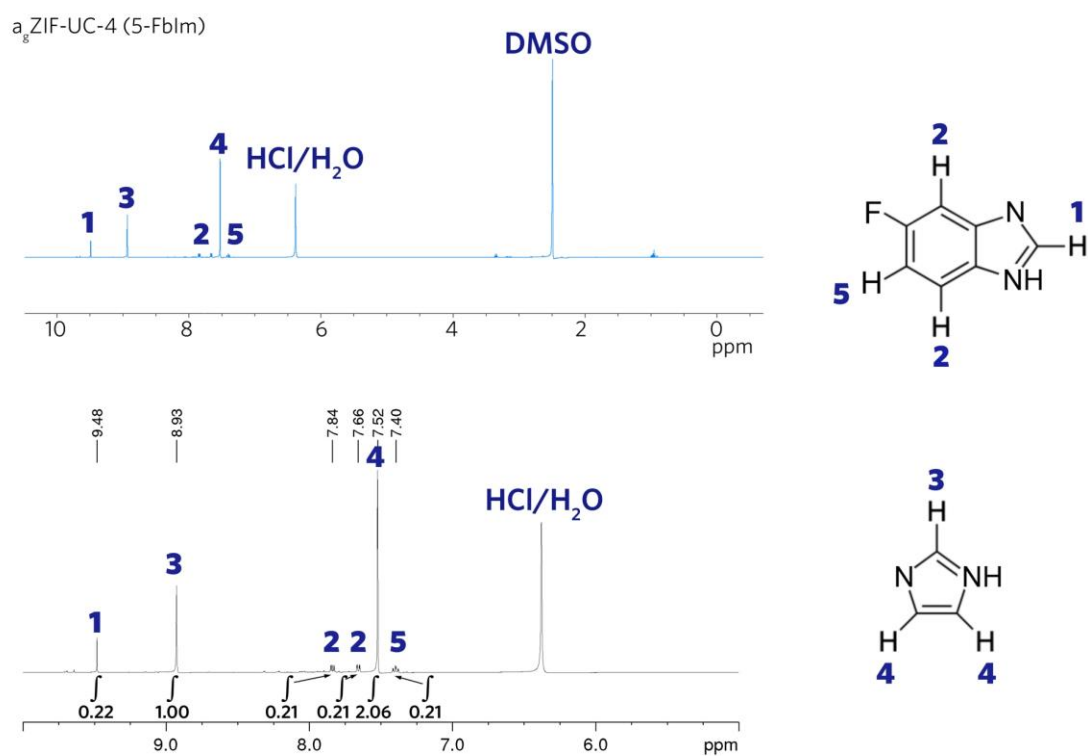


Figure S24. ^1H NMR spectrum of a_g ZIF-UC-4 (5-FbIm) digested in DCl/DMSO- d_6 . Top figure is the full spectrum and the bottom is a partially enlarged region.

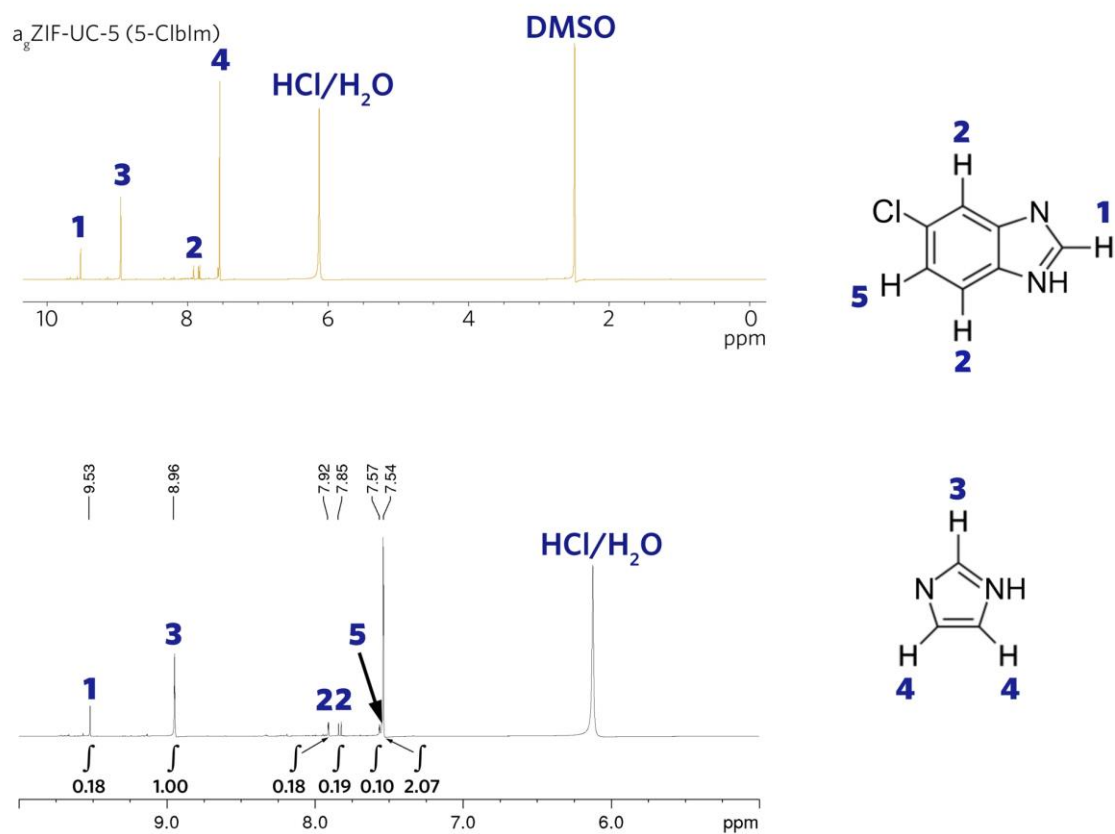


Figure S25. ¹H NMR spectrum of a_gZIF-UC-5 (5-ClbIm) digested in DCl/DMSO-*d*₆. Top figure is the full spectrum and the bottom is a partially enlarged region.

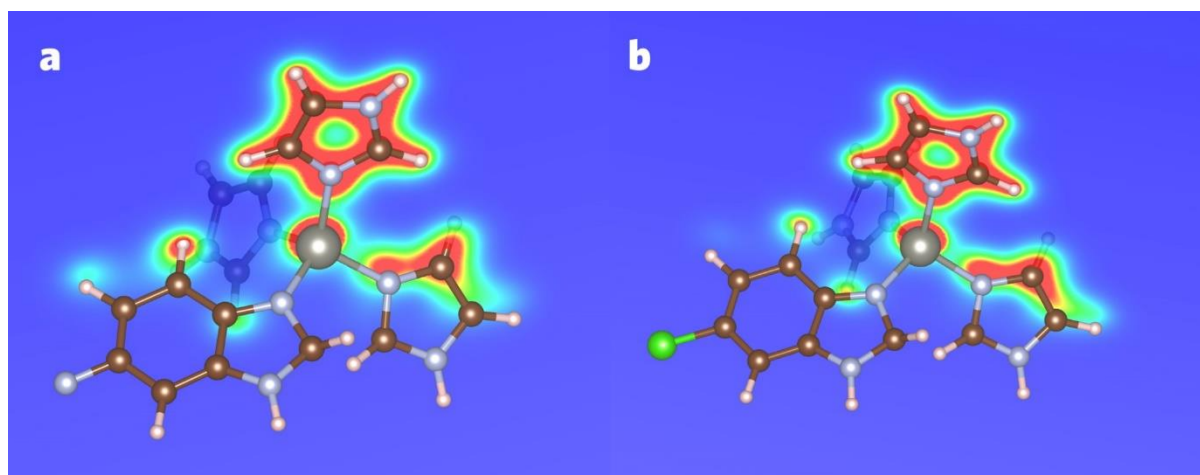


Figure S26. Charge density 2D contour plots of Zn-N tetrahedral cluster with (a) 5-FbIm and (b) 5-ClbIm ligands. Colours are in the RGB scale showing the charge density isosurface values between 0 (blue) to 0.2 (red) in e/bohr^3 .

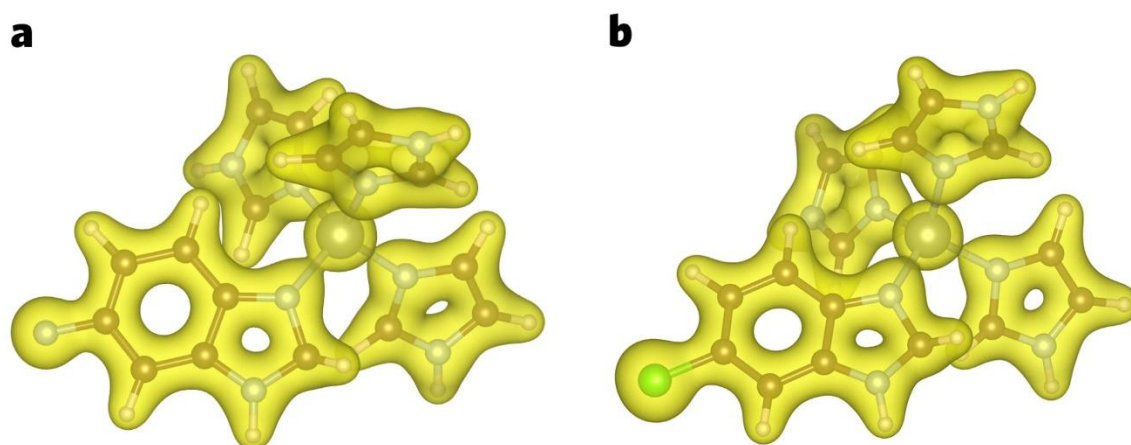


Figure S27. Charge density (3D) plots of Zn-N tetrahedral cluster with (a) 5-FbIm and (b) 5-ClbIm ligands. Isosurface value is set as $0.085 e/\text{bohr}^3$.

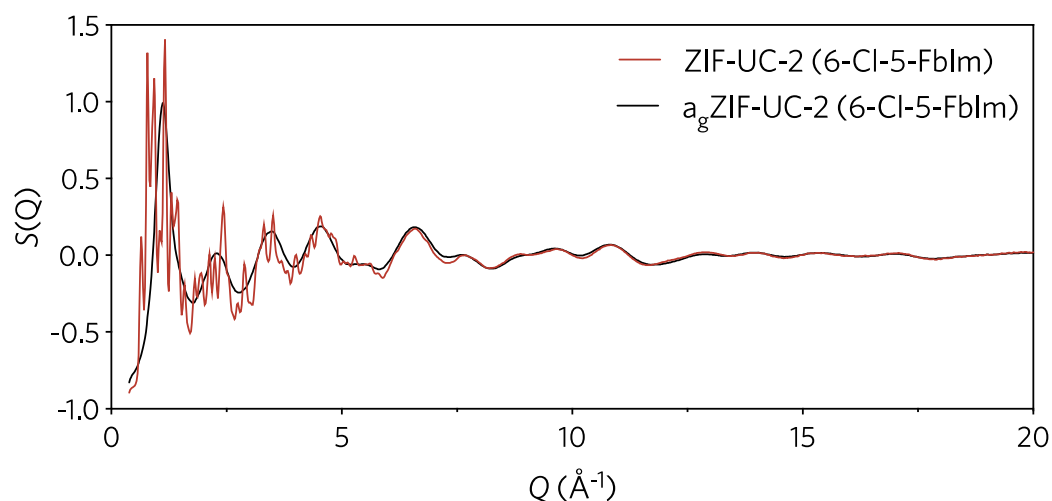


Figure S28. Synchrotron X-ray total scattering structure factors for ZIF-UC-2 (6-Cl-5-FbIm) crystal and a_g ZIF-UC-2 (6-Cl-5-FbIm).

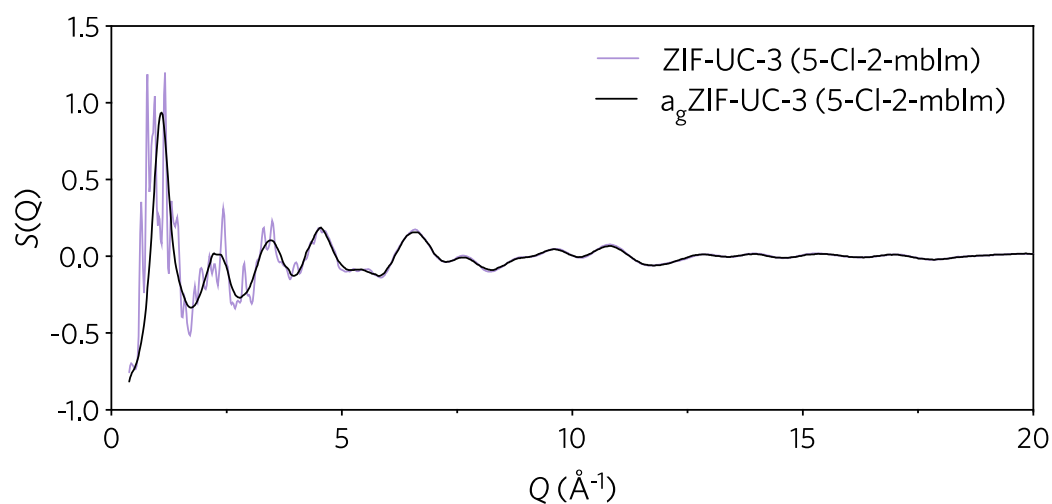


Figure S29. Synchrotron X-ray total scattering structure factors for ZIF-UC-3 (5-Cl-2-mbIm) crystal and a_g ZIF-UC-3 (5-Cl-2-mbIm).

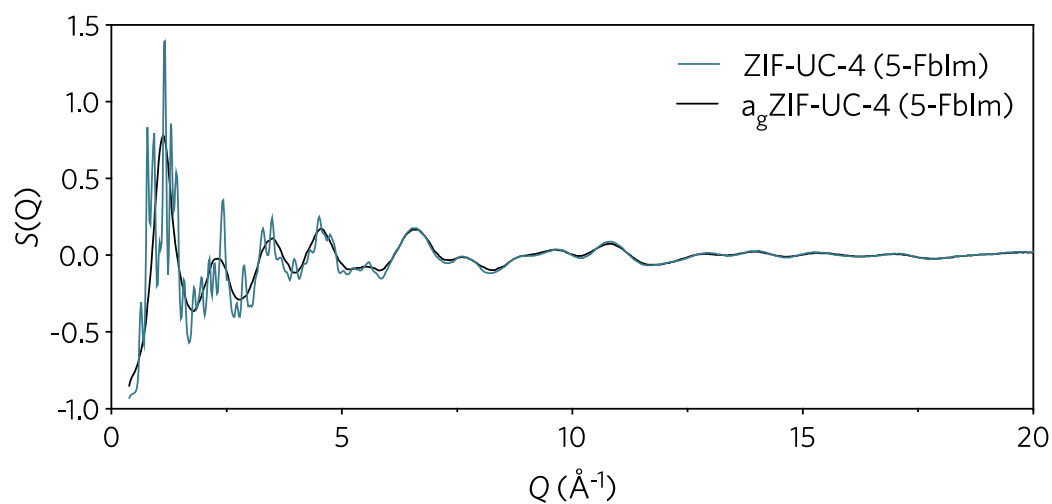


Figure S30. Synchrotron X-ray total scattering structure factors for ZIF-UC-4 (5-FbIm) crystal and a_g ZIF-UC-4 (5-FbIm).

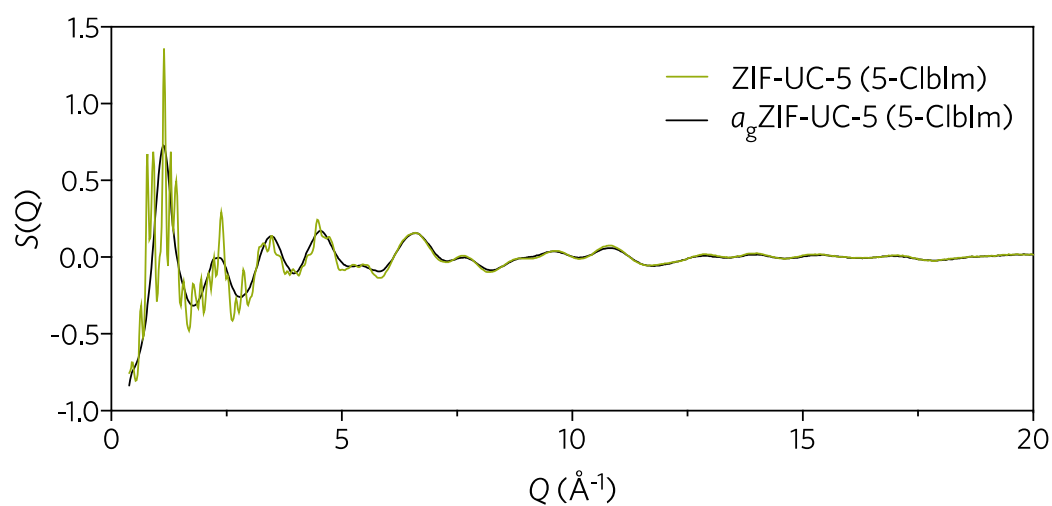


Figure S31. Synchrotron X-ray total scattering structure factors for ZIF-UC-5 (5-ClbIm) crystal and a_g ZIF-UC-5 (5-ClbIm).

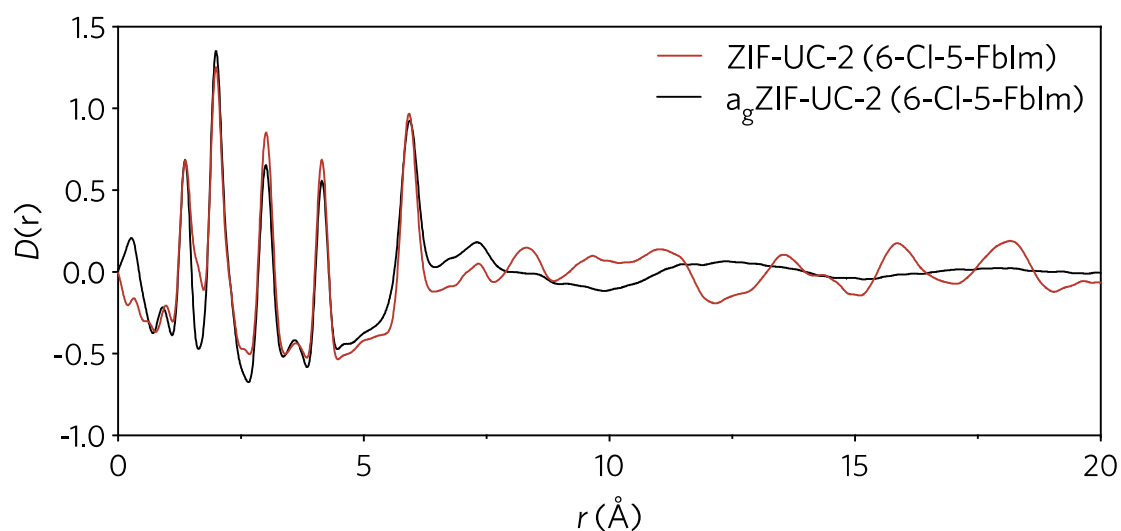


Figure S32. Pair distribution function $D(r)$ calculated *via* Fourier transform of the X-ray total scattering function $S(Q)$ for ZIF-UC-2 (6-Cl-5-FbIm) crystal and a_g ZIF-UC-2 (6-Cl-5-FbIm).

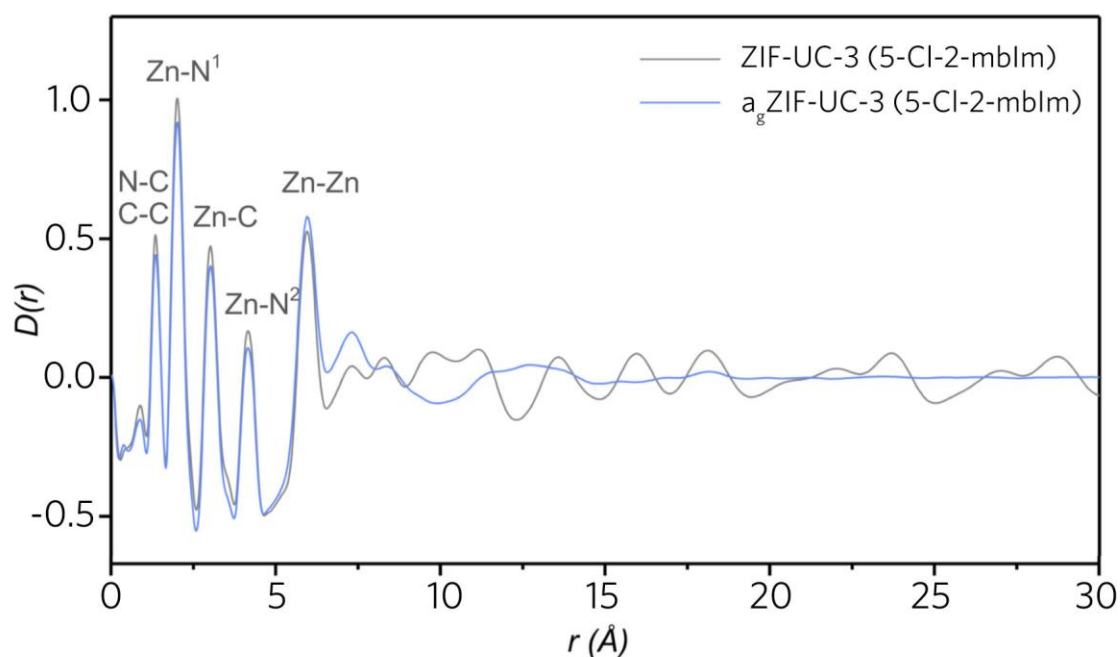


Figure S33. Pair distribution function $D(r)$ calculated *via* Fourier transform of the X-ray total scattering function $S(Q)$ for ZIF-UC-3 (5-Cl-2-mbIm) crystal and a_g ZIF-UC-3 (5-Cl-2-mbIm).

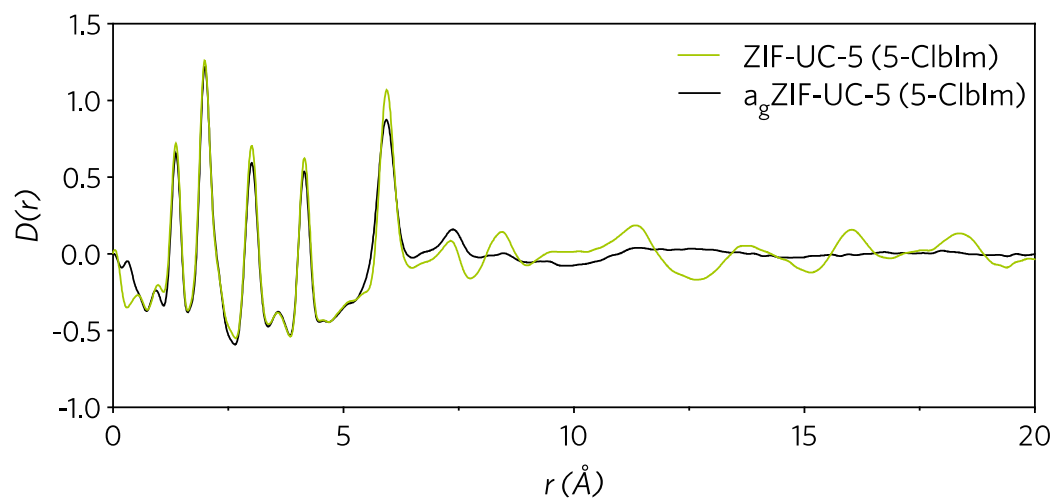


Figure S34. Pair distribution function $D(r)$ calculated *via* Fourier transform of the X-ray total scattering function $S(Q)$ for ZIF-UC-5 (5-ClbIm) crystal and a_g ZIF-UC-5 (5-ClbIm).

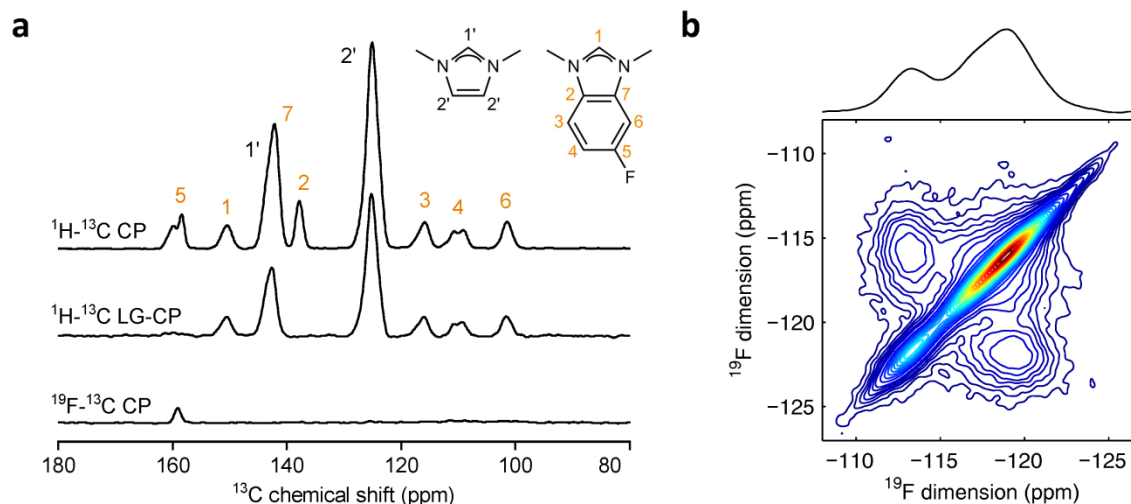


Figure S35. (a) Various ^{13}C MAS NMR spectra of crystalline ZIF-UC-4 (5-FbIm), corroborating the assignment of the individual signals in the spectra. ^1H - ^{13}C CPMAS spectrum exhibits signals of all C sites of Im and 5-FbIm. ^1H - ^{13}C Lee-Goldburg CPMAS spectrum shows only the signals of those C atoms that are directly bonded to H atoms (therefore signals of C5, C2 and C7 are missing). ^{19}F - ^{13}C CPMAS spectrum shows only the signal of C5, which is directly bonded to F. In the latter spectrum, C5 exhibits a single sharp signal, because ^{19}F decoupling is employed during detection. In the ^1H - ^{13}C CPMAS spectrum, where ^{19}F decoupling was not performed, the signal belonging to C5 is split into two signals by the strong ^{19}F - ^{13}C J-coupling interaction. (b) Two dimensional ^{19}F spin-diffusion NMR spectrum of ZIF-UC-4 (5-FbIm) crystals, recorded with the mixing time of 100 ms. Cross peaks between the signal resonating at -114 ppm and the signal resonating at -119 ppm indicate that ^{19}F nuclei occupying different sites are homogeneously dispersed within the same crystallites, and are not segregated into different domains or particles.

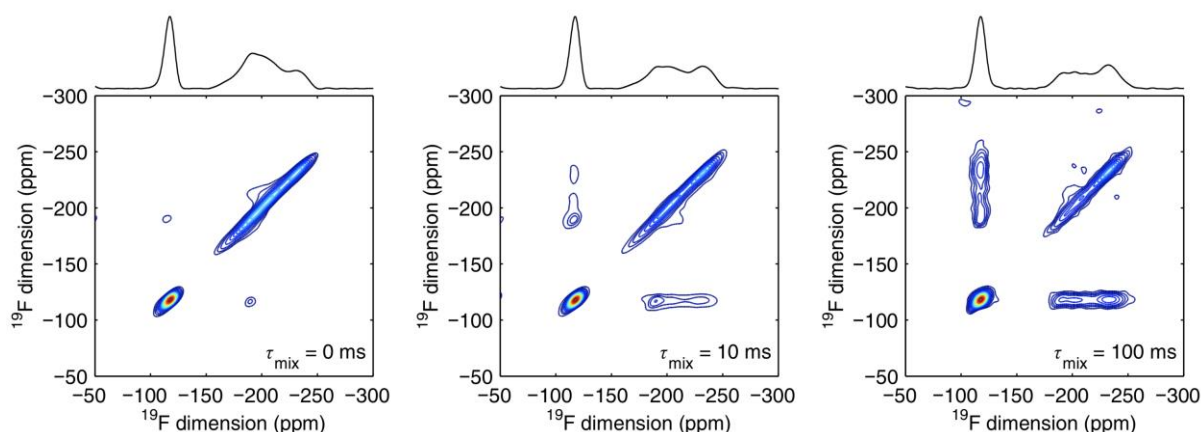


Figure S36. Two dimensional ^{19}F spin-diffusion NMR spectra of agZIF-UC-4 (5-FbIm), recorded with the mixing times of 0 ms, 10 ms, and 100 ms. Cross peaks between the signals resonating at about -115 ppm and the signals resonating between -170 and -240 ppm indicate that ^{19}F nuclei that give rise to the broad, shifted signals are within the same particles as ^{19}F nuclei that give rise to the unshifted narrow signal.

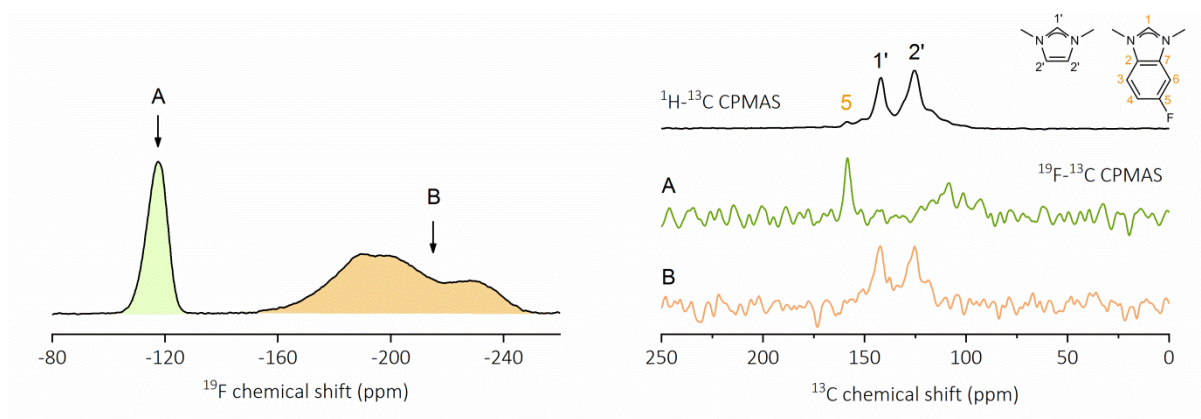


Figure S37. Selective ^{19}F - ^{13}C CPMAS NMR measurements in agZIF-UC-4 (5-FbIm). Selective cross-polarization from ^{19}F nuclei, contributing to the narrow ^{19}F signal of the glass, enhances the signal of C5 carbon nuclei. On the contrary, selective cross-polarization from ^{19}F nuclei, contributing to the broad ^{19}F signal of the glass, gives rise to a C spectrum in which only the two signals of Im can be detected. This observation suggests that F that contributed to the broad, shifted signal, is not bonded to benzene rings (it does not belong to 5-FbIm ligands of agZIF-UC-4 (5-FbIm)). Black arrows above the ^{19}F MAS NMR spectrum indicate the carrier frequencies used for selective excitation of F nuclei.

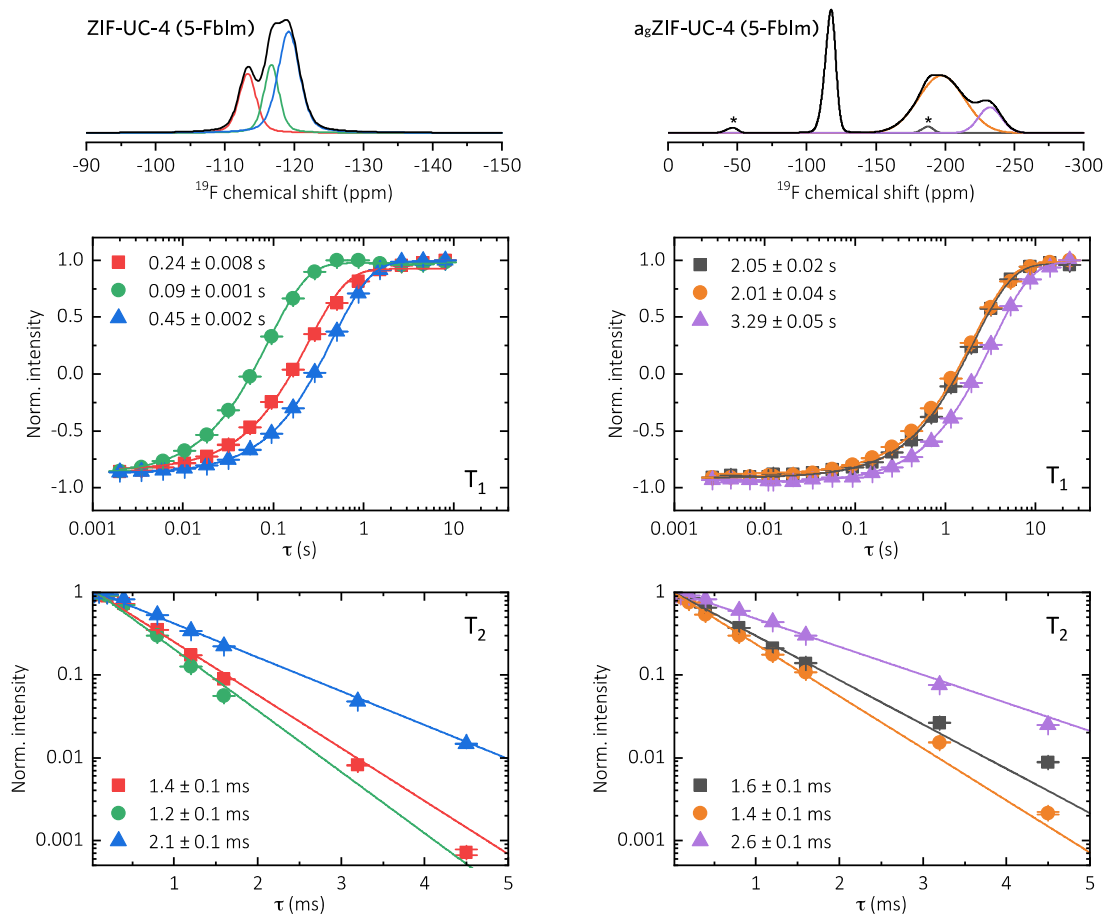


Figure S38. ^{19}F spin-lattice (T_1) and spin-spin (T_2) relaxation-time measurements in ZIF-UC-4 (5-Fblm) and agZIF-UC-4 (5-Fblm). The results show that relaxation in the glass is slower than in the crystalline material. They also show that there is no significant difference between the relaxation times of the non-shifted narrow and the shifted broad signals of the ^{19}F MAS NMR spectrum of the glass. This clearly indicates that paramagnetic species are not present in the glass and that they cannot be responsible for the large shift of the ^{19}F signal. T_1 and T_2 measurements were carried out under MAS using the inversion-recovery and spin-echo pulse sequences, respectively.

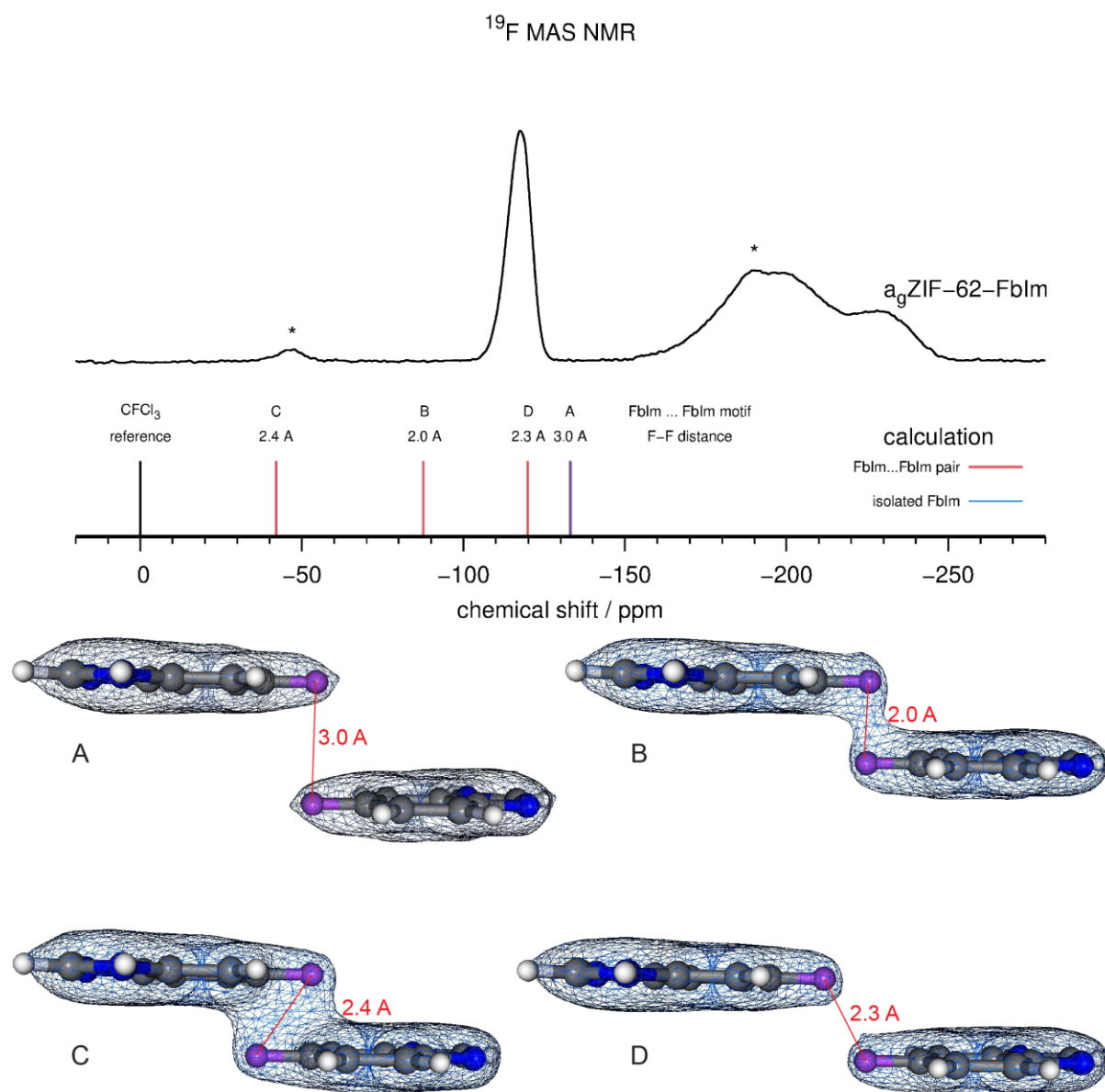


Figure S39. Comparison of ^{19}F chemical shifts measured in $\text{a}_g\text{ZIF-UC-4}$ (5-FbIm) and ^{19}F chemical shifts calculated for selected arrangements of pairs of 5-FbIm molecules. In different DFT-based calculations two 5-FbIm molecules are placed in different positions with respect to each other. The calculations suggest that in cases, when two 5-FbIm molecules are close one to another (and especially when the F atom of one molecule approaches the benzyl ring of the other molecule), large changes in chemical shifts can be expected. However, the predicted changes in chemical shifts have opposite sign than the experimentally detected ones. Calculations thus indicate that the detected broad, shifted ^{19}F signals with chemical shifts ranging between -170 and -240 ppm is not due to the formation of halogen bonds between pairs of 5-FbIm ligands within $\text{a}_g\text{ZIF-UC-4}$ (FbIm). Calculations of chemical shifts were carried out with Orca 4.0.1.2 package at the PBE0-D3/def2-QZVPP level.

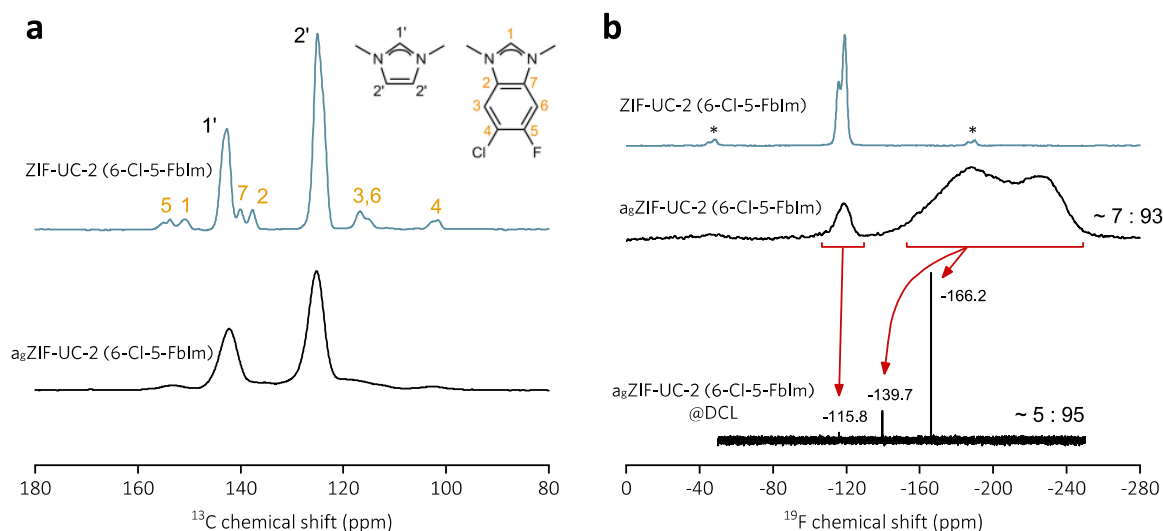


Figure S40. ^{13}C (a) and ^{19}F (b) MAS NMR spectra of ZIF-UC-2 (6-Cl-5-FbIm) and agZIF-UC-2 (6-Cl-5-FbIm). The spectrum at the bottom of (b) is the ^{19}F solution NMR spectrum of agZIF-UC-2 (6-Cl-5-FbIm), digested in DCL/DMSO- d_6 . Approximate intensity ratios for the signals at about -116 ppm and the shifted signals in the ^{19}F spectra of the glass and digested glass are presented. Arrows denote correlations between the ^{19}F NMR signals of the glass and digested glass. Asterisks mark positions of the spinning sidebands. Features that can be observed in the ^{19}F NMR spectra of ZIF-UC-2 (6-Cl-5-FbIm), agZIF-UC-2 (6-Cl-5-FbIm), and digested agZIF-UC-2 (6-Cl-5-FbIm) are almost identical to the ones detected in the ZIF-UC-4 (5-FbIm) series of samples.

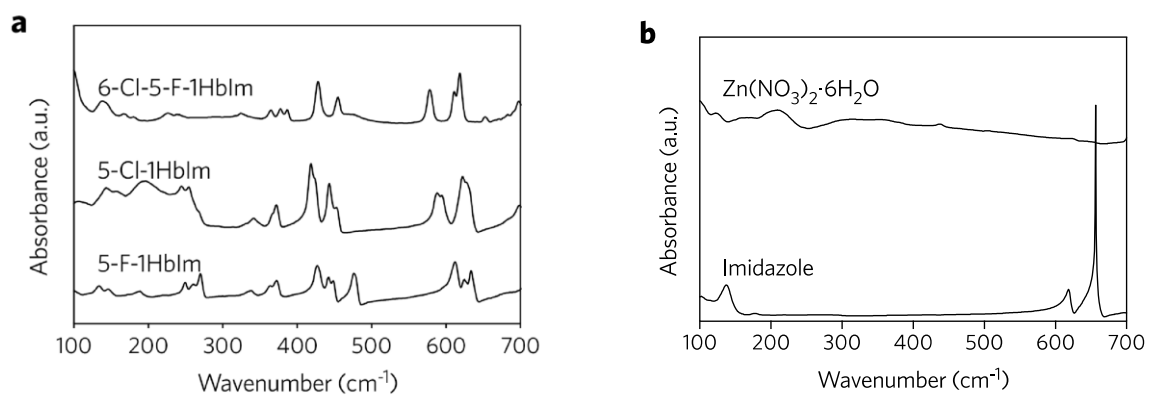


Figure S41. THz/Far-IR spectra of the ZIF precursors. Data were collected at room temperature.

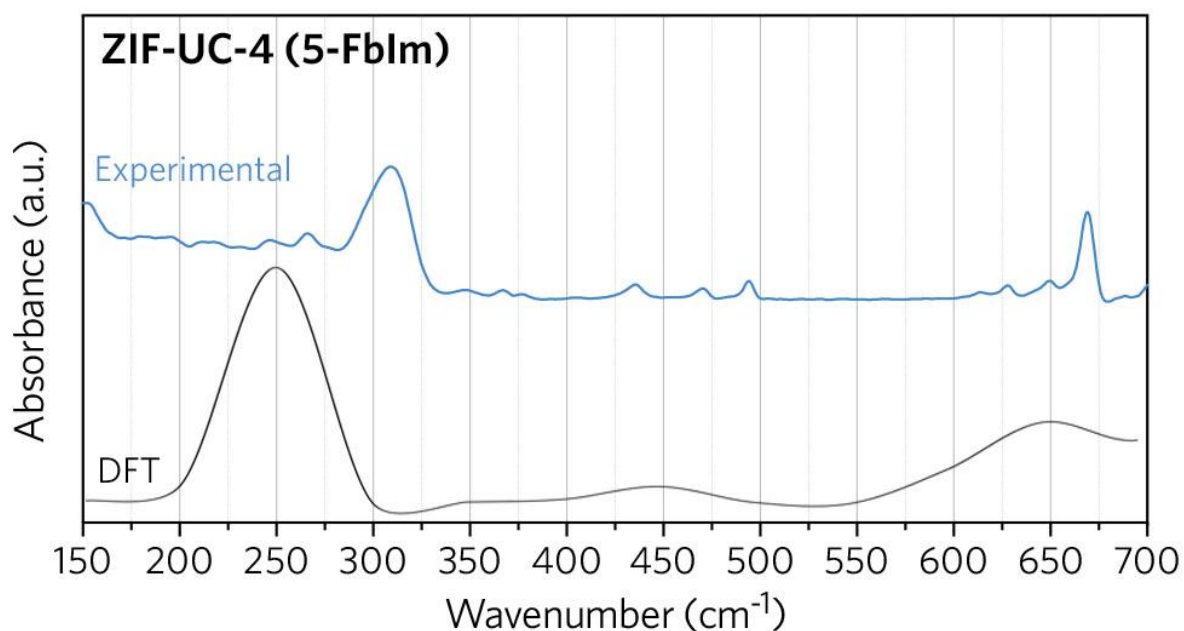


Figure S42. Comparison of the experimental ZIF-UC-4 (5-FbIm) and DFT calculated $\text{Zn}(\text{Im})_3(5\text{-FbIm})$ tetrahedron spectra in the wavenumber range of 150–700 cm^{-1} (measured & calculated at 298 K).

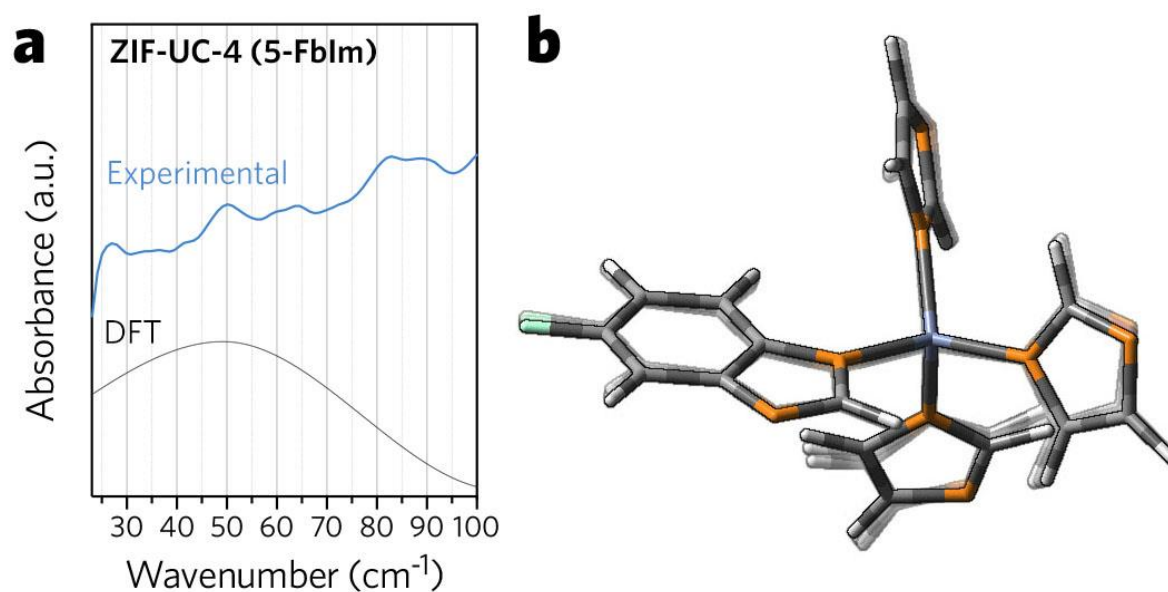


Figure S43. (a) Comparison of the experimental ZIF-UC-4 (5-FbIm) and DFT calculated $\text{Zn}(\text{Im})_3(5\text{-FbIm})$ tetrahedron spectra in the wavenumber range of 23–100 cm^{-1} . The corresponding tetrahedron motion for DFT calculation is demonstrated in panel (b). The vibrational movement of the single tetrahedron leads to the distortion movement of the 4 and 6 membered rings. Zn: blue; N: orange; F: green; C: grey; H: white.

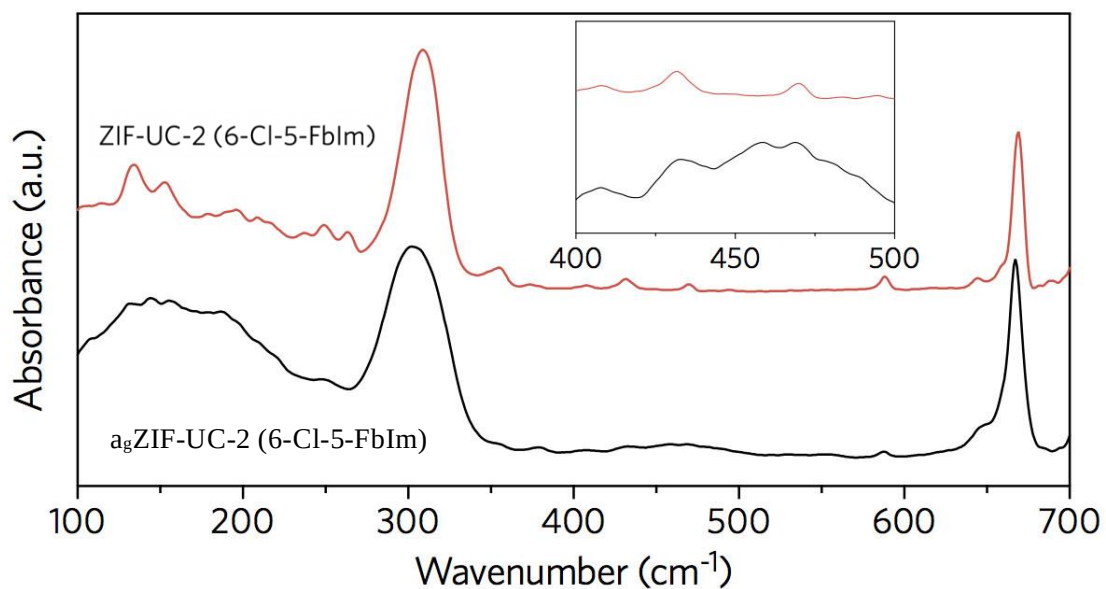


Figure S44. THz/Far-IR spectra of ZIF-UC-2 (6-Cl-5-FbIm) and agZIF-UC-2 (6-Cl-5-FbIm) measured at 298 K with the insert being the enlarged spectra highlighting the emergence of a broad peak at 400–500 cm^{-1} after melt quenching (intensity normalized against the Zn-N peak at ca. 300 cm^{-1}).

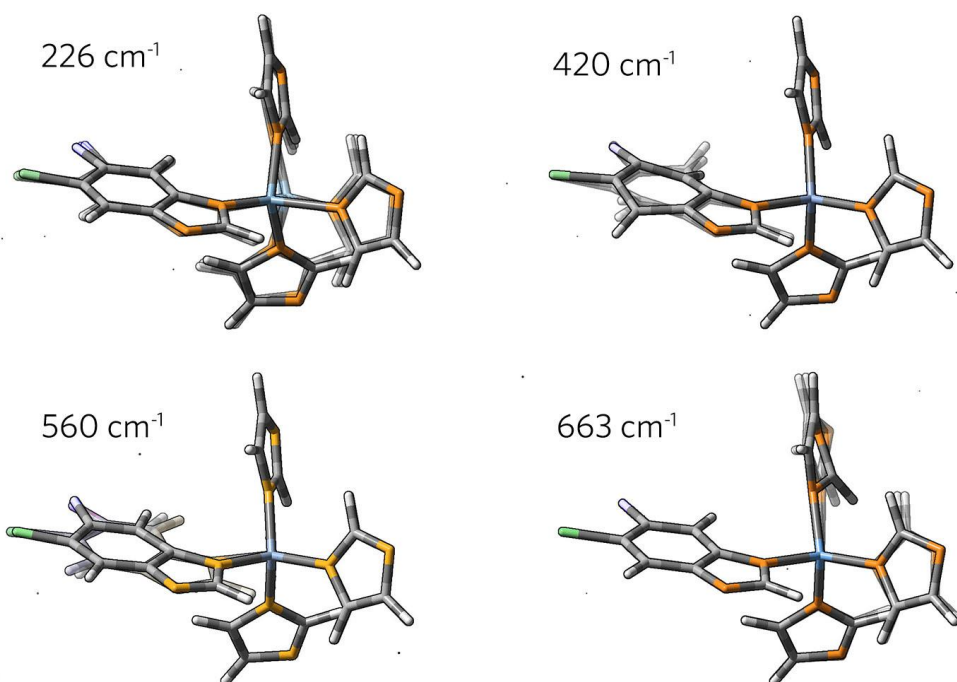
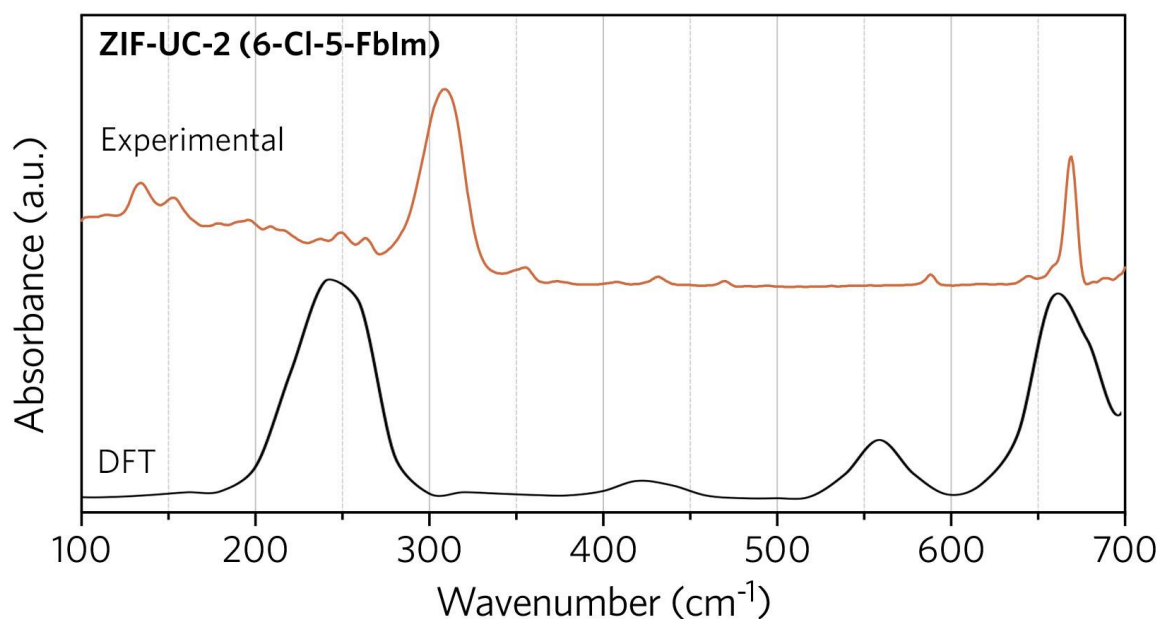


Figure S45. Comparison of the experimental ZIF-UC-2 (6-Cl-5-FbIm) and DFT calculated $\text{Zn}(\text{Im})_3(6\text{-Cl-5-FbIm})$ tetrahedron spectra in the wavenumber range of 100–700 cm^{-1} , with the schematic diagram of the corresponding vibrational movements for DFT calculation at 298 K. Zn: blue; N: orange; F: green; Cl: purple; C: grey; H: white.

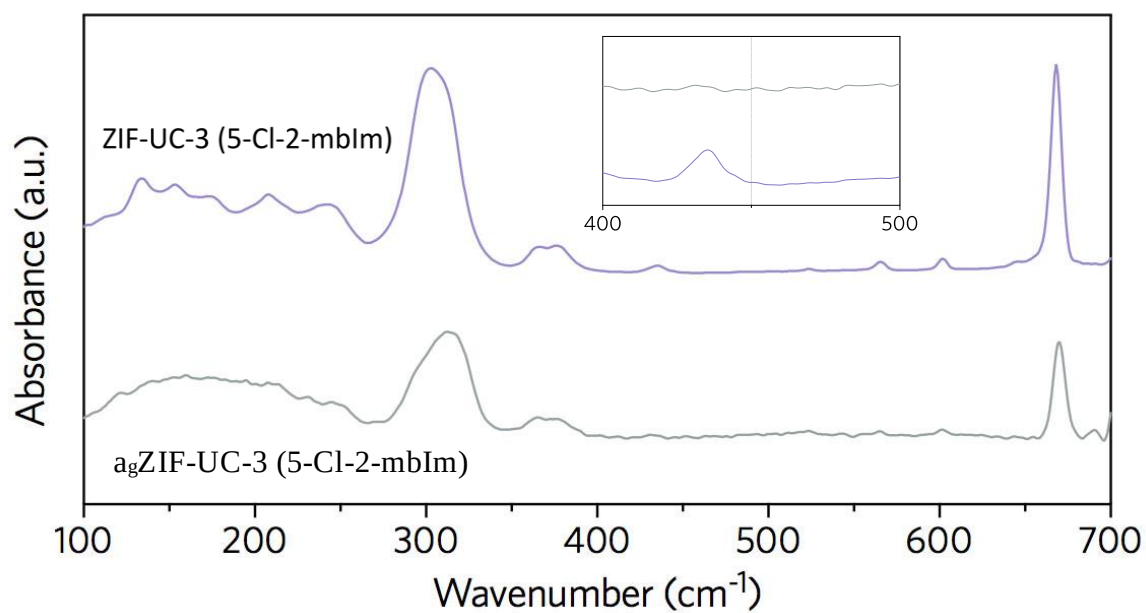


Figure S46. THz/Far-IR spectra of ZIF-UC-3 (5-Cl-2-mbIm) and agZIF-UC-3 (5-Cl-2-mbIm) measured at 298 K. Insert is the enlarged spectra showing the relatively feature less pattern after melting (intensity normalized against the Zn-N peak at *ca.* 300 cm^{-1}).

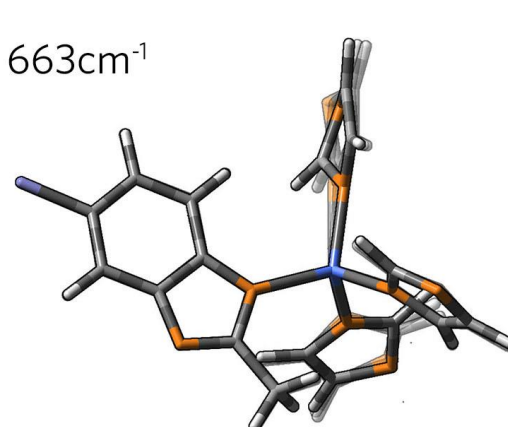
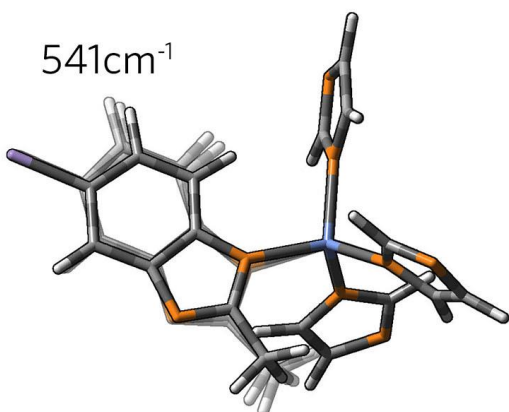
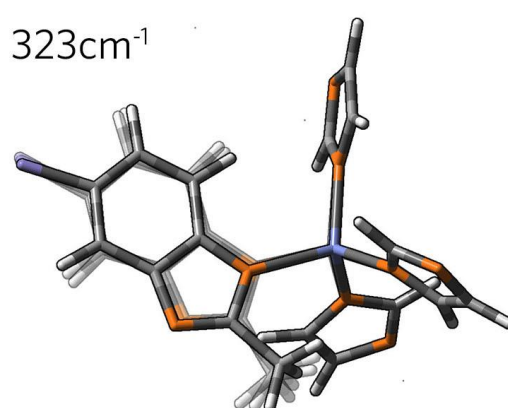
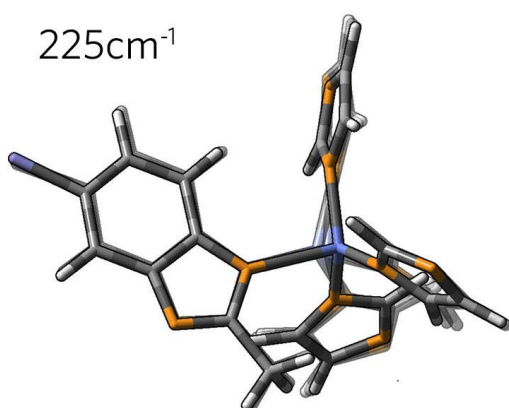
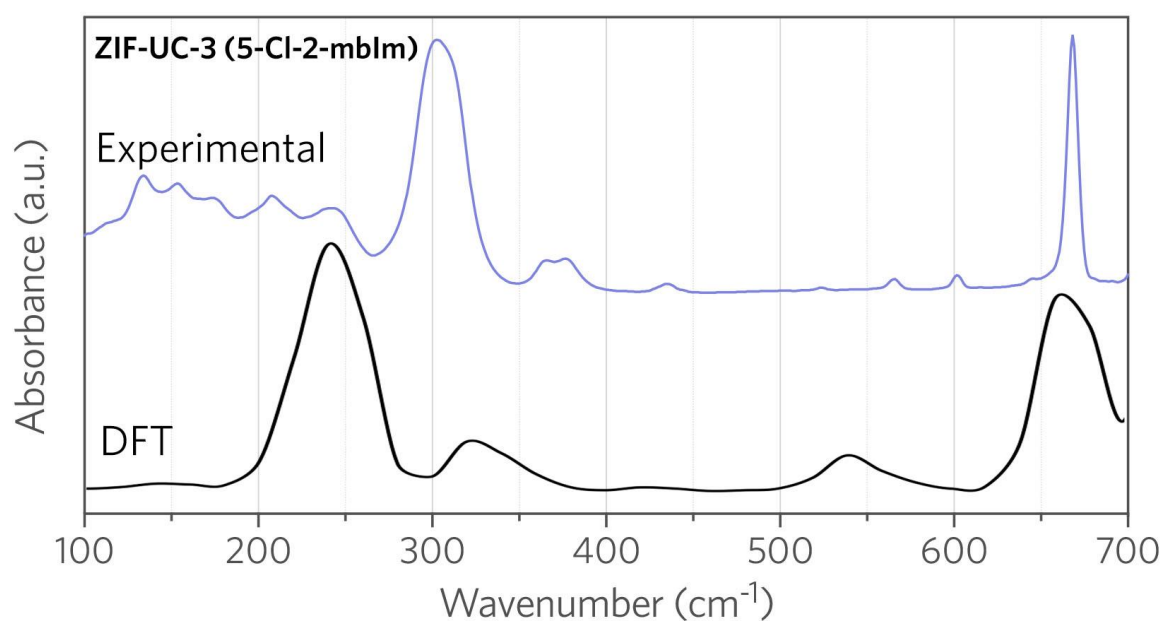


Figure S47. Comparison of the experimental ZIF-UC-3 (5-Cl-2-mblm) and DFT calculated $\text{Zn}(\text{Im})_3(5\text{-Cl-2-mblm})$ tetrahedron spectra in the wavenumber range of 100–700 cm^{-1} , with the schematic diagram of the corresponding vibrational movements for DFT calculation at 298 K. Zn: blue; N: orange; Cl: purple; C: grey; H: white.

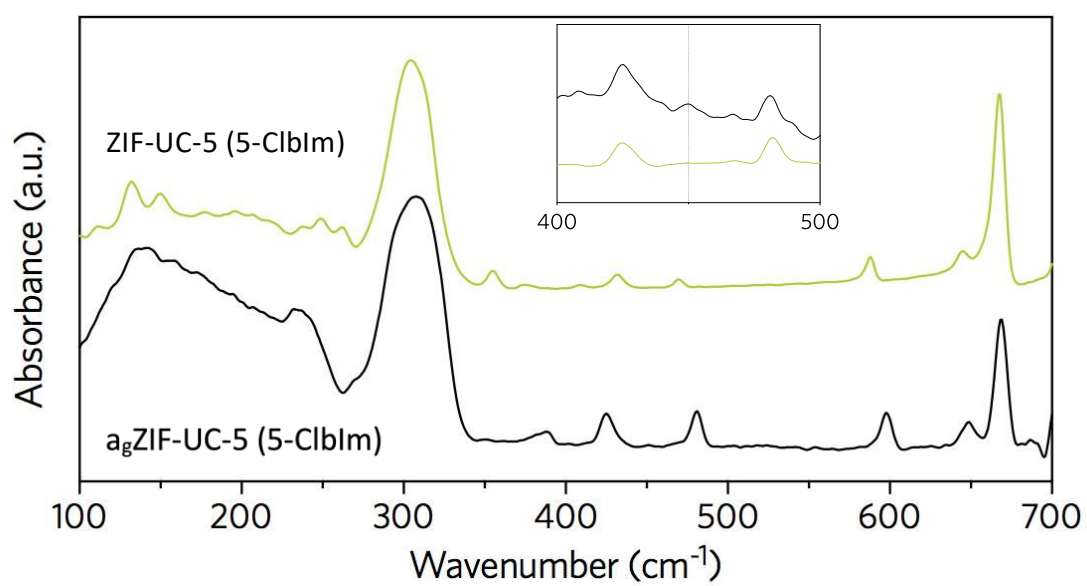


Figure S48. THz/Far-IR spectra of ZIF-UC-5 (5-ClbIm) and agZIF-UC-5 (5-ClbIm) measured at 298 K. Insert is the enlarged spectra (intensity normalized against the Zn-N peak at *ca.* 300 cm⁻¹).

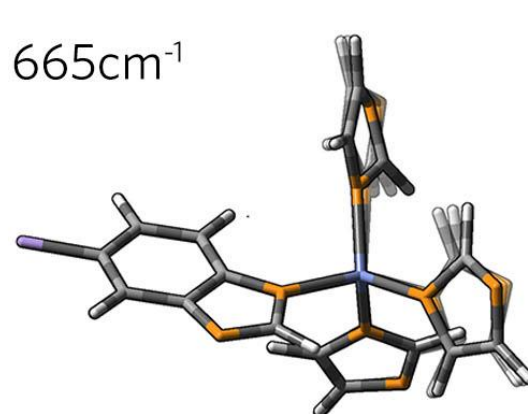
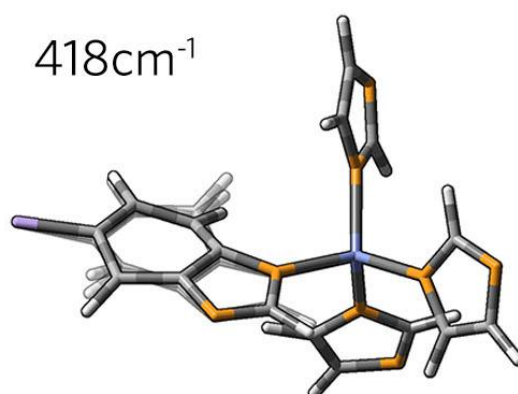
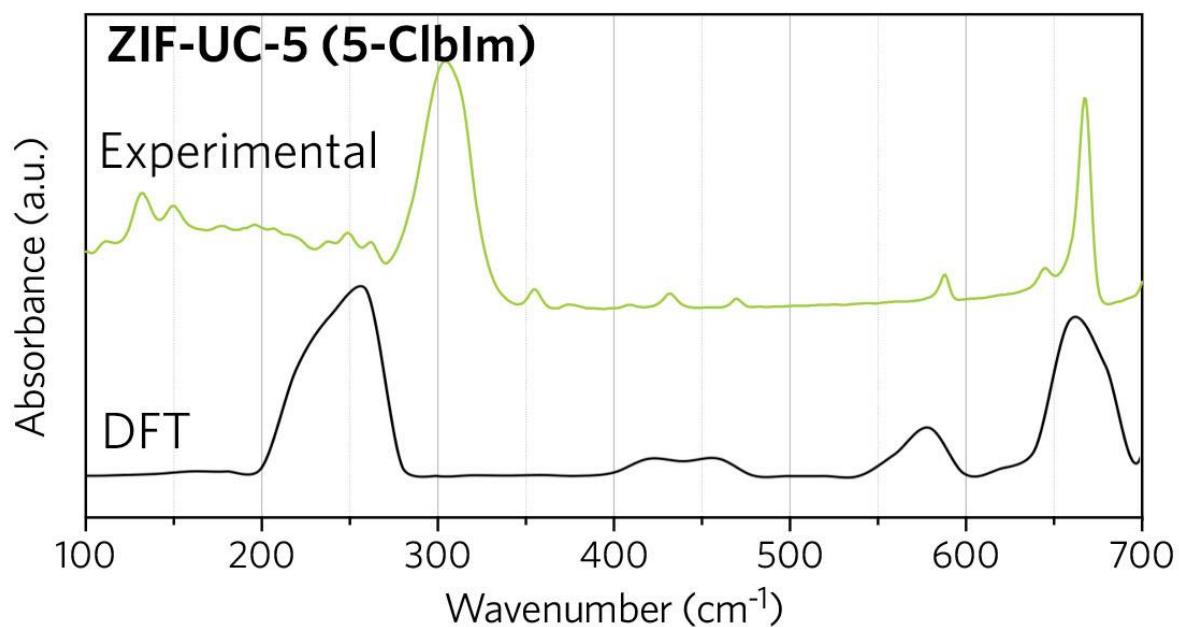


Figure S49. Comparison of the experimental ZIF-UC-5 (5-ClbIm) and DFT calculated $\text{Zn}(\text{Im})_3(5\text{-ClbIm})$ tetrahedron spectra in the wavenumber range of 100–700 cm^{-1} , with the schematic diagram of the corresponding vibrational movements for DFT calculation at 298 K. Zn: blue; N: orange; Cl: purple; C: grey; H: white.

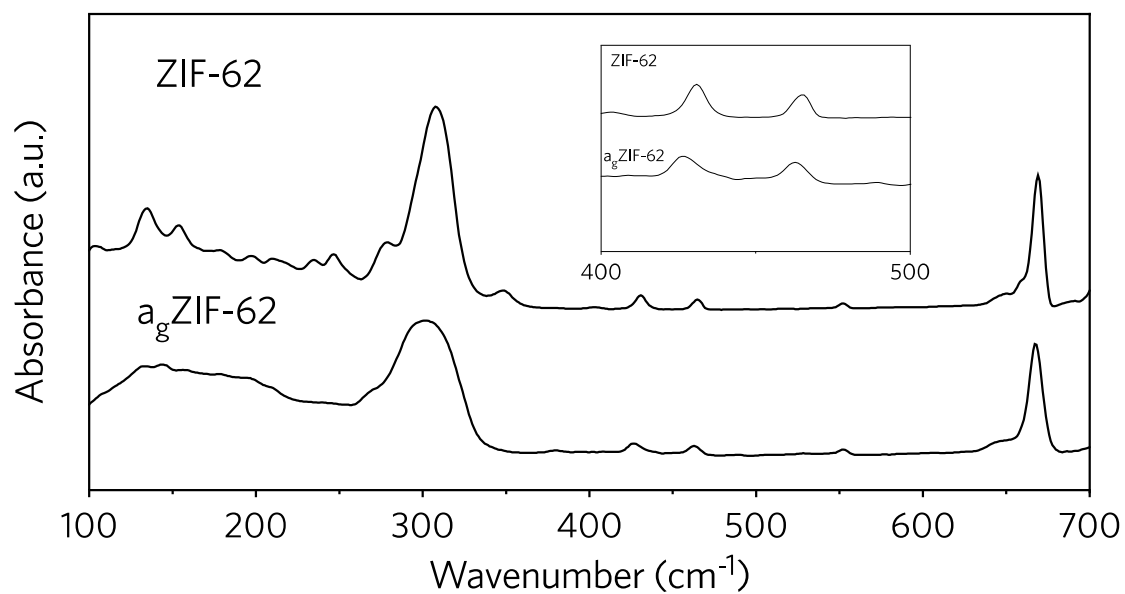
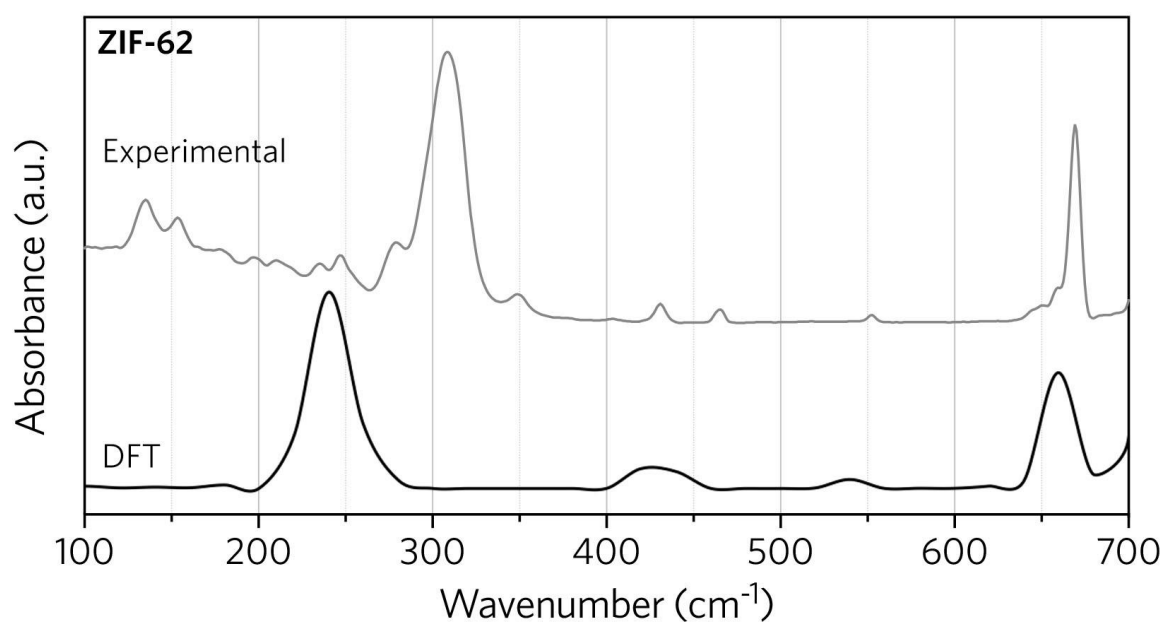
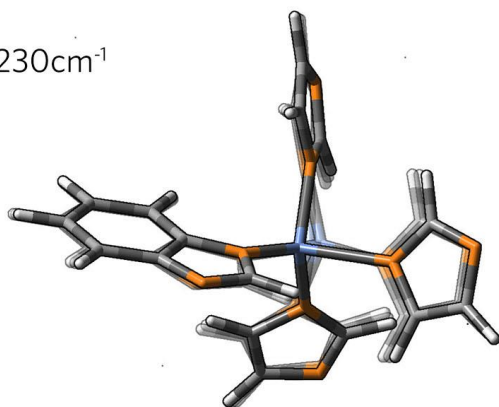


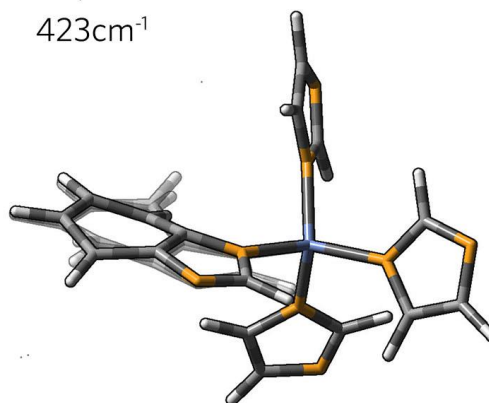
Figure S50. THz/Far-IR spectra of ZIF-62 and a_g ZIF-62 measured at 298 K. Insert is the enlarged spectra for crystal and glass (intensity normalized against the Zn-N peak at *ca.* 300 cm^{-1}).



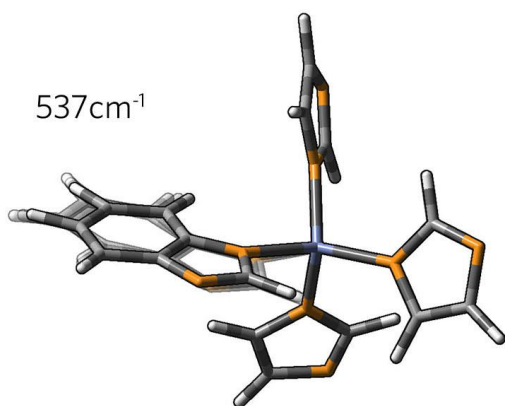
230cm⁻¹



423cm⁻¹



537cm⁻¹



665cm⁻¹

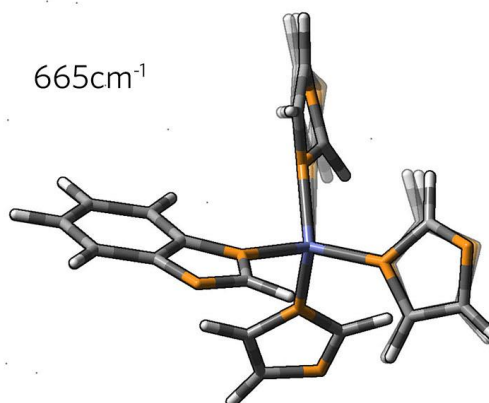


Figure S51. Comparison of the experimental ZIF-62 and DFT calculated $\text{Zn}(\text{Im})_3(\text{bIm})$ tetrahedron spectra in the wavenumber range of 100–700 cm^{-1} , with the schematic diagram of the corresponding vibrational movements for DFT calculation at 298 K. Zn: blue; N: orange; Cl: purple; C: grey; H: white.

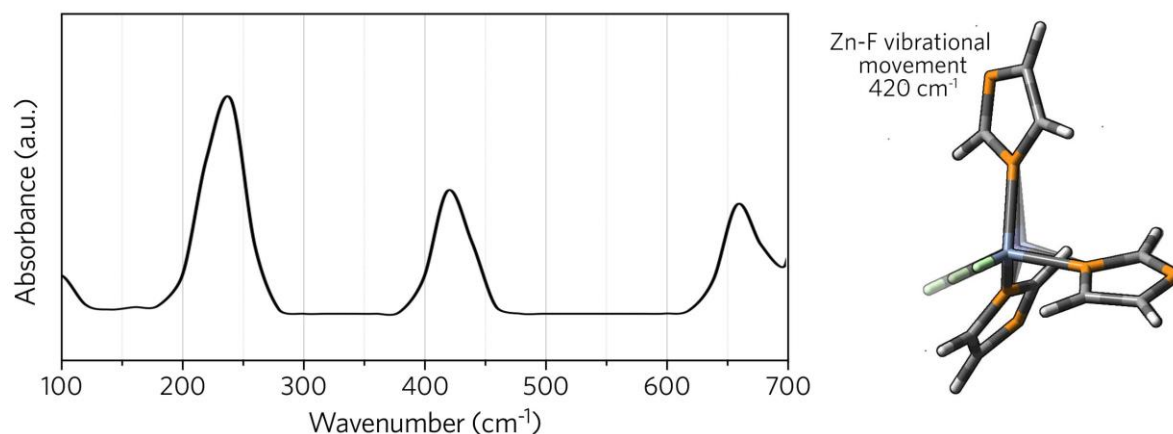


Figure S52. DFT calculated THz/FIR spectra of $\text{Zn}(\text{Im})_3\text{F}$ tetrahedron in the wavenumber range of 100–700 cm^{-1} (298 K), with the schematic diagram of the corresponding vibrational movement of Zn-F bond. Zn: blue; N: orange; F: green; C: grey; H: white.

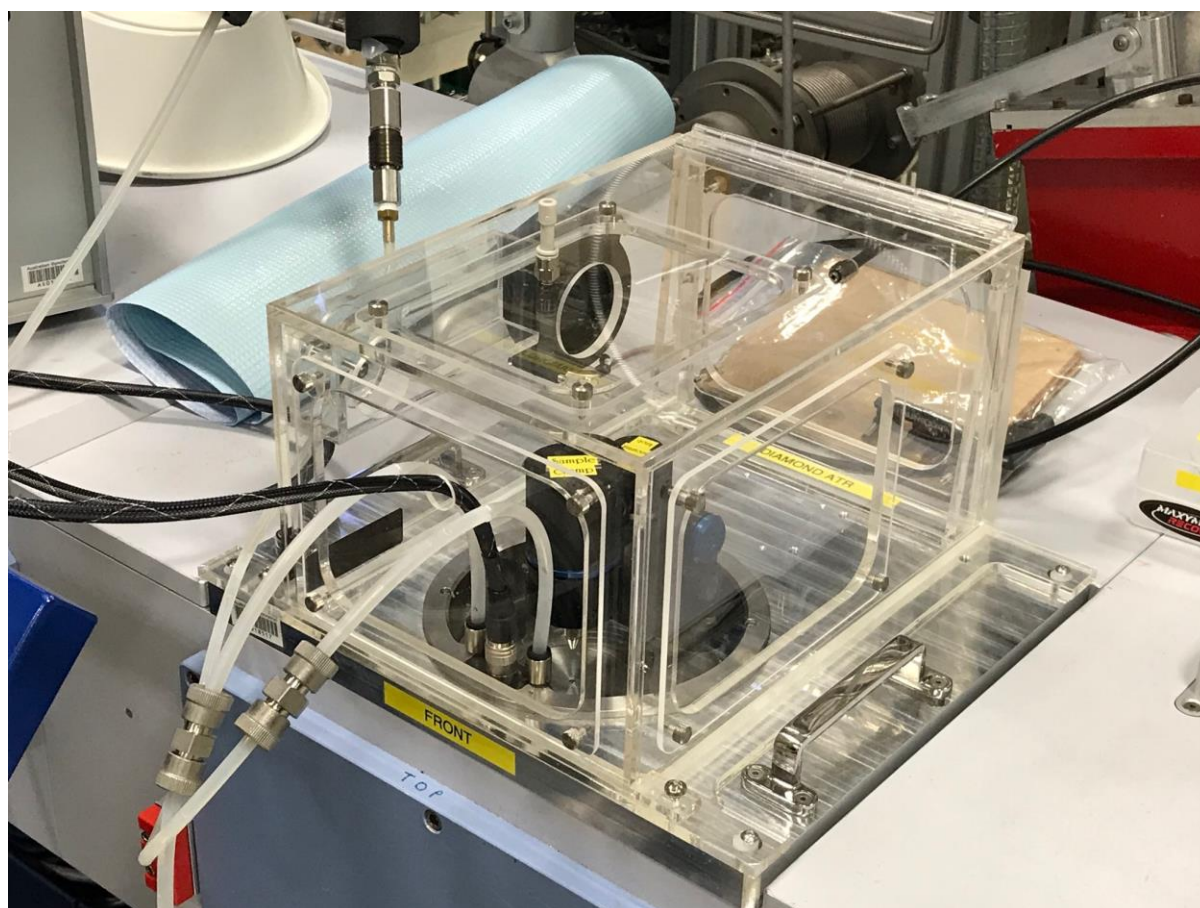


Figure S53. Experimental setup of the Synchrotron THz/Far-IR beamline for the *in situ* heating test at Australian Synchrotron.

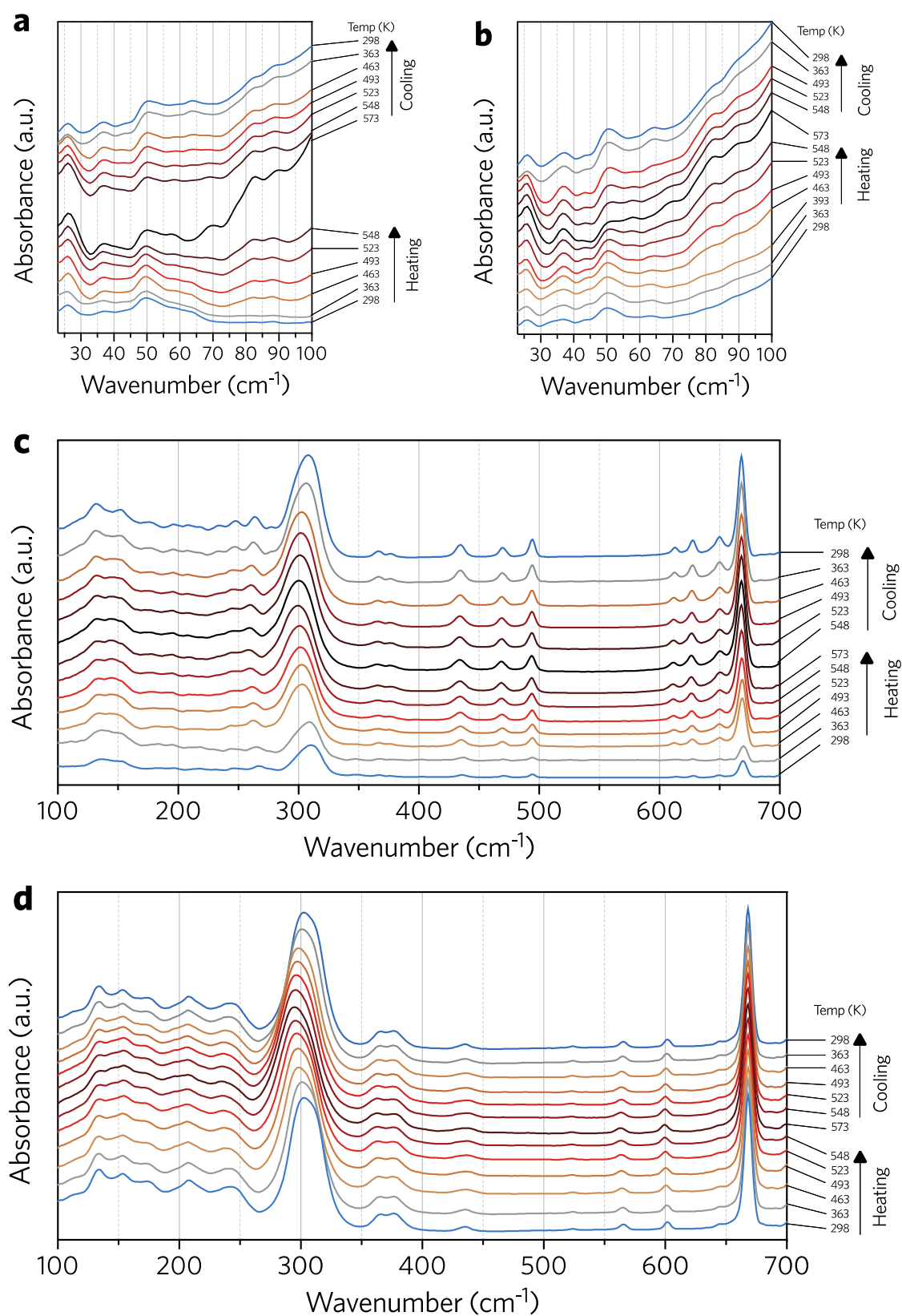


Figure S54. *In situ* THz/FIR spectra for ZIF-UC-4 (5-FbIm) (a and c) and ZIF-UC-3 (5-Cl-2-mbIm) (b and d) during the heating and subsequent cooling test. Each spectrum was collected after 3 min thermal equilibrium.

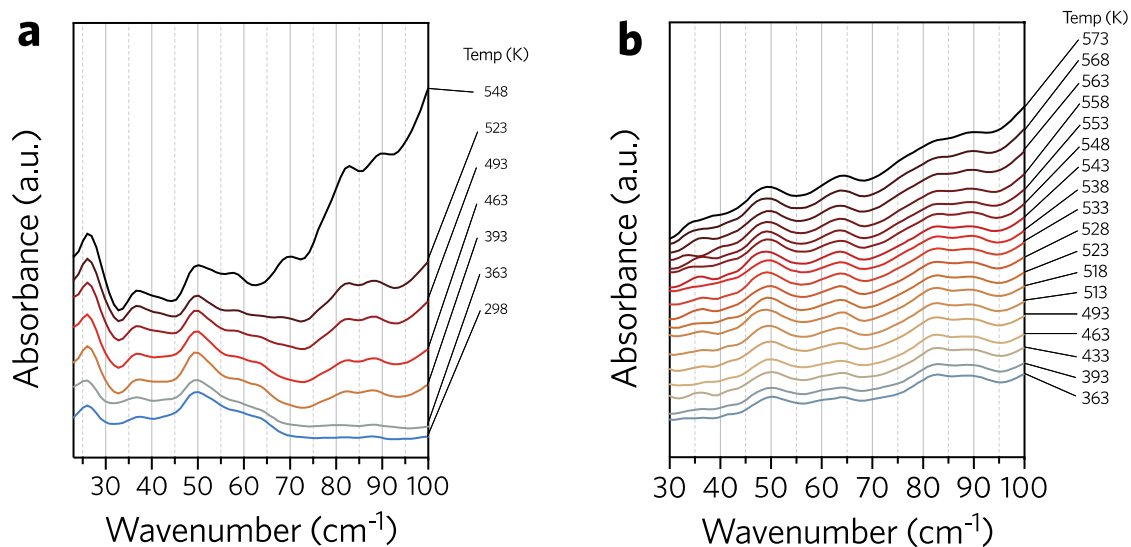


Figure S55. *In situ* THz/FIR spectra for (a) ZIF-UC-4 (5-FbIm) and (b) agZIF-UC-4 (5-FbIm) at the wavenumber range of 23–100 cm^{-1} .

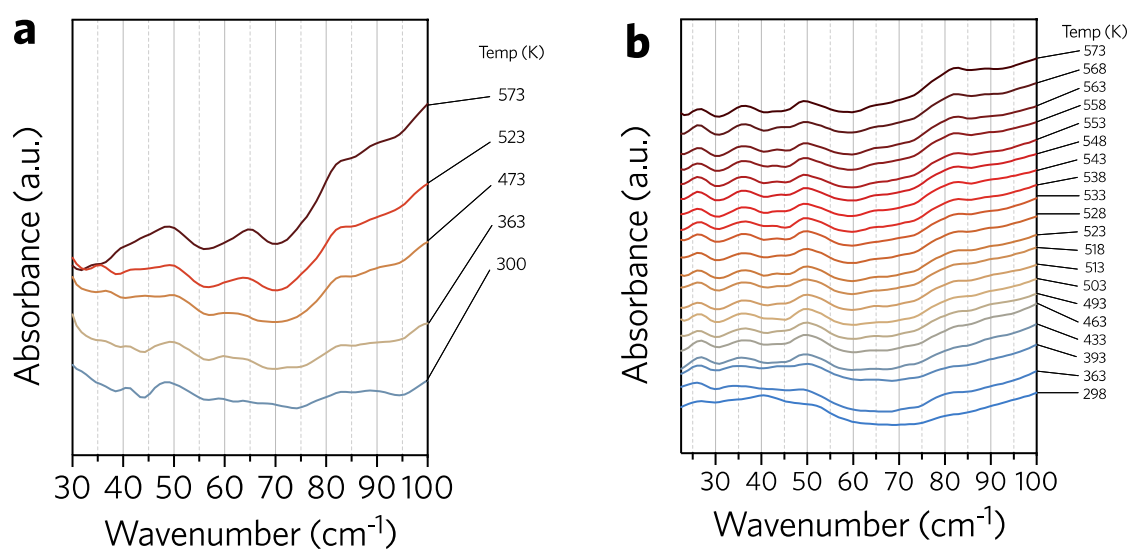


Figure S56. *In situ* synchrotron THz/FIR spectra for (a) ZIF-UC-2 (6-Cl-5-FbIm) and (b) agZIF-UC-2 (6-Cl-5-FbIm) at the wavenumber range of 23–100 cm^{-1} .

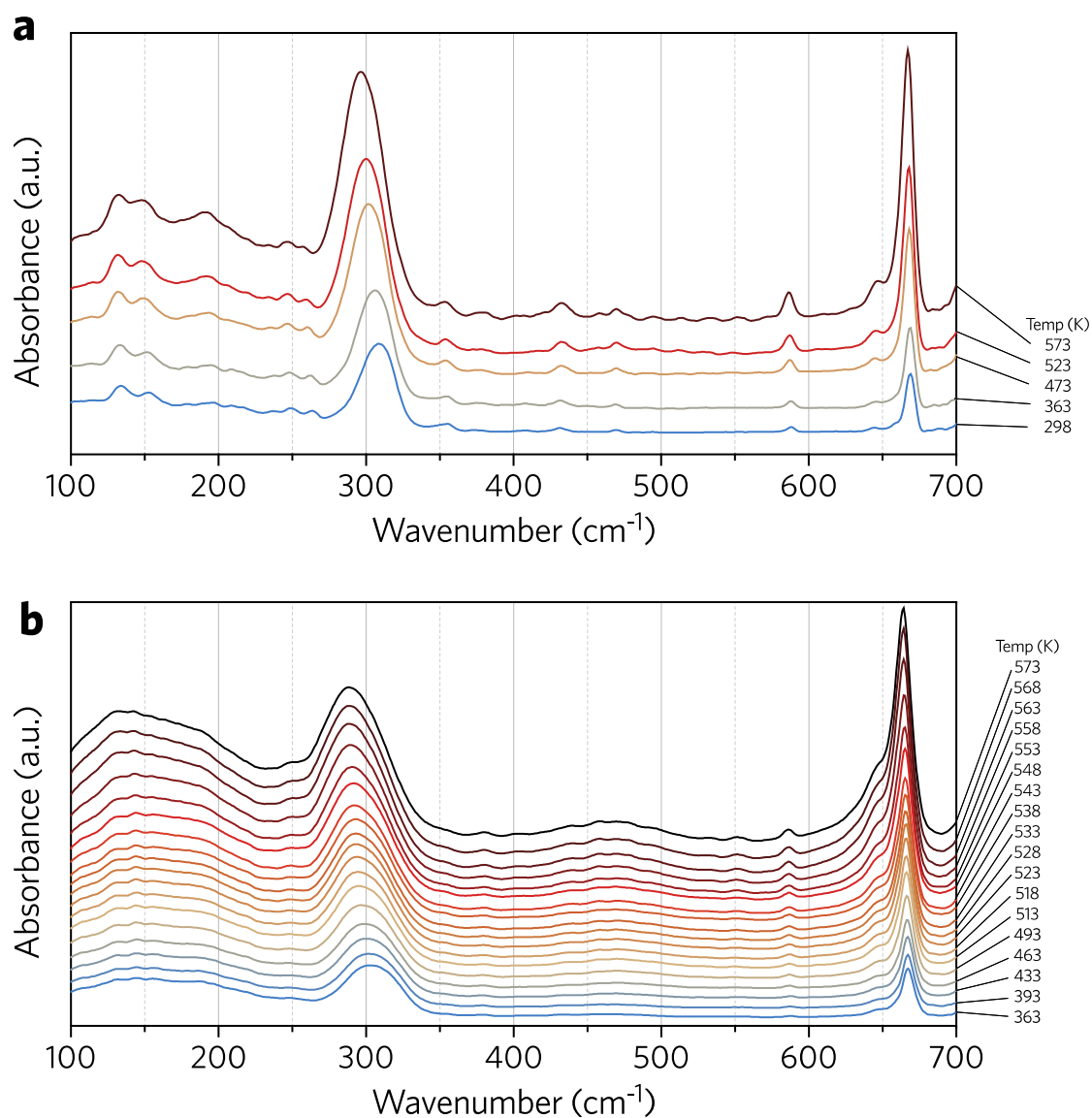


Figure S57. *In situ* synchrotron THz/FIR spectra for (a) ZIF-UC-2 (6-Cl-5-FbIm) and (b) agZIF-UC-2 (6-Cl-5-FbIm) at the wavenumber range of 100–700 cm^{-1} .

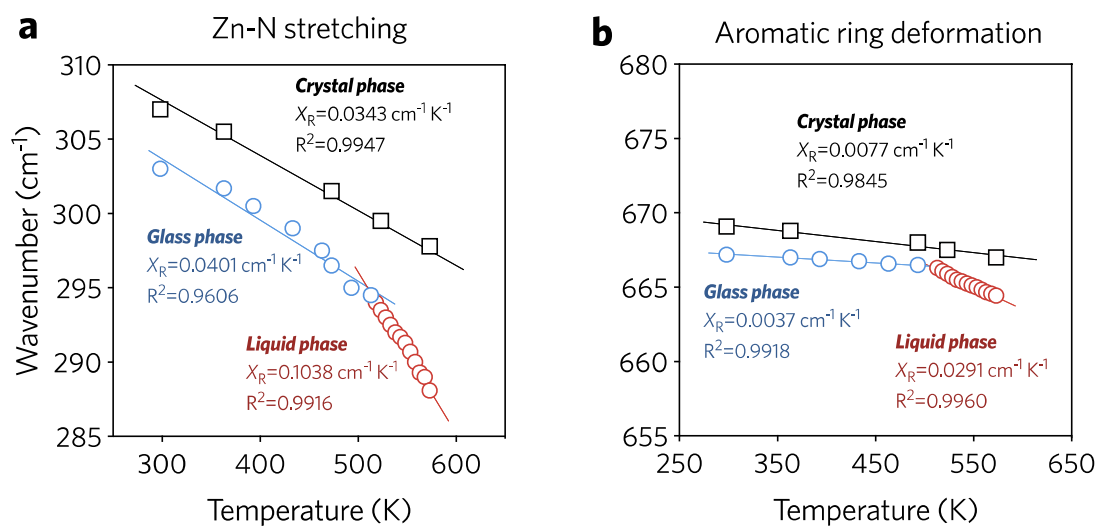


Figure S58. Deformation modes of the (a) Zn-N stretching and (b) aromatic ring deformation during the *in situ* THz/FIR heating test for ZIF-UC-2 (6-Cl-5-FbIm) and agZIF-UC-2 (6-Cl-5-FbIm).

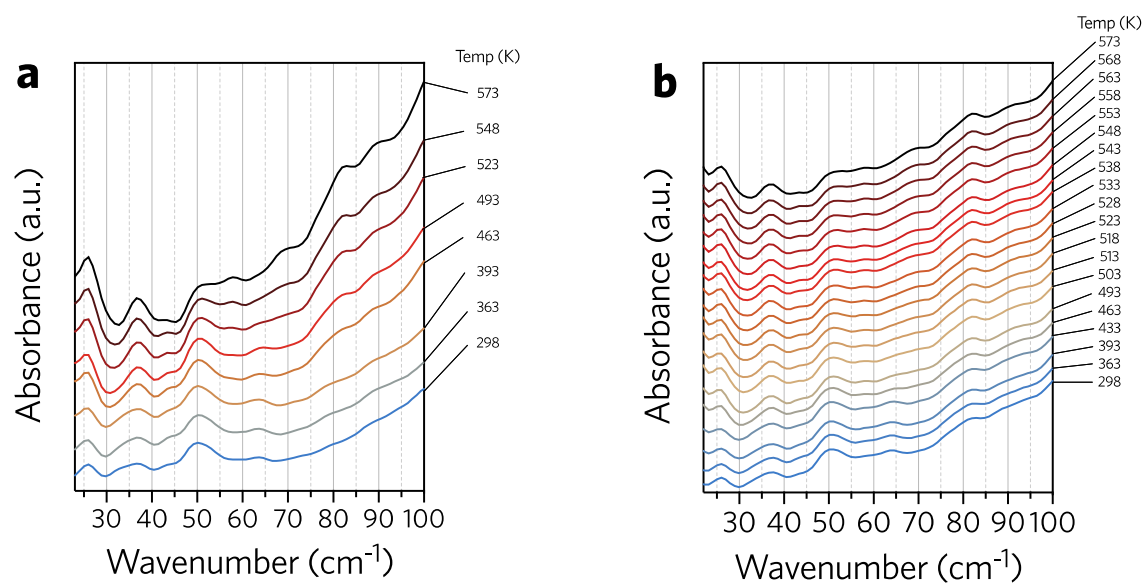


Figure S59. *In situ* synchrotron THz/FIR spectra for (a) ZIF-UC-3 (5-Cl-2-mbIm) and (b) agZIF-UC-3 (5-Cl-2-mbIm) at the wavenumber range of 23–100 cm^{-1} .

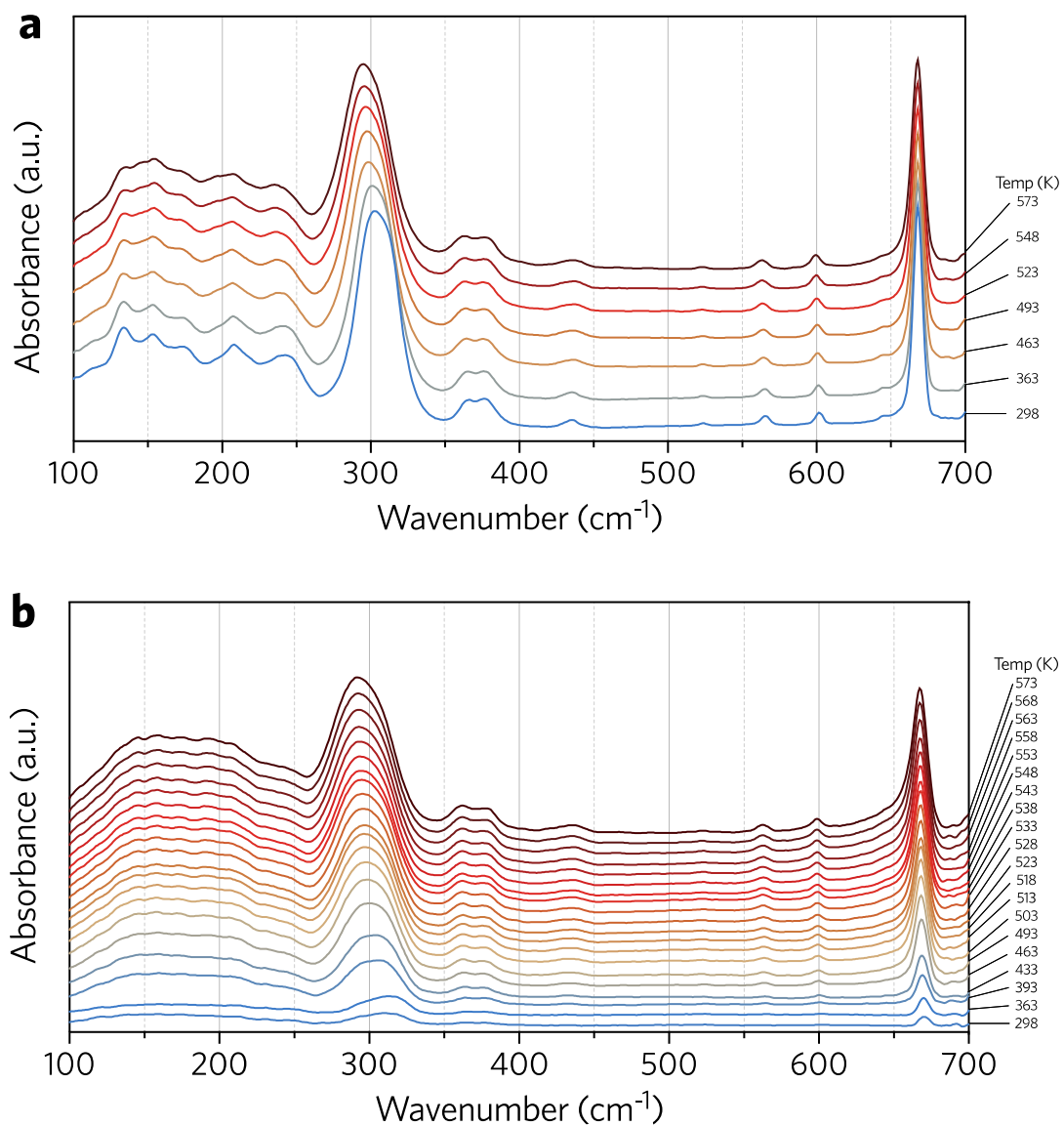


Figure S60. *In situ* synchrotron THz/FIR spectra for (a) ZIF-UC-3 (5-Cl-2-mbIm) and (b) agZIF-UC-3 (5-Cl-2-mbIm) at the wavenumber range of 100–700 cm^{-1} .

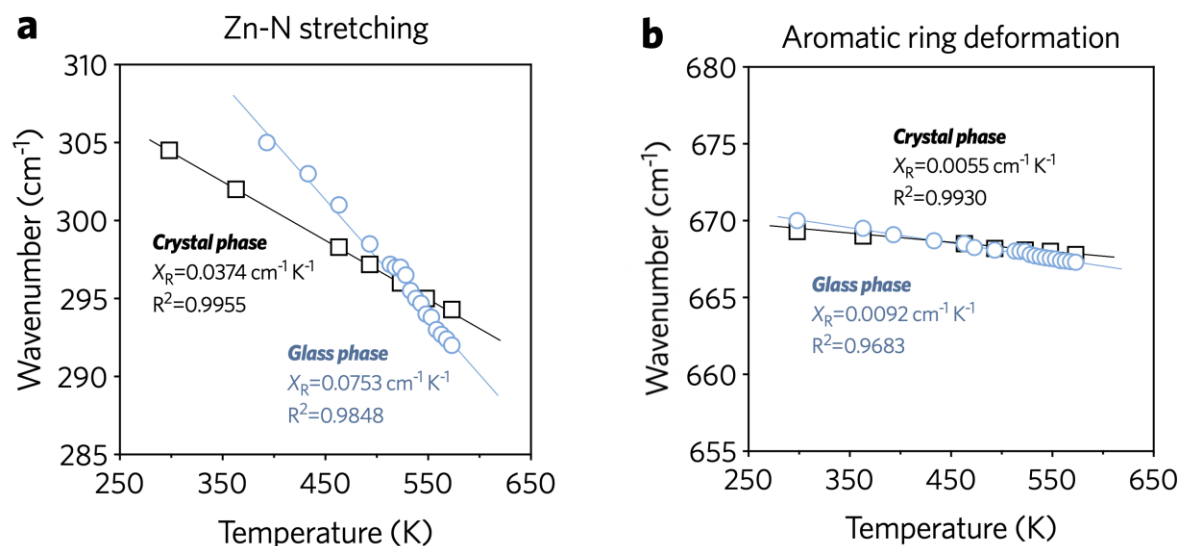


Figure S61. Deformation modes of the (a) Zn-N stretching and (b) aromatic ring deformation during the *in situ* THz/FIR heating test for ZIF-UC-3 (5-Cl-2-mbIm) and agZIF-UC-3 (5-Cl-2-mbIm).

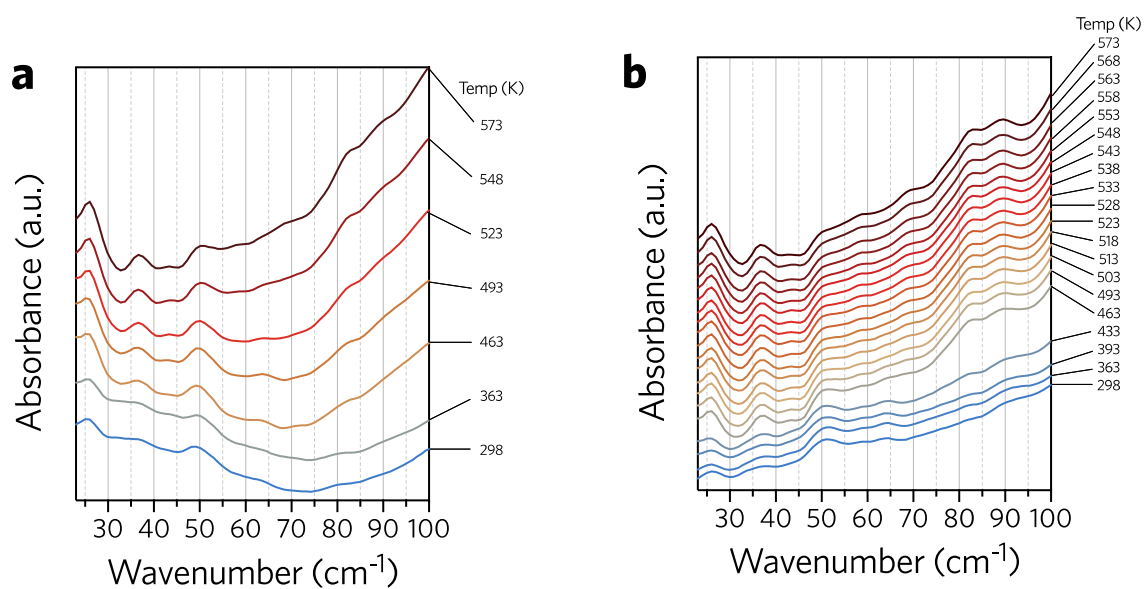


Figure S62. *In situ* synchrotron THz/FIR spectra for (a) ZIF-UC-5 (5-ClbIm) and (b) agZIF-UC-5 (5-ClbIm) at the wavenumber range of 23–100 cm^{-1} .

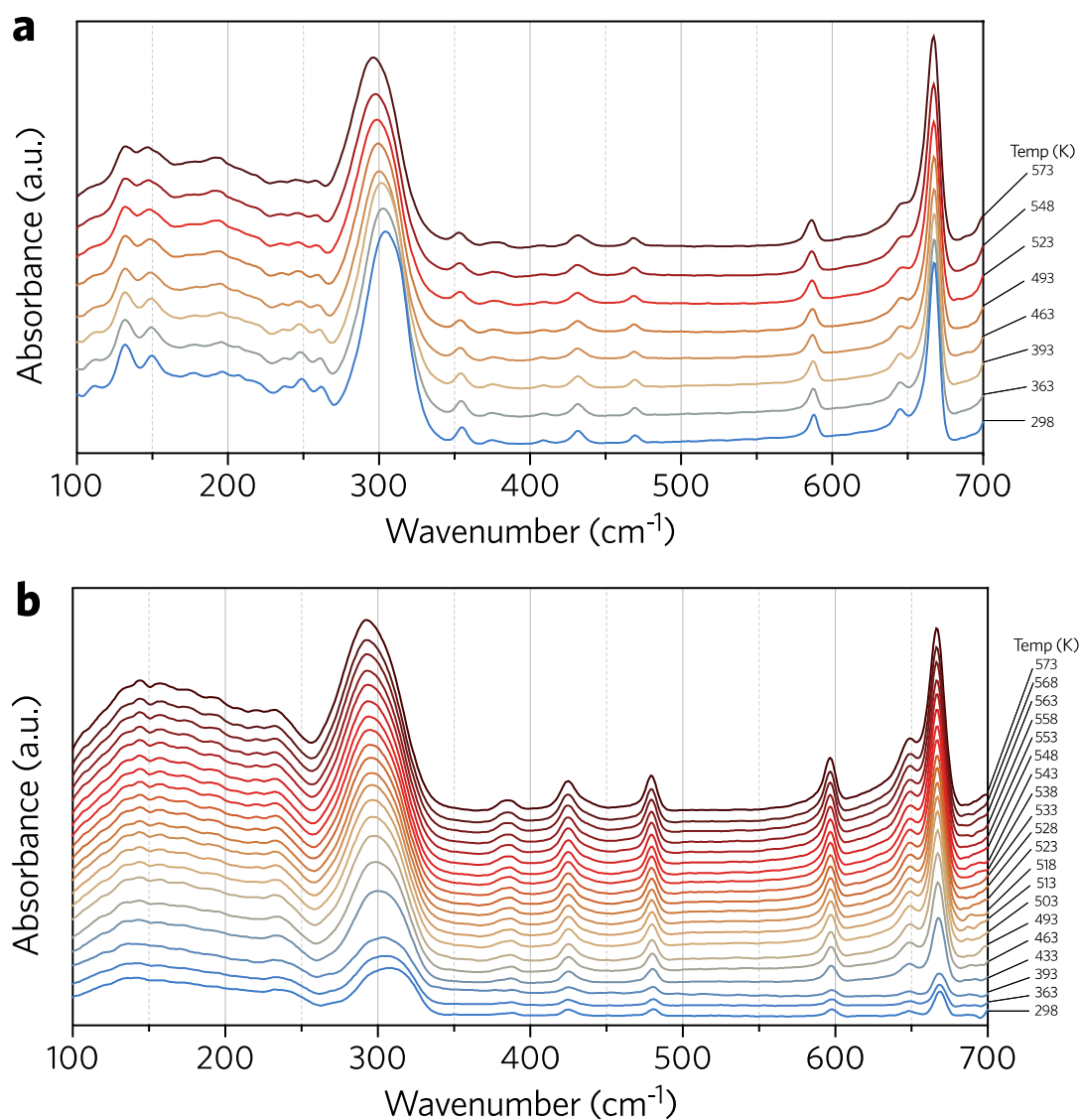


Figure S63. *In situ* synchrotron THz/FIR spectra for (a) ZIF-UC-5 (5-ClbIm) and (b) agZIF-UC-5 (5-ClbIm) at the wavenumber range of 100–700 cm^{-1} .

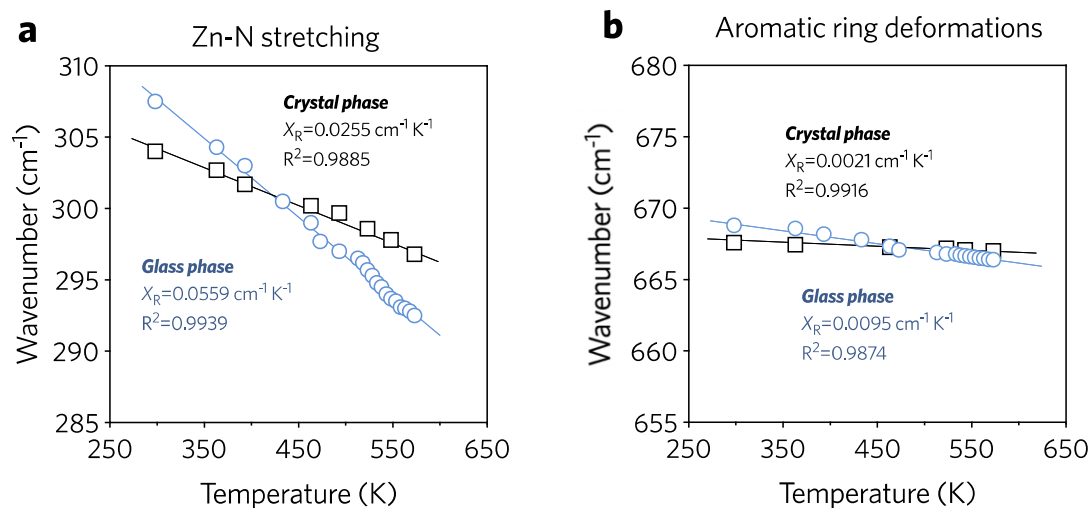


Figure S64. Deformation modes of the (a) Zn-N stretching and (b) aromatic ring deformation during the *in situ* THz FIR heating test for ZIF-UC-5 (5-ClbIm) and agZIF-UC-5 (5-ClbIm).

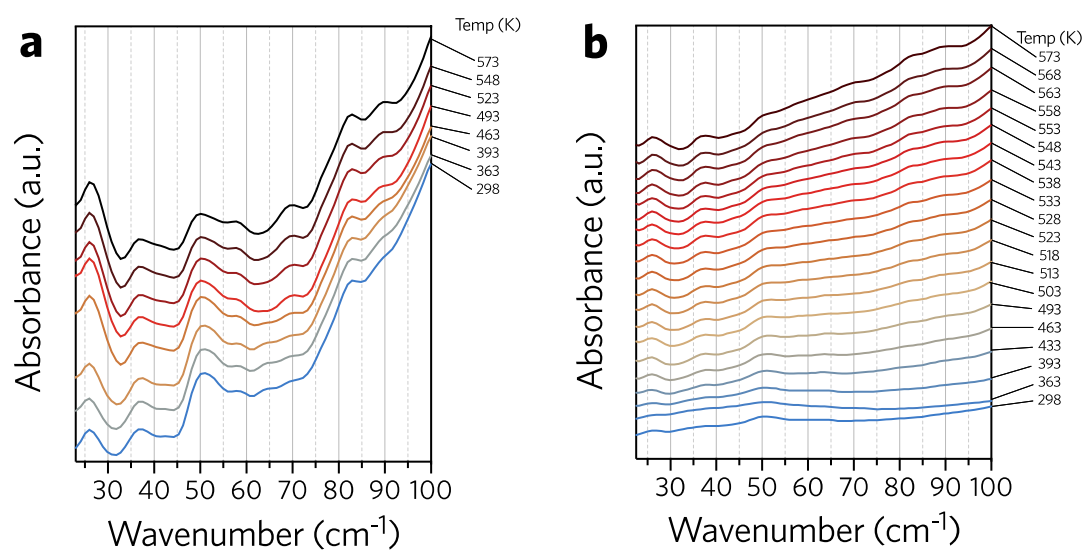


Figure S65. *In situ* synchrotron THz/FIR spectra for (a) ZIF-62 and (b) agZIF-62 at the wavenumber range of 23–100cm⁻¹.

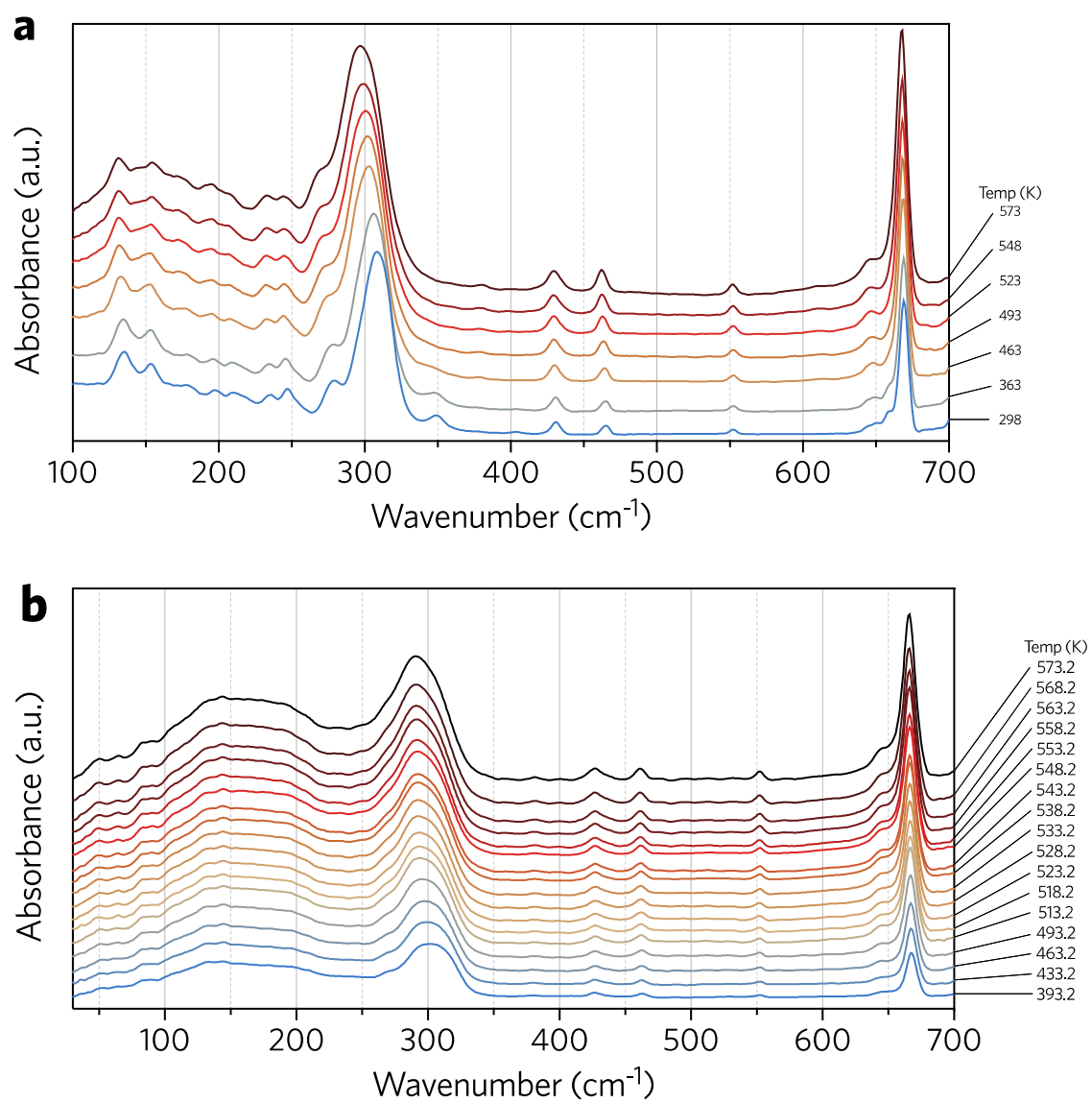


Figure S66. *In situ* synchrotron THz/FIR spectra for (a) ZIF-62 and (b) agZIF-62 at the wavenumber range of 100–700 cm^{-1} .

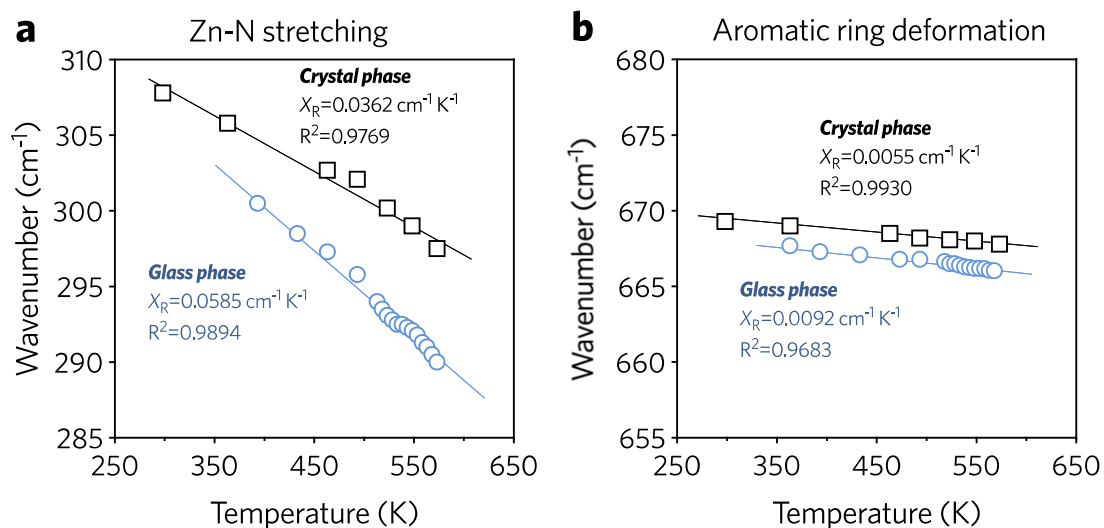


Figure S67. Deformation modes of the (a) Zn-N stretching and (b) aromatic ring deformation during the *in situ* THz FIR heating test for ZIF-62 and agZIF-62.

Table S9. Summary of curve fitting results (white: crystal phase, blue: glass phase and red: liquid phase).

Sample	Deformation mode	X_R	Standard Error of X_R	R^2	Reduced χ^2
ZIF-UC-2 (6-Cl-5-FbIm)	Zn-N stretching (Crystal phase)	0.0343	0.0014	0.9947	1.36
	Zn-N stretching (Glass phase)	0.0401	0.0036	0.9606	1.17
	Zn-N stretching (Liquid phase)	0.1038	0.0030	0.9916	1.21
	Aromatic ring deformation (Crystal phase)	0.0077	0.0008	0.9845	1.46
	Aromatic ring deformation (Glass phase)	0.0037	0.0002	0.9918	1.54
	Aromatic ring deformation (Liquid phase)	0.0291	0.0008	0.9960	1.26
ZIF-UC-3 (5-Cl-2-mbIm)	Zn-N stretching (Crystal phase)	0.0374	0.0003	0.9955	1.24
	Zn-N stretching (Glass phase)	0.0753	0.0024	0.9848	1.07
	Aromatic ring deformation (Crystal phase)	0.0055	0.0002	0.9930	1.55
	Aromatic ring deformation (Glass phase)	0.0092	0.0002	0.9683	1.57
ZIF-UC-4 (5-FbIm)	Zn-N stretching (Crystal phase)	0.0292	0.0005	0.9991	1.50
	Zn-N stretching (Glass phase)	0.0597	0.0013	0.9960	1.15
	Zn-N stretching (Liquid phase)	0.1377	0.0042	0.9891	1.35
	Aromatic ring deformation (Crystal phase)	0.0022	0.0001	0.9903	1.81
	Aromatic ring deformation (Glass phase)	0.0052	0.0003	0.9609	1.77
	Aromatic ring deformation (Liquid phase)	0.0496	0.0048	0.9429	1.43
ZIF-UC-5 (5-ClbIm)	Zn-N stretching (Crystal phase)	0.0255	0.0011	0.9885	1.17
	Zn-N stretching (Glass phase)	0.0559	0.0009	0.9939	1.07
	Aromatic ring deformation (Crystal phase)	0.0021	0.0002	0.9916	1.26
	Aromatic ring deformation (Glass phase)	0.0095	0.0003	0.9874	1.08
ZIF-62	Zn-N stretching (Crystal phase)	0.0362	0.0025	0.9769	1.20
	Zn-N stretching (Glass phase)	0.0585	0.0016	0.9894	1.08
	Aromatic ring deformation (Crystal phase)	0.0055	0.0002	0.9930	1.89
	Aromatic ring deformation (Glass phase)	0.0092	0.0003	0.9683	1.09

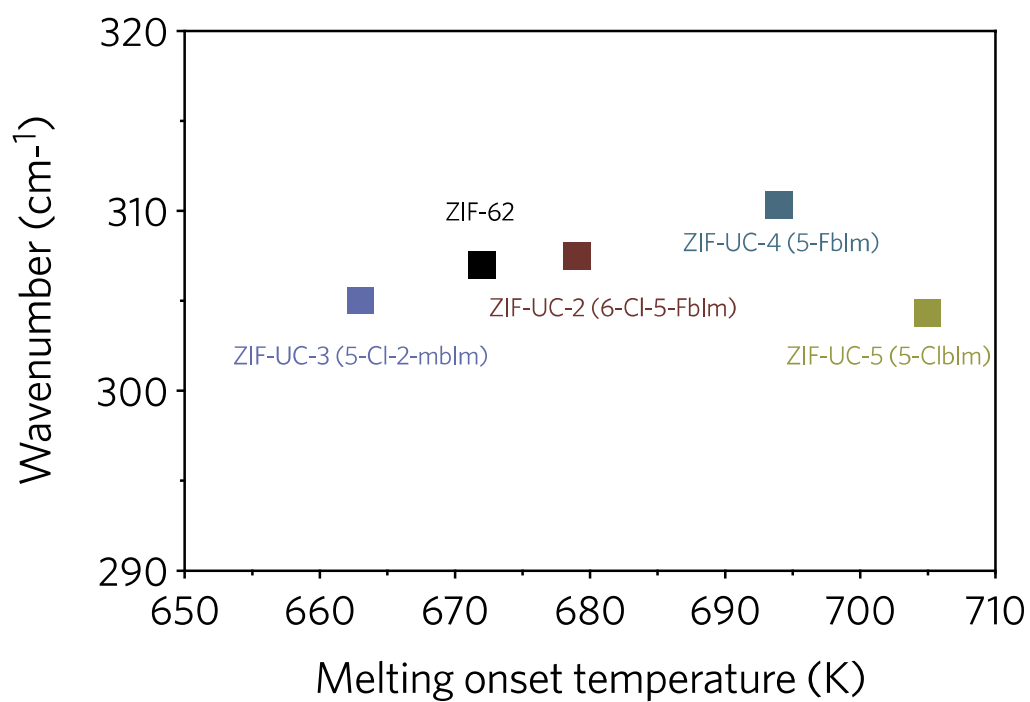


Figure S68. Correlation between the melting onset temperature and the Zn-N stretching wavenumber at room temperature (298 K).

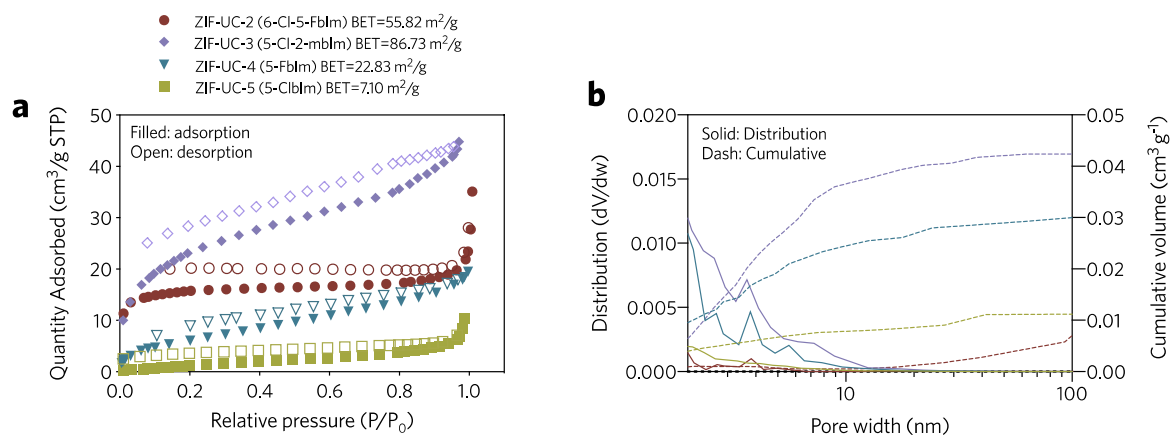


Figure S69. (a) Nitrogen adsorption-desorption isotherm at 77K for all halogenated crystalline ZIFs, and (b) pore size distribution of the ZIFs. BET surface area is calculated with the pressure range of 0.05-0.15 p/p_0 .

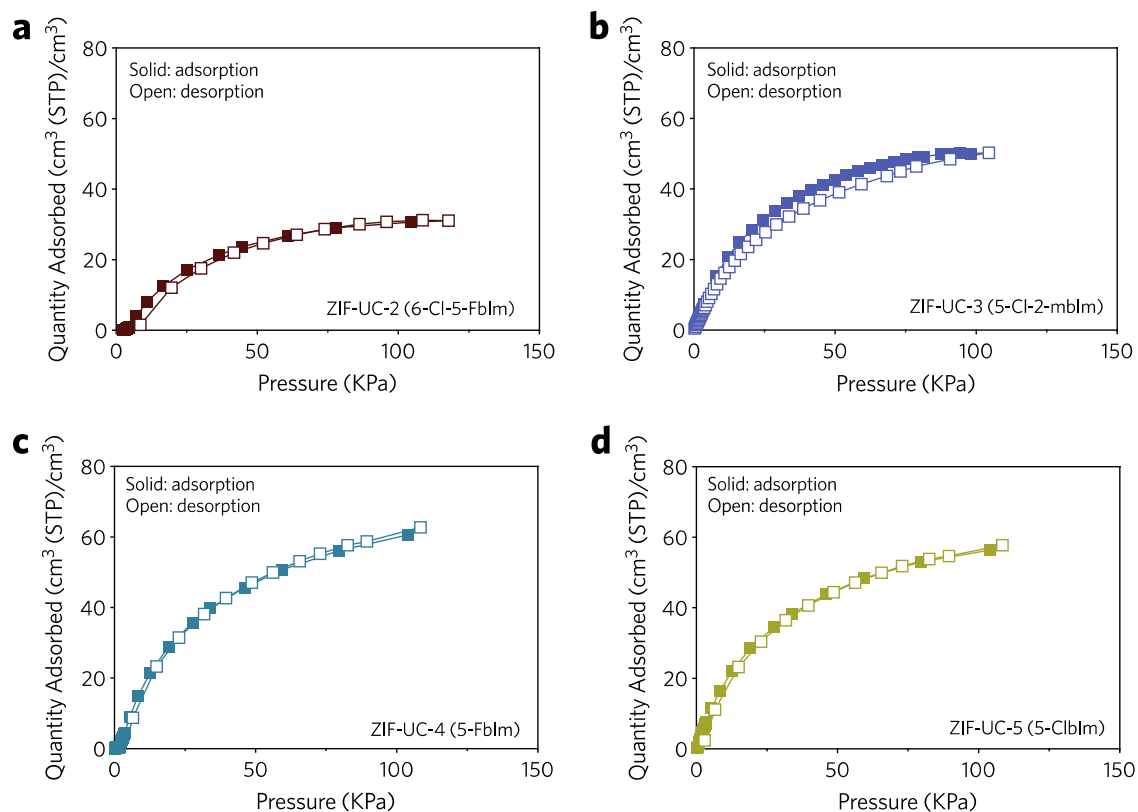
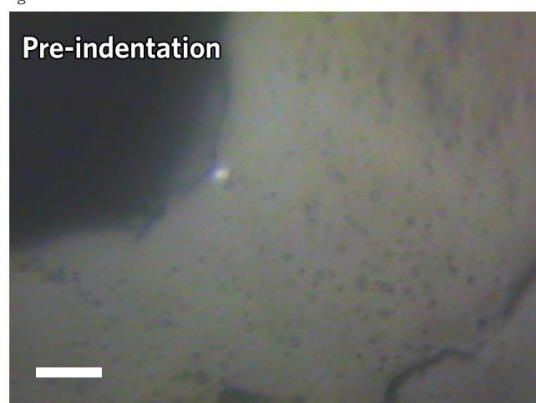
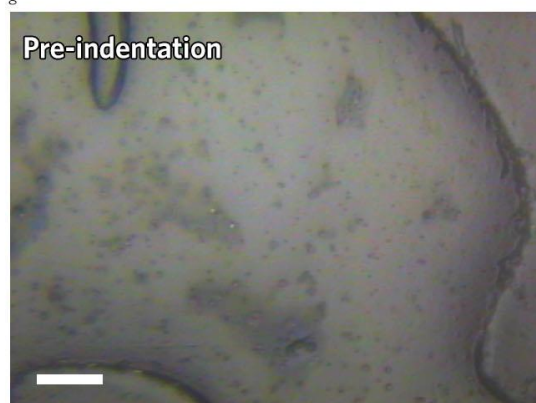


Figure S70. Carbon dioxide adsorption-desorption isotherms at 273 K for all halogenated ZIF crystals.

a_g ZIF-UC-2 (6-Cl-5-Fblm)



a_g ZIF-UC-4 (5-Fblm)



a_g ZIF-UC-5 (5-Clblm)

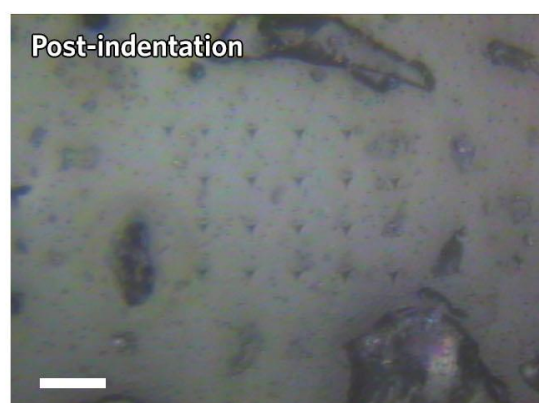
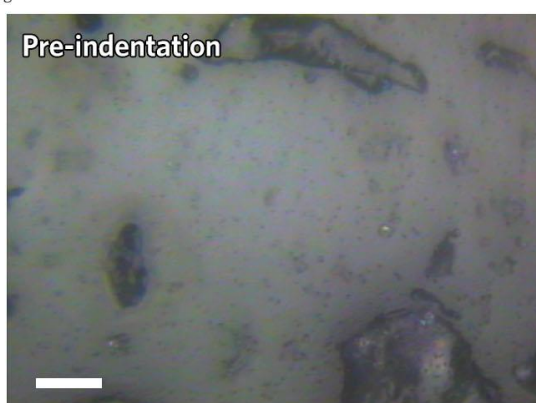


Figure S71. Optical images of the indentations for ZIF glass samples. Scale bars are 20 μm .

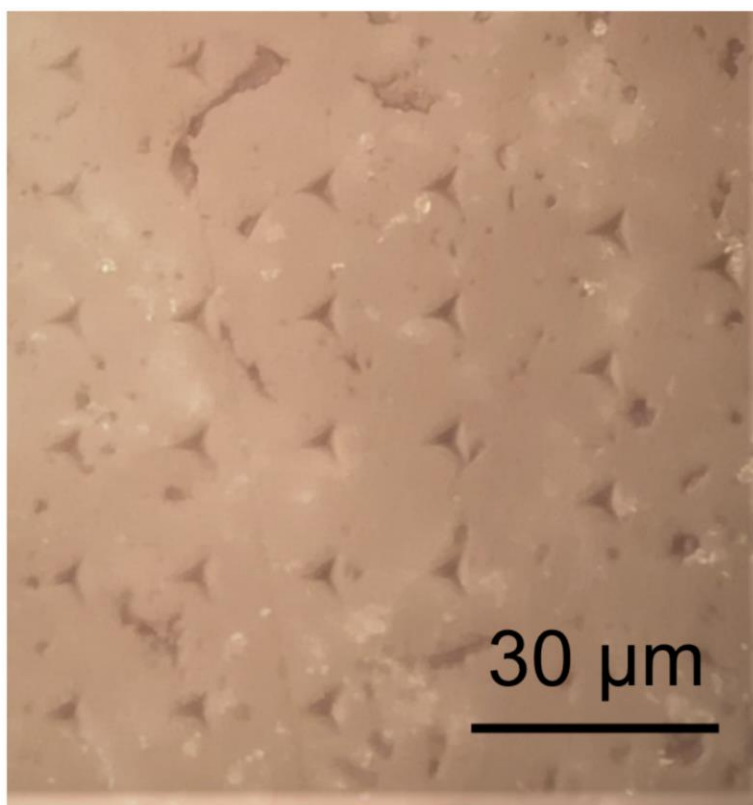


Figure S72. Optical image of the indentations for a_{gr}ZIF-UC-3 (5-Cl-2-mbIm) sample.

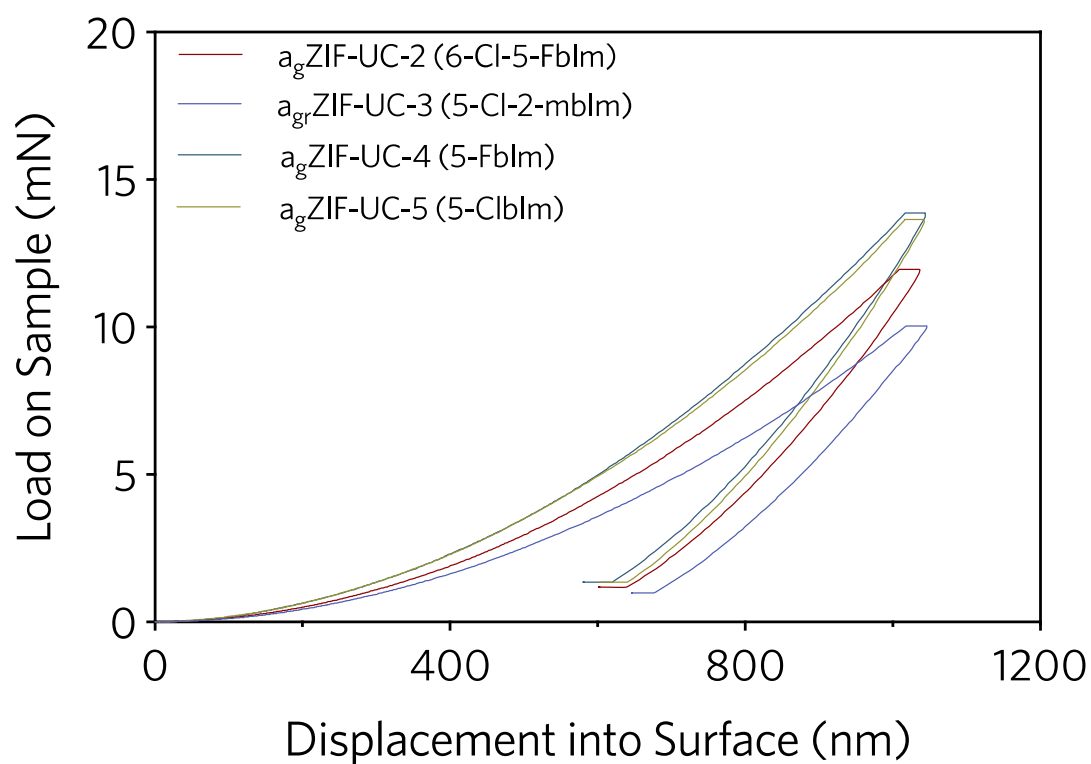


Figure S73. Typical load-displacement data for all glass samples.

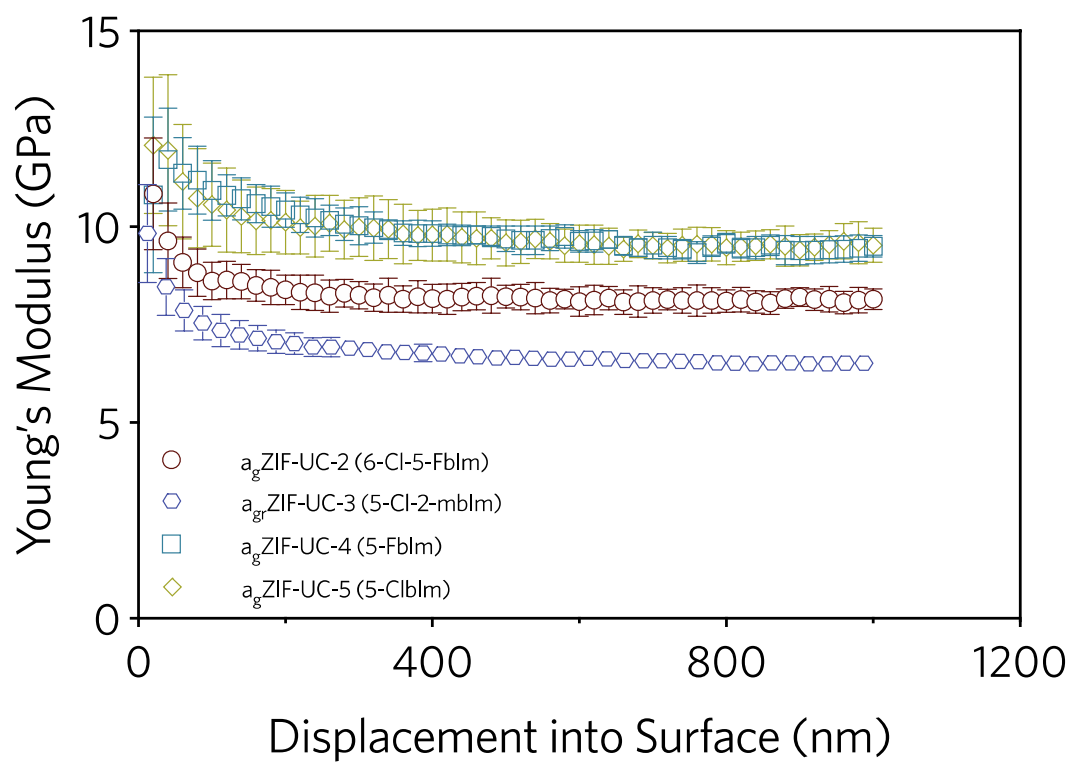


Figure S74. The variance of Young's modulus with indentation depth for ZIF glass samples, with standard deviation calculated based on 20 cycles of repeated tests.

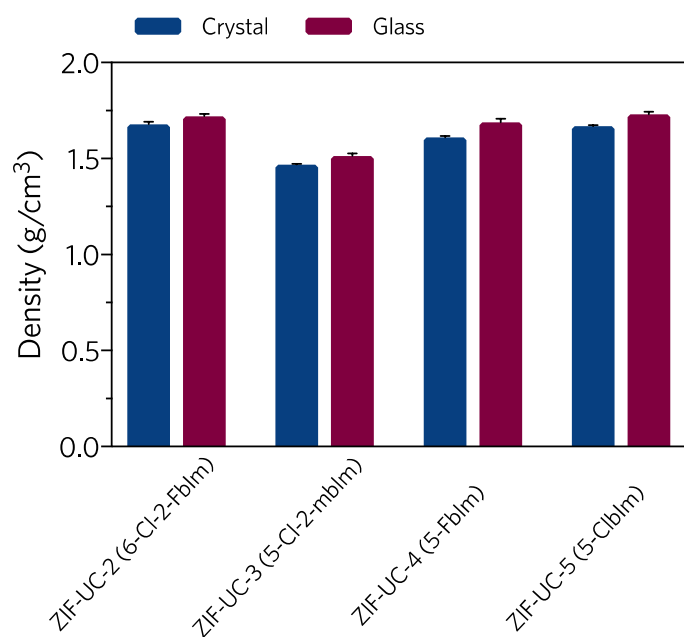


Figure S75. Density of MOF crystal and glass. Crystal samples are in blue and glass samples are in red.

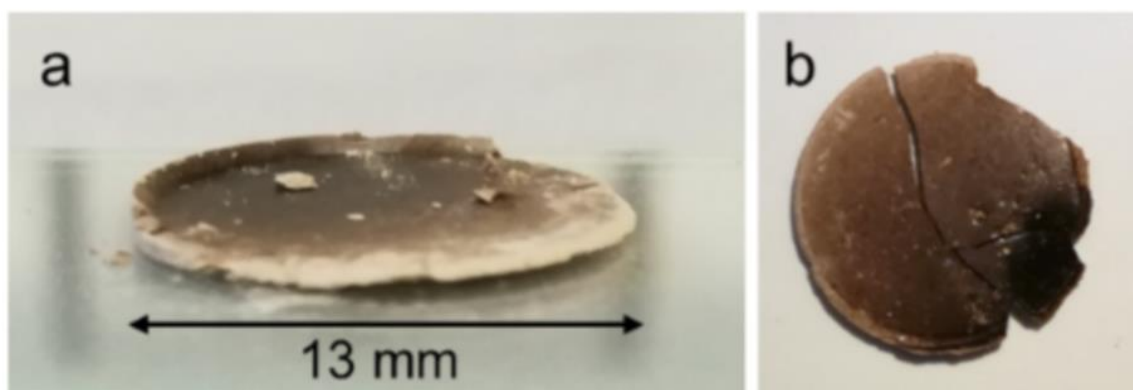


Figure S76. Photograph of a 13 mm diameter pellet of a_{gr} ZIF-UC-3 (5-Cl-2-mblm) (a) before and (b) after thermal treatment at 653 K for 5 hours under argon.

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